

FORMALIZATION OF LANGLANDS'S SECOND MAIN LEMMA FOR LOCAL EPSILON FACTORS

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ABSTRACT. We formalize in Lean 4 Langlands's Second Main Lemma for local epsilon factors over nonarchimedean local fields. The lemma compares local constants of characters of distinct intermediate fields in a bicyclic Galois extension. The proof follows the author's companion mathematical paper and uses the formalization of the First Main Lemma.

The First Main Lemma gives only a power relation, leaving a root-of-unity ambiguity. Resolving this ambiguity is the main part of the proof and requires further analysis according to ramification. The wild dyadic case is the mathematically new part of the companion proof, and its Lean formalization provides a machine-checked verification of this new case. Both mixed and equal characteristic are included. We give the exact Lean statement, identify the declarations used in the individual cases, and describe six simplifications of the mathematical proof suggested by the formalization.

ChatGPT assisted with transferring formulas and citations from the mathematical manuscript and with adding references to the corresponding Lean source, and suggested improvements to the wording of the paper.

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1. INTRODUCTION

1.1. The Second Main Lemma. Let F be a nonarchimedean local field with residue characteristic p , and let K/F be a finite Galois extension with

$$\mathrm{Gal}(K/F) \simeq C_\ell \times C_\ell,$$

where ℓ is prime. For a finite abelian extension E/F , put

$$(1) \quad S(E/F) = \{\nu \in \mathrm{Hom}_{\mathrm{cont}}(F^\times, \mathbf{C}^\times) : \nu \circ N_{E/F} = 1\}.$$

Here $\mathrm{Hom}_{\mathrm{cont}}$ denotes continuous group homomorphisms. The elements of (1) are called norm characters.

Fix a nontrivial continuous additive character $\psi_F : F \rightarrow \mathbf{C}^\times$. For every intermediate field E , including K , set

$$(2) \quad \psi_E = \psi_F \circ \mathrm{Tr}_{E/F}.$$

A quasi-character of $E^\times$ means a continuous homomorphism $\theta : E^\times \rightarrow \mathbf{C}^\times$.

Definition 1.1 (Primitive compatible families). A family (θ_L) , indexed by the degree- ℓ intermediate fields L , of quasi-characters of $L^\times$ is *compatible* if there is a quasi-character Θ of $K^\times$ such that

$$(3) \quad \theta_L \circ N_{K/L} = \Theta \quad ([L : F] = \ell).$$

It is *primitive* if there is no quasi-character λ of $F^\times$ such that $\Theta = \lambda \circ N_{K/F}$.

Compatibility implies that Θ is $\mathrm{Gal}(K/F)$ -invariant; see Lemma 3.2. Define

$$(4) \quad \mathcal{A}_L(\theta_L) = \Delta_L(\theta_L, \psi_L) \prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F).$$

The local constant Δ_E is defined in Section 2.2. The norm-character groups in (4) are finite, by Lemma 3.1.

Theorem 1.2 (Second Main Lemma). *For every primitive compatible family (θ_L) ,*

$$(5) \quad \mathcal{A}_{L_1}(\theta_{L_1}) = \mathcal{A}_{L_2}(\theta_{L_2}) \quad ([L_1 : F] = [L_2 : F] = \ell).$$

The assertion holds in mixed and equal characteristic, for every residue characteristic p and every prime ℓ .

This is [4, Theorem 1.1]. Under local class field theory, the characters in a primitive compatible family give different induction descriptions of the same irreducible degree- ℓ Weil representation. The equality (5) is the identity between local constants of characters required by those descriptions.

1.2. The formal proof. The proposition expressing Theorem 1.2 is `Basic.SecondMainIdentity`, and the theorem proving it is `secondMainLemma`. The fields, norms, traces, characters, and local constants in this statement are those used in Sections 2 and 3. Section 11 gives the statement and explains its assumptions.

The mathematical proof is given in [4]. Its inputs are the First Main Lemma and the ramification and stationary formulas of [2], whose formalization is described in [3]. The present development uses a fixed version of that formalization. The exact source revision, together with the versions of Lean and Mathlib, is recorded in Section 12.

The formal proof uses Lean 4 [7] and Mathlib [8] and follows the case division in the companion paper. The argument first proves the consequences of compatibility and primitivity. It then constructs the characters and stationary coefficients required in each ramification case and compares the resulting local constants.

The finite sums require particular care in residue characteristic two. Equality of their polar pairings does not determine their signs. The proof compares the functions themselves and accounts for the fibers of the maps used to reindex the sums. The relevant identities and their Lean proofs are discussed in Sections 8 and 9.

The revised version of [4] incorporates six simplifications suggested by the formalization. We explain them at the points where they are used and collect the references in Section 11.4.

1.3. Plan of the paper. Section 2 recalls local constants and the First Main Lemma. Section 3 develops the consequences of compatibility and primitivity, culminating in the power relation. Sections 4 and 5 then recall Lamprecht's formula and the norm and trace calculations used in the case-by-case proof.

The tame case $\ell \neq p$ is treated in Section 6, and the wild odd-prime case $\ell = p > 2$ in Section 7. Sections 8 and 9 deal with the case $\ell = p = 2$, first establishing the finite identities and then carrying out the separate local calculations. Section 10 assembles these cases to prove the theorem.

Section 11 gives the exact theorem formalized in Lean and compares its assumptions and proof with the mathematical argument. Finally, Section 12 records the source revision and the commands needed to rebuild the formalization.

2. LANGLANDS'S FIRST MAIN LEMMA

2.1. Local fields and conductors. For a nonarchimedean local field E , normalize the valuation by $v_E(E^\times) = \mathbf{Z}$. Write

$$\mathcal{O}_E = \{x : v_E(x) \geq 0\}, \quad \mathfrak{p}_E = \{x : v_E(x) \geq 1\}, \quad k_E = \mathcal{O}_E / \mathfrak{p}_E,$$

Put

$$U_E^0 = \mathcal{O}_E^\times, \quad U_E^j = 1 + \mathfrak{p}_E^j \quad (j \geq 1).$$

The conductor $m_E(\theta)$ of a quasi-character θ is the least integer $m \geq 0$ for which $\theta|_{U_E^m} = 1$. In particular, $m_E(\theta) = 0$ means that θ is unramified. For a nontrivial additive character ψ define $n_E(\psi)$ by requiring $\mathfrak{p}_E^{-n_E(\psi)}$ to be the largest fractional ideal on which ψ is trivial.

If E/F has ramification index $e_{E/F}$ and residue degree $f_{E/F}$, then

$$(6) \quad v_E|_{F^\times} = e_{E/F}v_F, \quad v_F(N_{E/F}x) = f_{E/F}v_E(x).$$

An integer occurring in a field denotes its image under $\mathbf{Z} \rightarrow E$. Thus the image of p is zero when $\text{char } E = p$.

2.2. The local constant. For $z \in \mathbf{C}^\times$ put $\text{ph}(z) = z/|z|$. Suppose first that $m = m_E(\theta) > 0$, and put $n = n_E(\psi)$. An element $\Gamma \in E^\times$ is *admissible* for (θ, ψ) if

$$(7) \quad v_E(\Gamma) = m + n.$$

Define

$$(8) \quad \sum_{u \in U_E^0/U_E^m} \psi(u/\Gamma)\theta(u)^{-1}.$$

The summand is independent of the chosen representative of u .

Proposition 2.1 (Finite-sum local constants). *The sum (8) is nonzero and has absolute value $|k_E|^{m/2}$. The number*

$$(9) \quad \Delta_E(\theta, \psi) = \theta(\Gamma) \text{ph} \left(\sum_{u \in U_E^0/U_E^m} \psi(u/\Gamma)\theta(u)^{-1} \right)$$

is independent of the admissible element Γ . It also equals

$$(10) \quad \theta(\Gamma) \text{ph} \left(\int_{\mathcal{O}_E^\times} \psi(u/\Gamma)\theta(u)^{-1} du \right)$$

for any positive Haar measure on $\mathcal{O}_E^\times$.

Proof. The additive character $x \mapsto \psi(x/\Gamma)$ induces a primitive character of $\mathcal{O}_E/\mathfrak{p}_E^m$. Extend θ^{-1} by zero on its nonunits. Its finite Fourier transform vanishes at nonunit frequencies; at unit frequencies its values differ by character values of absolute value one. The finite Fourier norm identity gives

$$\left| \sum_{u \in U_E^0/U_E^m} \psi(u/\Gamma)\theta(u)^{-1} \right|^2 = |k_E|^m.$$

This proves nonvanishing before a phase is taken. Multiplication by a unit compares two choices of Γ in (8). The integral is the finite sum multiplied by the positive measure of a coset of U_E^m . These arguments are given in [2, Proposition 2.4]. $\square$

If $m_E(\theta) = 0$ and π_E is a uniformizer, then

$$\Delta_E(\theta, \psi) = \theta(\pi_E)^{n_E(\psi)}.$$

Lemma 2.2 (Elementary local-constant identities). *For $a \in E^\times$ and a nontrivial additive character ψ ,*

$$(11) \quad \Delta_E(1, \psi) = 1, \quad \Delta_E(\theta, \psi(a \cdot)) = \theta(a)\Delta_E(\theta, \psi).$$

If $\sigma \in \text{Aut}(E)$ preserves ψ , then $\Delta_E(\theta^\sigma, \psi) = \Delta_E(\theta, \psi)$, where $\theta^\sigma(x) = \theta(\sigma^{-1}x)$. If L/F is cyclic of odd prime degree, then

$$(12) \quad \prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F) = 1.$$

Proof. The first assertions follow by changes of variables and finite character orthogonality; see [2, Lemmas 2.8, 2.10–2.11] and [4, Section 2]. For (12), every nontrivial $\nu \in S(L/F)$ has odd order, hence $\nu(-1) = 1$. Pair ν with ν^{-1} and use $\Delta_F(1, \psi_F) = 1$. The inverse-pairing identity is [2, Lemma 2.11]. The product identity is [2, Corollary 2.12]. $\square$

2.3. The First Main Lemma.

Theorem 2.3 (First Main Lemma). *Let E/F be cyclic of prime degree. For every quasi-character χ of $F^\times$ and every nontrivial additive character ψ_F ,*

$$(13) \quad \Delta_E(\chi \circ N_{E/F}, \psi_F \circ \text{Tr}_{E/F}) \prod_{\nu \in S(E/F)} \Delta_F(\nu, \psi_F) = \prod_{\nu \in S(E/F)} \Delta_F(\chi\nu, \psi_F).$$

See [2, Theorem 1.1] for a complete local proof. Its formalization is the subject of [3].

3. COMPATIBLE CHARACTERS AND THE POWER RELATION

Lean names linked by `\lean` belong to the namespace `LanglandsSecondMainLemma`, while names linked by `\fmllean` belong to `LanglandsFirstMainLemma`. The links point to the corresponding sources at the revisions specified in Section 12.

3.1. Norm subgroups. Recall the setup from Section 1.

Lemma 3.1 (Norm-character groups). *For every degree- ℓ intermediate field L , the groups $S(L/F)$ and $S(K/L)$ have order ℓ , and $S(K/F)$ has order ℓ^2 . Norm pullback gives an exact sequence*

$$(14) \quad 1 \longrightarrow S(L/F) \longrightarrow S(K/F) \xrightarrow{\xi \mapsto \xi \circ N_{L/F}} S(K/L) \longrightarrow 1.$$

In particular, the group $\text{Gal}(L/F)$ acts trivially on $S(K/L)$.

For distinct degree- ℓ intermediate fields L_1, L_2 ,

$$(15) \quad N_{L_1/F}(L_1^\times) N_{L_2/F}(L_2^\times) = F^\times.$$

Proof. The norm-index and norm-subgroup arguments used for these assertions are in [4, Appendix D]. For example, the kernel in (14) is $S(L/F)$ by its definition. The orders give surjectivity. The two norm subgroups in (15) are distinct subgroups of index ℓ ; their product is consequently $F^\times$. $\square$

3.2. Compatibility and Primitivity.

Lemma 3.2 (Descent of invariant characters). *If Θ is $\text{Gal}(K/F)$ -invariant, there is a compatible family of characters with common pullback Θ . Conversely, the common norm pullback of characters of two distinct degree- ℓ intermediate fields is $\text{Gal}(K/F)$ -invariant. Such a prescribed pair can therefore be extended to a compatible family.*

Proof. Fix a degree- ℓ intermediate field L . Hilbert 90 says that the kernel of $N_{K/L}$ consists of the elements $\sigma(x)/x$, for a generator $\sigma \in \text{Gal}(K/L)$. Invariance of Θ shows that

$$N_{K/L}(x) \longmapsto \Theta(x)$$

is a well-defined character of $N_{K/L}(K^\times) \subset L^\times$. This norm subgroup is open of finite index. Combining it with the continuity of Θ , we see the character above is continuous and extends to a continuous character of $L^\times$. See [4, Lemmas C.1 and C.2].

For the converse, a norm pullback from L is invariant under $\text{Gal}(K/L)$. For two distinct L , these subgroups generate $\text{Gal}(K/F)$. $\square$

The descent in Lemma 3.2 is proved by `Characters.invariantCharacter_inducing_exists`. The converse is `Characters.compatible_invariant`.

Lemma 3.3 (Conjugate twists). *Let (θ_i) be a primitive compatible family of quasi-characters. For a degree- ℓ intermediate field L_i , if σ_i generates $\text{Gal}(L_i/F)$, the character*

$$\mu_i = \theta_i^{\sigma_i} / \theta_i$$

generates $S(K/L_i)$. Consequently,

$$(16) \quad \{\theta_i \nu : \nu \in S(K/L_i)\} = \{\theta_i^{\sigma_i^j} : 0 \leq j < \ell\}.$$

As a result, the quantity $\mathcal{A}_{L_i}(\theta_i)$ is independent of the choice of θ_i .

Proof. Compatibility and invariance of Θ imply $\mu_i \circ N_{K/L_i} = 1$. If $\mu_i = 1$, then θ_i is $\text{Gal}(L_i/F)$ -invariant. Apply the proof of Lemma 3.2 to L_i/F : there would be a character $N_{L_i/F}(x) \mapsto \theta_i(x)$ on $N_{L_i/F}(L_i^\times)$ which extends to a quasi-character λ of $F^\times$ with $\theta_i = \lambda \circ N_{L_i/F}$. Then $\Theta = \lambda \circ N_{K/F}$, contrary to primitivity. Thus $\mu_i \neq 1$, and Lemma 3.1 shows that it is a generator. Its conjugates equal itself, again by that lemma, hence $\theta_i^{\sigma_i^j} = \theta_i \mu_i^j$.

For fixed Θ and L_i , the characters whose norm pullback is Θ are precisely $\theta_i \nu$, with $\nu \in S(K/L_i)$. By (16), they are the Galois conjugates of θ_i . Lemma 2.2 therefore shows that $\mathcal{A}_{L_i}(\theta_i)$ is independent of this choice of character. $\square$

For Lemma 3.3, `Characters.ConjugateTwistData` records the isomorphism

$$\text{Gal}(L_i/F) \longrightarrow S(K/L_i), \quad \sigma_i \longmapsto \theta_i^{\sigma_i} / \theta_i.$$

The theorem `Characters.conjugacy` constructs this isomorphism from a primitive compatible pair. The resulting independence of the inducing character is `Characters.inducingChoice_independent`.

Lemma 3.4 (Transitivity of norm-character products). *For every degree- ℓ intermediate field L ,*

$$(17) \quad \prod_{\xi \in S(K/F)} \Delta_F(\xi, \psi_F) = \left(\prod_{\mu \in S(K/L)} \Delta_L(\mu, \psi_L) \right) \left(\prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F) \right)^\ell.$$

Proof. Let $R \subset S(K/F)$ be a set of representatives for $S(K/F)/S(L/F)$. Applying (13) to a quasi-character $\xi \in R$ gives

$$\Delta_L(\xi \circ N_{L/F}, \psi_L) \prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F) = \prod_{\nu \in S(L/F)} \Delta_F(\xi \nu, \psi_F).$$

Multiplying over $\xi \in R$, we obtain

$$\left(\prod_{\xi \in R} \Delta_L(\xi \circ N_{L/F}, \psi_L) \right) \left(\prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F) \right)^\ell = \prod_{\xi \in S(K/F)} \Delta_F(\xi, \psi_F).$$

The first product is

$$\prod_{\mu \in S(K/L)} \Delta_L(\mu, \psi_L),$$

which proves (17). $\square$

Proposition 3.5 (The common power relation). *For a primitive compatible family (θ_L) and every degree- ℓ intermediate field L ,*

$$(18) \quad \mathcal{A}_L(\theta_L)^\ell = \Delta_K(\Theta, \psi_K) \prod_{\xi \in S(K/F)} \Delta_F(\xi, \psi_F).$$

Consequently,

$$(19) \quad \left(\frac{\mathcal{A}_{L_1}(\theta_{L_1})}{\mathcal{A}_{L_2}(\theta_{L_2})} \right)^\ell = 1.$$

Proof. Apply (13) to K/L and θ_L . By (16) and automorphism invariance in Lemma 2.2, its right side is $\Delta_L(\theta_L, \psi_L)^\ell$. Thus

$$\Delta_L(\theta_L, \psi_L)^\ell = \Delta_K(\Theta, \psi_K) \prod_{\mu \in S(K/L)} \Delta_L(\mu, \psi_L).$$

Then multiply by $\left(\prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F) \right)^\ell$ and use (17). All local constants are nonzero, so division gives the last assertion. $\square$

The odd-degree norm-character product identity (12) and the odd-prime specialization of the common power relation (18)–(19) are proved by `Characters.oddPower`. These are also [4, Corollary 4.3].

3.3. Change of additive character. Put

$$(20) \quad \epsilon_L = \prod_{\nu \in S(L/F)} \nu.$$

This is the trivial character for odd ℓ and is the nontrivial quadratic character ω_L for $\ell = 2$.

Lemma 3.6 (Common restriction to the base field). *For a primitive compatible family (θ_L) , the character $\theta_L|_{F^\times} \epsilon_L$ is independent of L . Consequently the quotient (19) is unchanged when ψ_F is replaced by $\psi_F(a \cdot)$, for $a \in F^\times$.*

Proof. By (15), it suffices to compare the restrictions on $N_{L_i/F}(L_i^\times)$, for $i = 1, 2$. Let $a = N_{L_1/F}(z)$, with $z \in L_1^\times$. Compatibility gives

$$\theta_{L_2}(a) = \Theta(z) = \theta_{L_1}(z)^\ell.$$

On the other hand, (16) gives

$$\theta_{L_1}(a) = \theta_{L_1}(z)^\ell \mu_{L_1}(z)^{\ell(\ell-1)/2}.$$

For odd ℓ , the last factor is one and the ϵ_{L_i} are trivial.

For $\ell = 2$, $\omega_{L_2} \in S(K/F)$. Its image in $S(K/L_1)$ is $\omega_{L_2} \circ N_{L_1/F}$ is nontrivial; otherwise $\omega_{L_2} \in S(L_1/F)$, hence $\omega_{L_2} = \omega_{L_1}$, which would imply $N_{L_2/F}(L_2^\times) = N_{L_1/F}(L_1^\times)$, a contradiction to (15). Thus

$$\mu_{L_1} = \omega_{L_2} \circ N_{L_1/F}.$$

Since $a = N_{L_1/F}(z)$, we have $\omega_{L_1}(a) = 1$. Hence

$$\begin{aligned} \theta_{L_1}(a) \epsilon_{L_1}(a) &= \theta_{L_1}(z)^2 \mu_{L_1}(z) \\ &= \theta_{L_1}(z)^2 \omega_{L_2}(a) \\ &= \theta_{L_2}(a) \epsilon_{L_2}(a). \end{aligned}$$

Finally, if ψ_F is replaced by $\psi_F(a \cdot)$, then by (11) $\mathcal{A}_L(\theta_L)$ is multiplied by

$$\theta_L(a) \prod_{\nu \in S(L/F)} \nu(a) = \theta_L(a) \epsilon_L(a),$$

which is independent of L by the first assertion. Therefore the quotient in (19) is unchanged. $\square$

The restriction identity in Lemma 3.6 is also proved in [Characters.conjugacy](#).

4. STATIONARY CLASSES AND FINITE SUMS

4.1. Lamprecht's formula. Let E be a nonarchimedean local field, let $\theta : E^\times \rightarrow \mathbf{C}^\times$ be a quasi-character, and let $\psi_E : E \rightarrow \mathbf{C}^\times$ be a nontrivial continuous additive character. Suppose that $m = m_E(\theta) > 1$, and write

$$m = 2d + \varepsilon, \quad \varepsilon \in \{0, 1\}.$$

In particular $d \geq 1$. Put $n = n_E(\psi_E)$.

Definition 4.1 (Stationary representatives and critical functions). Fix an admissible element Γ for (θ, ψ_E) . A *stationary representative* is an element $b \in \mathcal{O}_E^\times$ such that

$$(21) \quad \theta(1+z) = \psi_E(bz/\Gamma) \quad (z \in \mathfrak{p}_E^{d+\varepsilon}).$$

Put

$$g = b/\Gamma, \quad v_E(g) = -m - n.$$

Define the *critical function*

$$(22) \quad H_E : V_E \longrightarrow \mathbf{C}^\times, \quad V_E = \mathfrak{p}_E^d / \mathfrak{p}_E^{d+\varepsilon}, \quad H_E(x) = \psi_E(gx) \theta(1+x)^{-1}.$$

Set

$$(23) \quad g_E = |V_E|^{-1/2} \sum_{x \in V_E} H_E(x).$$

Well-definedness of (22). For $x \in \mathfrak{p}_E^d$ and $z \in \mathfrak{p}_E^{d+\varepsilon}$,

$$\frac{1+x+z}{1+x} = 1 + \frac{z}{1+x}, \quad \frac{z}{1+x} \in \mathfrak{p}_E^{d+\varepsilon}.$$

By (21),

$$\frac{H_E(x+z)}{H_E(x)} = \frac{\psi_E(gz)}{\theta(1+z/(1+x))} = \psi_E\left(\frac{gxz}{1+x}\right) = 1,$$

because $v_E(gxz) \geq -m-n+d+(d+\varepsilon) = -n$. Thus the value depends only on the class of x in V_E . For $\varepsilon = 0$, $V_E = \mathfrak{p}_E^d/\mathfrak{p}_E^d$ is the trivial group, and $H_E(0) = 1$, so (23) gives $g_E = 1$. $\square$

Theorem 4.2 (Lamprecht's formula). *Stationary representatives exist. For every b satisfying (21),*

$$(24) \quad \Delta_E(\theta, \psi_E) = \theta(\Gamma/b)\psi_E(b/\Gamma)g_E, \quad |g_E| = 1.$$

Proof. Put $s = d + \varepsilon$, so $m - s = d$ and $2s \geq m$. The map $x \mapsto \theta(1+x)$ is an additive character of $\mathfrak{p}_E^s/\mathfrak{p}_E^m$: the product term xy lies in $\mathfrak{p}_E^m$. The pairing

$$\frac{\mathfrak{p}_E^{-m-n}}{\mathfrak{p}_E^{-s-n}} \times \frac{\mathfrak{p}_E^s}{\mathfrak{p}_E^m} \longrightarrow \mathbf{C}^\times, \quad (g, x) \longmapsto \psi_E(gx),$$

is perfect. Indeed, by definition $\mathfrak{p}_E^{-n}$ is the largest fractional ideal on which ψ_E is trivial. Hence

$$\psi_E(gx) = 1 \quad \text{for all } x \in \mathfrak{p}_E^s \iff g\mathfrak{p}_E^s \subset \mathfrak{p}_E^{-n} \iff g \in \mathfrak{p}_E^{-s-n},$$

and similarly

$$\psi_E(gx) = 1 \quad \text{for all } g \in \mathfrak{p}_E^{-m-n} \iff x \in \mathfrak{p}_E^m.$$

Thus the two displayed subgroups are exactly the annihilators, and the pairing identifies $\mathfrak{p}_E^{-m-n}/\mathfrak{p}_E^{-s-n}$ with the character group of $\mathfrak{p}_E^s/\mathfrak{p}_E^m$. Therefore there is a unique class

$$g \in \mathfrak{p}_E^{-m-n}/\mathfrak{p}_E^{-s-n}$$

such that $\theta(1+x) = \psi_E(gx)$ for $x \in \mathfrak{p}_E^s$; choose a representative $g \in \mathfrak{p}_E^{-m-n}$. If $v_E(g) > -m-n$, then for $x \in \mathfrak{p}_E^{m-1}$ one has $gx \in \mathfrak{p}_E^{-n}$, so $\theta(1+x) = 1$. This would make θ trivial on U_E^{m-1} , contrary to $m_E(\theta) = m$. Hence $v_E(g) = -m-n$. Put $b = \Gamma g$; then $b \in \mathcal{O}_E^\times$ and (21) holds. Every class in U_E^0/U_E^m is uniquely of the form $t(1+x)$, where t runs through U_E^0/U_E^s and x through $\mathfrak{p}_E^s/\mathfrak{p}_E^m$. For fixed t , the contribution of the classes $u = t(1+x)$ to the sum in (8) is

$$\psi_E(t/\Gamma)\theta(t)^{-1} \sum_{x \in \mathfrak{p}_E^s/\mathfrak{p}_E^m} \psi_E((t/\Gamma - g)x).$$

By character orthogonality, the inner sum vanishes unless $t/\Gamma - g \in \mathfrak{p}_E^{-s-n}$, or equivalently $t - b \in \mathfrak{p}_E^d$. If this condition holds, the inner sum is $|k_E|^d$. The surviving classes are $t = b(1+y)$ with $y \in \mathfrak{p}_E^d/\mathfrak{p}_E^s = V_E$. Consequently

$$(25) \quad \sum_{\bar{u} \in U_E^0/U_E^m} \psi_E(u/\Gamma)\theta(u)^{-1} = |k_E|^d \psi_E(g)\theta(b)^{-1} \sum_{y \in V_E} H_E(y).$$

For even m , the last sum is 1. For odd m , the identity (26), proved below, gives

$$\left| \sum_{x \in V_E} H_E(x) \right|^2 = \sum_{t \in V_E} H_E(t) \sum_{y \in V_E} \psi_E(gty) = |V_E|.$$

For $t \neq 0$ in V_E , a representative has valuation d , so $gt\mathfrak{p}_E^d = \mathfrak{p}_E^{-n-1}$. The character $y \mapsto \psi_E(gty)$ is therefore nontrivial, and its sum is zero. For $t = 0$, the sum is $|V_E|$. Thus (23) has absolute value 1 in both cases. Taking phases in (25) and multiplying by $\theta(\Gamma)$ gives (24), since $\theta(\Gamma)\theta(b)^{-1} = \theta(\Gamma/b)$.

The finite duality and stationary-class construction used above are proved in [2, Lemmas 4.1–4.2]. Lamprecht's formula in this normalization is [2, Theorem 4.5]. The form with the residual critical function used later is [4, Lemma 4.1]. $\square$

For fixed admissible Γ , the class of the stationary representative b in $\mathcal{O}_E/\mathfrak{p}_E^d$ is unique. This is the class denoted `stationaryNumeratorClass` in the First Main Lemma development. Its existence and uniqueness are proved by `stationaryNumeratorClass_existsUnique`. Theorems `lamprechtEven` and `lamprechtOdd` give the two parities of Theorem 4.2.

4.2. The polar pairing. Suppose $m = 2d + 1$. Multiplication of principal units gives the following identity on $V_E = \mathfrak{p}_E^d/\mathfrak{p}_E^{d+1}$.

Lemma 4.3 (Nondegeneracy of the polar pairing). *For $x, y \in V_E$,*

$$(26) \quad H_E(x + y) = H_E(x)H_E(y)\psi_E(gxy).$$

The map

$$(x, y) \mapsto \psi_E(gxy)$$

is the polar pairing of H_E ; it is a nondegenerate symmetric bicharacter of the additive group V_E .

Proof. Choose representatives $x, y \in \mathfrak{p}_E^d$. Since

$$\frac{(1+x)(1+y)}{1+x+y} = 1 + \frac{xy}{1+x+y} \equiv 1 + xy \pmod{\mathfrak{p}_E^{3d}},$$

and $3d \geq 2d + 1 = m$, the two units have the same value under θ . Also $xy \in \mathfrak{p}_E^{2d} \subset \mathfrak{p}_E^{d+1}$, so (21) gives $\theta(1 + xy) = \psi_E(gxy)$. Dividing the three values in (22) now gives (26).

If x or y is changed by an element of $\mathfrak{p}_E^{d+1}$, the change in gxy belongs to $\mathfrak{p}_E^{-n}$. Thus the polar pairing is well defined on $V_E \times V_E$. It is symmetric and is a character in each variable. For a nonzero class $x \in V_E$, its representative has valuation d . Hence $gx\mathfrak{p}_E^d = \mathfrak{p}_E^{-n-1}$. Since ψ_E is not trivial on this ideal, there exists $y \in \mathfrak{p}_E^d$ for which $\psi_E(gxy) \neq 1$. This proves nondegeneracy. $\square$

The functional equation for the residual critical function is part of [2, Theorem 4.5]. The same polar-pairing identity is equation (16) in [4, Lemma 4.1].

The identity for the polar pairing is formalized by `lamprechtHasseValue_map_add`. The nontrivial additive character defining this pairing is formalized by `lamprechtResidualAddChar_n_e_one`.

Equation (26) determines H_E only up to multiplication by an additive character of V_E . More precisely, if H' is another function satisfying (26) with the same polar pairing, then

$$\chi(x) = H'(x)H_E(x)^{-1}$$

satisfies $\chi(x + y) = \chi(x)\chi(y)$. By nondegeneracy, every additive character of V_E is of the form

$$x \mapsto \psi_E(gax)$$

for a unique $a \in V_E$.

Lemma 4.4 (Translation of critical sums). *Let V be a finite abelian group and let $H : V \rightarrow \mathbf{C}^\times$ satisfy $H(0) = 1$ and $H(x + y) = H(x)H(y)B(x, y)$ for a symmetric bicharacter B . For $a \in V$ put $H_a(x) = H(x)B(a, x)$. Then*

$$(27) \quad \sum_{x \in V} H_a(x) = H(a)^{-1} \sum_{x \in V} H(x).$$

Proof. Note $H_a(x) = H(a)^{-1}H(x + a)$; translating the finite sum gives (27). $\square$

For finite fields, this translation identity is formalized in Lean as `HasseFunction.sum_translate`. Since $B(a, \cdot)$ is an additive character,

$$H_a(x + y) = H_a(x)H_a(y)B(x, y).$$

Thus two functions with the same polar pairing B can have different sums.

4.3. Odd residue characteristic. When $p > 2$, the stationary representative can be chosen so that $H_E(x)$ is purely quadratic. In residue characteristic two there is no corresponding normalization in general, so the complete critical function must be retained.

Lemma 4.5 (Quadratic normalization). *Suppose $p > 2$. If $m = 2d + 1$, there is a stationary representative b for the fixed admissible Γ such that, with $g = b/\Gamma$,*

$$(28) \quad \theta(1 + x) = \psi_E(g(x - x^2/2)) \quad (x \in \mathfrak{p}_E^d).$$

For this choice,

$$(29) \quad H_E(x) = \psi_E(gx^2/2), \quad g_E^4 = 1.$$

For even m , the same fourth-power assertion holds because $g_E = 1$.

Proof. The map

$$U_E^d/U_E^m \longrightarrow \mathfrak{p}_E^d/\mathfrak{p}_E^m, \quad 1 + x \longmapsto x - x^2/2,$$

is an isomorphism of abelian groups. Its additivity follows by expansion, since the omitted terms have degree at least three and $3d \geq m$. Its inverse is given by $x \mapsto 1 + x + x^2/2$ on these quotients. Via this isomorphism, $\theta|_{U_E^d/U_E^m}$ becomes an additive character of $\mathfrak{p}_E^d/\mathfrak{p}_E^m$. Use the perfect pairing

$$\frac{\mathfrak{p}_E^{-m-n}}{\mathfrak{p}_E^{-d-n}} \times \frac{\mathfrak{p}_E^d}{\mathfrak{p}_E^m} \longrightarrow \mathbf{C}^\times, \quad (g, x) \longmapsto \psi_E(gx).$$

The argument in the proof of Theorem 4.2 gives a coefficient g with $v_E(g) = -m - n$ satisfying (28); put $b = \Gamma g$. Equation (29) then follows from (22). The remaining sum is a nondegenerate quadratic Gauss sum over k_E , divided by $|k_E|^{1/2}$. The classical quadratic Gauss-sum identity says that its square is the quadratic character of -1 , hence its fourth power is one. $\square$

The exact choice of stationary representative in Lemma 4.5 is not packaged as a single Lean theorem. The finite-field theorem `CriticalPolarFunction.eq_quadratic` is more general: after identifying V_E with the residue field k and writing the polar pairing in (26) as

$$\psi_E(gxy) = \psi_0(Axy)$$

for a fixed nontrivial additive character ψ_0 of k , every critical function with this polar pairing has the form

$$H(x) = \psi_0\left(\frac{A}{2}x^2 + cx\right)$$

for a unique $c \in k$; uniqueness is `CriticalPolarFunction.affineCoefficient_unique`.

Lemma 4.5 uses the special stationary representative chosen above, so that the coefficient c is zero. Writing ν for the quadratic character of k , `quadraticPhase_eq_basic` then gives

$$\text{ph}\left(\sum_{x \in k} \psi_0(Ax^2/2)\right) = \nu(A) \text{ph}\left(\sum_{x \in k} \psi_0(x^2/2)\right),$$

and `quadraticPhase_basic_sq` gives

$$\text{ph}\left(\sum_{x \in k} \psi_0(x^2/2)\right)^2 = \nu(-1).$$

Hence the phase in (29) has fourth power 1.

The general case is used later in the odd-prime calculation. Completing the square separates the resulting finite sum into the same pure quadratic Gauss sum as above and an additional additive-character factor. This step is formalized by `quadraticPhase_completeSquare`. The general quadratic-phase identity, including the linear term, is [2, Lemma 5.4]. The additional additive-character factor produced by completing the square is used in the later odd-prime calculation in [4, Lemma 6.5].

4.4. Stable twists.

Lemma 4.6 (Stable twisting). *Let χ have conductor $m > 1$, let η have conductor at most $\lfloor m/2 \rfloor$, let Γ be admissible for (χ, ψ_E) , and let b be a stationary representative for χ . Then the same Γ, b are admissible and stationary for $\eta\chi$, and*

$$(30) \quad \Delta_E(\eta\chi, \psi_E) = \eta(\Gamma/b)\Delta_E(\chi, \psi_E).$$

Proof. Write $m = 2d + \varepsilon$ and put $g = b/\Gamma$. Since $m_E(\eta) \leq d$, the character η is trivial on U_E^d . For $x \in \mathfrak{p}_E^{d+\varepsilon}$ one has $1 + x \in U_E^d$, so

$$(\eta\chi)(1 + x) = \chi(1 + x) = \psi_E(gx).$$

Thus the same b satisfies (21) for $\eta\chi$. The critical functions in (22), and hence their normalized sums in (23), are equal. Applying (24) to $\eta\chi$ and to χ gives (30). $\square$

The stable-twist formula is [2, Lemma 4.7]. The same formula is equation (13) of [4].

The Lean proof separates the two assertions in Lemma 4.6; `stationaryNumeratorClass_stableTwist` proves that χ and $\eta\chi$ have the same stationary numerator class, and the theorem `stableTwist` gives (30).

5. NORMS, TRACES, AND CONGRUENCES

5.1. Trace ideals and conductors under a norm. Let E/F be finite separable. Define $D_{E/F}$ by

$$\mathfrak{D}_{E/F}^{-1} := \{x \in E : \text{Tr}_{E/F}(x\mathcal{O}_E) \subset \mathcal{O}_F\} = \mathfrak{p}_E^{-D_{E/F}},$$

and put $e = e_{E/F}$.

The trace and norm functions are the field trace and field norm, so their tower identities are inherited from Mathlib.

Lemma 5.1 (Trace ideals and additive conductors). *For every integer j ,*

$$(31) \quad \text{Tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}_F^{\lfloor (j+D_{E/F})/e \rfloor}, \quad n_E(\psi_F \circ \text{Tr}_{E/F}) = en_F(\psi_F) + D_{E/F}.$$

Proof. The inverse different is the trace dual of $\mathcal{O}_E$. If $\text{Tr}(\mathfrak{p}_E^j) = \mathfrak{p}_F^c$, its dual ideal consists of the $y \in F$ satisfying $y\mathfrak{p}_E^j \subset \mathfrak{p}_E^{-D_{E/F}}$, or $ev_F(y) + j \geq -D_{E/F}$. This determines c and proves the first formula. The second follows by applying the first to the largest trivial ideal of ψ_F . See [4, equation (7)]. $\square$

If E/F is unramified, then for every $r \geq 0$,

$$N_{E/F}(U_E^r) = U_F^r,$$

and for every character χ of $F^\times$,

$$m_E(\chi \circ N_{E/F}) = m_F(\chi).$$

These unramified norm and conductor identities are proved in [2, Lemma 3.11].

Suppose now that E/F is totally ramified and cyclic of prime degree ℓ . Its lower ramification break t is the integer for which the lower ramification groups equal $\text{Gal}(E/F)$ through t and are trivial after t . Thus $t = 0$ in the tame case. Let $\psi_{E/F}$ denote the inverse Herbrand function. The different exponent defined above and the inverse Herbrand function satisfy

$$(32) \quad D_{E/F} = (\ell - 1)(t + 1), \quad \psi_{E/F}(s) = \begin{cases} s, & 0 \leq s \leq t, \\ t + \ell(s - t), & s \geq t. \end{cases}$$

These ramification formulas are recorded in [4, Section 3].

Proposition 5.2 (Conductors under norm pullback). *Assume that E/F is totally ramified and cyclic of prime degree ℓ , with lower ramification break t . Every nontrivial character in $S(E/F)$ has conductor $t + 1$. If χ has conductor $m > t + 1$, then*

$$(33) \quad m_E(\chi \circ N_{E/F}) = \psi_{E/F}(m - 1) + 1 = \ell m - D_{E/F}.$$

For $m \leq t$, norm pullback preserves m . At $m = t + 1$, its conductor is at most m ; a strict decrease occurs precisely when twisting χ by a norm character lowers its conductor.

Proof. These are the cyclic-prime results recalled in [4, Section 3]; their proofs are in [2, Theorem 3.7, Corollary 3.9 and Proposition 3.10]. $\square$

The proof of Proposition 5.2 uses the imported `cyclicPrimeNormFiltration`.

5.2. Norm approximation below the break.

Lemma 5.3 (Norm approximation). *Let E/F be totally ramified cyclic of prime degree ℓ , with lower ramification break t . For $A \in F^\times$ and $0 \leq j \leq t$, there is $x \in E^\times$ such that*

$$v_E(x) = v_F(A), \quad \frac{N_{E/F}(x)}{A} \in U_F^j.$$

Proof. Norm induces an isomorphism

$$E^\times / U_E^t \xrightarrow{\sim} F^\times / U_F^t.$$

Choose x whose norm has the same class as A modulo U_F^t . Then $N_{E/F}(x)/A \in U_F^t \subset U_F^j$. Since E/F is totally ramified, $v_F(N_{E/F} x) = v_E(x)$, hence $v_E(x) = v_F(A)$. $\square$

The norm-filtration isomorphism used in the proof is [2, Theorem 3.7(c)].

5.3. Approximation by an element of a subfield. The following lemma has two later uses. First, in the totally ramified C_p^2 -extension of Section 7.2, with $p > 2$, it is used in Lemma 7.12 to choose $\Delta \in K$ and $a \in L_2$ such that

$$\Delta^p - \Delta = a, \quad v_K(\Delta) = -t, \quad v_{L_2}(a) = -t, \quad K = L_2(\Delta).$$

Here t is the lower break of K/L_2 ; in mixed characteristic the construction uses the bound (64). This application is [4, Proposition 7.4].

Second, let M/A be a totally ramified C_p^2 -extension, let E/A be an intermediate extension of degree p , and let σ generate $\text{Gal}(E/A)$. If $x = \sigma(y)/y$ with $y \in E^\times$, the lemma is used to choose $a \in A$ such that

$$v_E(y - a) = \max_{b \in A} v_E(y - b).$$

This application is [4, Lemma 9.32]. The approximation statement itself is [4, Lemma 7.2].

Lemma 5.4 (Best additive approximation). *Let L/M be a finite totally ramified extension with ramification index e , let σ be a M -automorphism of L , and let $y \in L$ satisfy $\sigma y \neq y$. There is $a \in M$ such that*

$$q := v_L(y - a) = \max_{b \in M} v_L(y - b), \quad e \nmid q, \quad q \leq v_L(\sigma y - y).$$

Proof. For every $b \in M$, valuation invariance gives

$$v_L(y - b) \leq v_L(\sigma(y - b) - (y - b)) = v_L(\sigma y - y).$$

The values on the left are finite integers and form a nonempty set bounded above. Choose a attaining the maximum q . If $q = ek$, choose a uniformizer π_M . The element $(y - a)\pi_M^{-k}$ is a unit in L . Since the residue fields agree, it has the same residue as some $u \in \mathcal{O}_M^\times$. Then $v_L(y - a - u\pi_M^k) > q$, a contradiction. $\square$

The original proof obtained a best approximation from the closedness of the subfield. Here $v_L(y - b) \leq v_L(\sigma y - y)$ bounds all approximation valuations by a finite integer, so their nonempty set has a maximum directly; see [4, Remark 7.3]. The best-approximation step in Lemma 5.4 is used inside the formal proof of the Artin–Schreier construction `oddAS_approximate_generator`.

5.4. Symmetric functions and truncated logarithms. The formulas in this subsection are used later in the odd-prime calculation for two distinct purposes. The norm expansion and the valuation bound on its intermediate symmetric terms control the error terms occurring in norms and traces. The truncated logarithm then converts multiplicative expressions on deep unit groups into additive expressions modulo the precise ideals on which the relevant additive characters are trivial. These are the inputs used in the totally ramified calculation of Section 7.

For a finite Galois extension E/F of degree s , let $E_j(x)$ be the j th elementary symmetric function of the s conjugates of x . Then

$$(34) \quad N_{E/F}(1+x) = 1 + \text{Tr}_{E/F}(x) + E_2(x) + \cdots + N_{E/F}(x).$$

If E/F is wildly ramified cyclic of degree p with lower break t , then for $1 \leq j < p$,

$$(35) \quad v_F(E_j(x)) \geq \left\lfloor \frac{jv_E(x) + (p-1)(t+1)}{p} \right\rfloor.$$

This is equation (11) of [4]. The estimate remains valid for negative $v_E(x)$; this is used later for stationary coefficients of negative valuation and for the error terms in the totally ramified odd-prime calculation. See [4, Sections 7 and 9].

For $p > 2$, put

$$(36) \quad P(Z) = \sum_{j=1}^{p-1} \frac{Z^j}{j}.$$

Its coefficients belong to $\mathbf{Z}_{(p)}$, so it can be evaluated in both mixed and equal characteristic p . It is the truncation of $-\log(1-Z)$.

Lemma 5.5 (Truncated logarithms of products). *The polynomial*

$$P(Z+W-ZW) - P(Z) - P(W)$$

has no monomial of total degree less than p . For a Galois extension E/F , the difference

$$(37) \quad P(1 - N_{E/F}(1-z)) - \text{Tr}_{E/F}(P(z))$$

is therefore a polynomial in the conjugates of z whose monomials have total degree at least p .

Proof. The formal identity $-\log((1-Z)(1-W)) = -\log(1-Z) - \log(1-W)$ gives equality in every total degree less than p , where all denominators are invertible in $\mathbf{Z}_{(p)}$. Iteration gives the assertion for a product of conjugates. This is a polynomial calculation; convergence of an infinite logarithm is not being assumed. $\square$

Lemma 5.6 (Norm-logarithm congruences). *Put $T = t+1$ and $q_0 = \lceil T/p \rceil$. If E/F is totally ramified cyclic of degree p with lower break t , $h \geq 0$, and $z \in \mathfrak{p}_E^{h+q_0}$, then*

$$(38) \quad P(1 - N_{E/F}(1-z)) \equiv \text{Tr}_{E/F}(P(z)) + N_{E/F}(P(z)) \pmod{\mathfrak{p}_F^{T+h}}.$$

If E/F is unramified of degree p and $z \in \mathfrak{p}_E^{q_0}$, then

$$(39) \quad P(1 - N_{E/F}(1-z)) \equiv \text{Tr}_{E/F}(P(z)) \pmod{\mathfrak{p}_F^T}.$$

Proof. Lemma 5.5 gives the first cancellation: after expanding in the p conjugates of z , every term of degree less than p vanishes. For the required modulus one uses the finer symmetric form of this error. Write e_i for the elementary symmetric functions of p indeterminates and put

$$u = 1 - \prod_{a=1}^p (1 - z_a) = e_1 - e_2 + \cdots + e_p.$$

After subtracting both $\sum_a P(z_a)$ and $\prod_a P(z_a)$, every remaining monomial has symmetric weight at least p and contains at least two positive-degree elementary-symmetric factors; the coefficients of pure powers of e_p are divisible by p . Thus Lemma 5.5 accounts for the disappearance of the terms of degree $< p$, while this refinement identifies the error terms that remain.

Specializing the z_a to the conjugates of z , the bound (35) gives the required depth for every error monomial containing a factor e_i with $i < p$. For a pure power of e_p , its coefficient is

divisible by p ; in equal characteristic the term is zero, and in mixed characteristic the trace-ideal formula applied to $\text{Tr}_{E/F}(1) = p$ supplies the additional valuation. Hence the whole error lies in $\mathfrak{p}_F^{T+h}$ in the ramified case. In the unramified case every remaining term has degree at least p , and $pq_0 \geq T$, so it lies in $\mathfrak{p}_F^T$. This gives (38) and (39). $\square$

The refined symmetric-polynomial calculation and the two resulting congruences are [4, Lemma 6.3]. The ramified congruence is formalized by `Odd.normLog`, and the unramified congruence by `Odd.unramifiedNormLog`.

The totally ramified odd-prime calculation later indexes the conjugates Δ_ξ of an Artin–Schreier generator Δ by

$$\xi \in \mathcal{T} := \{0\} \cup \mu_{p-1}.$$

To the precision required in the subsequent norm and trace calculation, Δ_ξ is replaced by $\Delta + \xi$. One then has to evaluate sums in the variable $\xi \in \mathcal{T}$. The following identities are the ones used there. The conjugate parametrization and this replacement are carried out in [4, Section 9].

Lemma 5.7 (Finite interpolation). *Let A be a field of characteristic zero or p containing μ_{p-1} , and put $\mathcal{T} = \{0\} \cup \mu_{p-1} \subset A$. For $0 \leq i < p$, $\Delta, Z \in A$, and $Z \notin -\mathcal{T}$,*

$$(40) \quad \sum_{\xi \in \mathcal{T}} \frac{(\Delta + \xi)^i}{Z + \xi} = \frac{p\Delta^i}{Z} + (p-1) \frac{(\Delta - Z)^i}{Z^p - Z}.$$

For $1 \leq i < p$ one also has

$$(41) \quad \begin{aligned} \sum_{\xi \in \mathcal{T}} (\Delta + \xi)^i &= p\Delta^i + (p-1)\mathbf{1}_{\{i=p-1\}}, \\ \sum_{\xi \in \mu_{p-1}} \xi(\Delta + \xi)^i &= (p-1)(\mathbf{1}_{\{i=p-2\}} + i\Delta\mathbf{1}_{\{i=p-1\}}). \end{aligned}$$

Proof. Since p is odd,

$$\prod_{\xi \in \mathcal{T}} (Z + \xi) = Z^p - Z.$$

The two sides of (40), viewed as rational functions of Z , have the same simple poles at $Z = -\xi$ with the same residues, and both vanish at infinity. Their difference is therefore zero. The identities in (41) follow by expanding $(\Delta + \xi)^i$ and using

$$\sum_{\xi \in \mu_{p-1}} \xi^m = 0 \quad \text{unless } p-1 \mid m,$$

in which case the sum is $p-1$. $\square$

The three identities are [4, Lemma 9.3]. The rational identity is formalized by `Finite.interpolation`; the two power sums are formalized in the same file.

In characteristic p the first term $p\Delta^i/Z$ in (40) is zero. In mixed characteristic its valuation is estimated before it is discarded.

5.5. Comparison of the proof cases. The common power relation (19) leaves an ℓ th root of unity to determine. For each comparison, choose a primitive compatible family (χ_L) of minimal conductors with explicit principal-unit formulas. The model characters χ_L are chosen so that, for one character λ of $F^\times$,

$$\theta_L = \chi_L(\lambda \circ N_{L/F}).$$

Where (30) applies, the model contributes only an explicit character value, and (13) compares the norm pullbacks over F .

Case	Models and coefficients	How equality is obtained
Tame $\ell \neq p$ Section 6	The restrictions of conductor-one models to $\mathcal{O}_L^\times$ factor through $k_L^\times$. For higher conductors, a stationary coefficient in F works in both norm pullbacks (Lemmas 6.4–6.5).	The conductor-one identity follows from finite-field Gauss sums and a leading-term congruence; $\ell \neq p$ makes the congruence distinguish the possible ℓ th roots (Proposition 6.2). All higher conductors fall in the stable range (Proposition 6.6).
Wild odd prime $\ell = p > 2$ Section 7	Model conductors depend on the breaks. The truncated logarithm P gives additive coordinates. Norm and trace estimates establish compatibility and control the corrections from changing stationary coefficients (Lemmas 7.4 and 7.13).	The explicit calculations leave a quotient of normalized quadratic Gauss sums, whose fourth power is 1 by Lemma 4.5. Its p th power is 1 by (19); since p is odd, the quotient is 1 (Propositions 7.11 and 7.19).
Wild quadratic $\ell = p = 2$ Sections 8–9	The models again depend on the breaks. Quadratic norm identities, together with norms and traces of one element, separate common factors from critical sums (Lemmas 8.2–8.4).	The remaining ambiguity is a sign. Polar pairings alone do not determine the critical sums (Lemma 4.4). Pointwise comparison of the full critical functions, with reindexing, fiber counting, and cancellation on omitted cosets, determines the sign (Lemma 8.5). Even-conductor critical sums are 1.

A recurrent reason for using norms of one $C \in K^\times$ in the wild cases is the exact identity $\theta_L(N_{K/L}(C)) = \Theta(C)$ from (3). The remaining work is to compare the additive-character factors and critical sums.

6. THE TAME CASE

6.1. The inertia subgroup.

Lemma 6.1 (Inertia in a prime-square extension). *Let I be the inertia subgroup of $\text{Gal}(K/F) \simeq C_\ell^2$. Then $|I|$ is either ℓ or ℓ^2 . If $\ell \neq p$, then $|I| = \ell$. In that case there is a unique unramified degree- ℓ intermediate field U , and every other degree- ℓ intermediate field E is totally ramified over F , and K/E is unramified. If $\ell = p$ and U/F is an unramified degree- p intermediate field, then*

$$I = \text{Gal}(K/U),$$

so K/U is totally and wildly ramified of degree p .

Proof. The quotient $\text{Gal}(K/F)/I$ is identified with the Galois group of the residue-field extension, hence is cyclic. Thus $I \neq 1$, so $|I| = \ell$ or ℓ^2 .

Assume $\ell \neq p$. The wild inertia subgroup is a p -group, hence is trivial, and tame inertia is cyclic. Therefore I is cyclic, and consequently $|I| = \ell$. Then $U = K^I$ is the maximal unramified subextension, with $[U : F] = \ell$.

Let $E = K^H$ be another degree- ℓ intermediate field. Then H and I are distinct subgroups of order ℓ , so

$$H \cap I = 1, \quad HI = \text{Gal}(K/F).$$

The inertia subgroup of E/F is $HI/H \simeq I$, so E/F is totally ramified. The inertia subgroup of K/E is $H \cap I = 1$, so K/E is unramified.

Now assume $\ell = p$ and let $H = \text{Gal}(K/U)$ for an unramified degree- p intermediate field U/F . The inertia subgroup of U/F is IH/H ; since U/F is unramified, $I \subset H$. As $I \neq 1$ and $|H| = p$, one has $I = H$. Hence the inertia subgroup of K/U is $I \cap H = H$, so K/U is totally ramified of degree p . Since the residue characteristic is p , this ramification is wild. $\square$

The inertia and residue-action facts used in the proof are [4, Lemma D.2].

For a degree- ℓ intermediate field L , put the norm-character product

$$\Lambda_L = \prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F).$$

Then (4) reads

$$\mathcal{A}_L(\theta_L) = \Delta_L(\theta_L, \psi_L) \Lambda_L.$$

For the tame case, the target is the equality (5) in Theorem 1.2. By Lemma 6.1, there is a unique unramified degree- ℓ intermediate field U . It is therefore enough to prove

$$(42) \quad \mathcal{A}_U(\theta_U) = \mathcal{A}_E(\theta_E) \quad ([E : F] = \ell, E \neq U).$$

6.2. The finite-field identity. For a finite field k , a nontrivial additive character ψ_k , and a multiplicative character χ of $k^\times$, put

$$(43) \quad \tau_k(\chi) = - \sum_{x \in k^\times} \chi(x)^{-1} \psi_k(x).$$

For a finite extension κ/k , the additive character on κ is $\psi_k \circ \text{Tr}_{\kappa/k}$.

Proposition 6.2 (Primitive finite-field Gauss-sum identity). *Let k have q elements, let $\ell \mid q-1$ be prime, and let κ/k have degree ℓ . Suppose that characters χ_κ of $\kappa^\times$ and χ_0 of $k^\times$ satisfy*

$$\chi_\kappa^\ell = \chi_0 \circ N_{\kappa/k}, \quad \chi_0|_{\mu_\ell} \neq 1.$$

Let μ be a character of $k^\times$ of order ℓ . Then

$$(44) \quad \tau_\kappa(\chi_\kappa) = \chi_0(\ell) \tau_k(\chi_0) \prod_{j=1}^{\ell-1} \tau_k(\mu^j).$$

Proof. Fix a primitive p th root of unity ζ_p and embed all character values into $\mathbf{Q}_p(\mu_{q^\ell-1}, \zeta_p)$, identifying its residue field with κ . Let $\omega_\kappa(x)$ be the unique $(q^\ell-1)$ st root of unity reducing to $x \in \kappa^\times$, and put $\omega_k = \omega_\kappa|_{k^\times}$. Write $u \equiv_\times v$ when u/v is a unit with residue 1 in this field. First take $\psi_k(x) = \zeta_p^{\text{Tr}_{\kappa/\mathbf{F}_p}(x)}$ and its trace pullback to κ . Put

$$S = \frac{q^\ell - 1}{q - 1},$$

and write $\chi_0 = \omega_k^a$, where $1 \leq a \leq q-2$ and $\ell \nmid a$. The exponent of χ_κ is aS/ℓ modulo $(q^\ell-1)/\ell$; the possible choices are Frobenius conjugate and have the same Gauss sum. Thus we may take the exponent to be exactly $a_* = aS/\ell$.

Put

$$L = \frac{\tau_\kappa(\chi_\kappa)}{\chi_0(\ell) \tau_k(\chi_0) \prod_{j=1}^{\ell-1} \tau_k(\mu^j)}.$$

The twists of χ_κ by the norm lifts of μ^j are its Frobenius conjugates. The Hasse–Davenport product and lifting formulas therefore give

$$L^\ell = 1.$$

For the canonical additive character on $\mathbf{F}_{p^f}$ and the Teichmüller character ω of $\mathbf{F}_{p^f}^\times$, the leading Gauss congruence, for $0 < a < p^f - 1$, is

$$\tau_{\mathbf{F}_{p^f}}(\omega^a) \equiv_\times \frac{\varpi^{\sum a_i}}{\prod a_i!}, \quad a = \sum a_i p^i,$$

where $\varpi = \zeta_p - 1$. Applying this to the numerator and denominator of L , the base- q digit calculation and the factorial congruence give

$$L \equiv_{\times} 1.$$

Since $L^\ell = 1$ and $\ell \neq p$, reduction is injective on μ_ℓ , so $L = 1$. Changing the canonical additive character by multiplication by $c \in k^\times$ multiplies the two sides of (44) by the same character value. Hence (44) holds for every nontrivial additive character. $\square$

The identity is [4, Theorem 5.1]. The equality $L^\ell = 1$ is equation (24), while the leading-term and factorial calculations are equations (21) and (27) of [4]. The leading Gauss congruence is [4, Theorem A.3].

Proposition 6.2 is formalized by `TameFiniteComparison`. The Hasse–Davenport identities are imported from the First Main Lemma development, and the leading Gauss congruence is formalized in `Gauss/LeadingTerm.lean`.

6.3. Characters of conductor one. Put

$$k = k_F = k_E, \quad \kappa = k_U = k_K,$$

and suppose $m_U(\theta_U) = 1$. By compatibility,

$$\theta_U \circ N_{K/U} = \theta_E \circ N_{K/E} = \Theta;$$

this is (3). By (16), primitivity implies that every twist of θ_U by $S(K/U)$ is a Galois conjugate of θ_U , hence has conductor 1. Therefore the critical conductor $1 = t + 1$ does not drop under norm pullback from U to K by [2, Proposition 3.10], so

$$m_K(\Theta) = 1.$$

Since K/E is unramified, $m_E(\theta_E) = 1$.

Let χ_κ and χ_0 be the residue characters of θ_U and θ_E . For $u \in \mathcal{O}_K^\times$, reduction of the two norm maps gives

$$\overline{N_{K/U}u} = \bar{u}^\ell, \quad \overline{N_{K/E}u} = N_{\kappa/k}(\bar{u}).$$

Thus compatibility gives

$$\chi_\kappa^\ell = \chi_0 \circ N_{\kappa/k}.$$

By Lemma 3.3, primitivity makes the Frobenius conjugate quotient of θ_U nontrivial of order ℓ . Writing the residue characters in Teichmüller exponents as in the proof of Proposition 6.2, this is equivalent to

$$\chi_0|_{\mu_\ell} \neq 1.$$

Thus Proposition 6.2 applies.

Let $n = n_F(\psi_F)$, choose a uniformizer π_F , and put

$$\Gamma = \pi_F^{n+1}.$$

Then Γ is admissible over both U and E . If ψ_k is the residual additive character defined by $\psi_F(x/\Gamma)$, the residual additive characters over U and E are

$$\psi_k \circ \text{Tr}_{\kappa/k}, \quad \psi_k(\ell \cdot),$$

respectively.

Proposition 6.3 (Conductor-one comparison). *If $\ell \neq p$ and $m_U(\theta_U) = 1$, then*

$$\mathcal{A}_U(\theta_U) = \mathcal{A}_E(\theta_E).$$

Proof. Choose a character μ of $k^\times$ of order ℓ . The conductor-one formulas are

$$(45) \quad \begin{aligned} \Delta_U(\theta_U, \psi_U) &= -\theta_U(\Gamma) \text{ph } \tau_\kappa(\chi_\kappa), \\ \Delta_E(\theta_E, \psi_E) &= -\theta_E(\Gamma) \chi_0(\ell) \text{ph } \tau_k(\chi_0). \end{aligned}$$

For the norm-character products,

$$(46) \quad \begin{aligned} \Lambda_U &= (-1)^{(\ell-1)n}, \\ \Lambda_E &= (-1)^{\ell-1} \epsilon_E(\Gamma) \prod_{j=1}^{\ell-1} \text{ph } \tau_k(\mu^j). \end{aligned}$$

Moreover Lemma 3.6 gives

$$(47) \quad \frac{\theta_U(\Gamma)}{\theta_E(\Gamma)} = \frac{\epsilon_E(\Gamma)}{\epsilon_U(\Gamma)}, \quad \epsilon_U(\Gamma) = (-1)^{(\ell-1)(n+1)}.$$

Substitution of (45)–(47) into (4) reduces the quotient $\mathcal{A}_U(\theta_U)/\mathcal{A}_E(\theta_E)$ to

$$\text{ph} \left(\frac{\tau_\kappa(\chi_\kappa)}{\chi_0(\ell) \tau_k(\chi_0) \prod_{j=1}^{\ell-1} \tau_k(\mu^j)} \right) = 1,$$

by (44). □

The formulas (45)–(46) and their substitution are the conductor-one calculation in [4, Section 5]. Proposition 6.3 is formalized by `Tame.comparison_of_conductor_one`.

6.4. Characters of higher conductor.

Lemma 6.4 (Reduction to conductor one). *Suppose $m = m_U(\theta_U) > 1$. There are characters χ_U, χ_E and a character λ of $F^\times$ such that*

$$(48) \quad \theta_U = \chi_U(\lambda \circ N_{U/F}), \quad \theta_E = \chi_E(\lambda \circ N_{E/F}),$$

and

$$(49) \quad m_U(\chi_U) = m_E(\chi_E) = 1, \quad m_F(\lambda) = m.$$

The pair (χ_U, χ_E) is primitive and compatible.

Proof. By Lemma 3.3,

$$\theta_U^\sigma / \theta_U \in S(K/U), \quad \theta_U^\sigma / \theta_U \neq 1.$$

Since K/U is tamely ramified, its break is 0, and therefore

$$m_U(\theta_U^\sigma / \theta_U) = 1.$$

This conductor statement is Proposition 5.2. Hence $\theta_U|_{U_U^1}$ is invariant. Hilbert 90 and surjectivity of the unramified unit norm give a character λ of $F^\times$ whose pullback agrees with θ_U on U_U^1 . Put

$$\chi_U = \theta_U / (\lambda \circ N_{U/F}), \quad \chi_E = \theta_E / (\lambda \circ N_{E/F}).$$

Then χ_U is trivial on U_U^1 . From (3) and $N_{U/F} \circ N_{K/U} = N_{K/F} = N_{E/F} \circ N_{K/E}$ one has

$$\chi_U \circ N_{K/U} = \chi_E \circ N_{K/E}.$$

Since K/E is unramified, $N_{K/E} : U_K^1 \rightarrow U_E^1$ is surjective; hence χ_E is trivial on U_E^1 . The common pullback of (χ_U, χ_E) is

$$\Theta(\lambda \circ N_{K/F})^{-1},$$

so the pair is primitive by Definition 1.1. Lemma 3.3 then excludes conductor zero for both characters. Hence

$$m_U(\chi_U) = m_E(\chi_E) = 1.$$

Since $m > 1$ and $\chi_U|_{U_U^1} = 1$, (48) gives

$$m_U(\lambda \circ N_{U/F}) = m_U(\theta_U) = m.$$

As U/F is unramified, $m_U(\lambda \circ N_{U/F}) = m_F(\lambda)$. □

The decomposition (48) is formalized by `Tame.tame_commonTwist`.

Lemma 6.5 (Stationary coefficients under norm pullback). *Let L/F be cyclic of prime degree ℓ , let $m > 1$, and let λ have conductor m . Suppose $g \in F^\times$ satisfies*

$$(50) \quad \lambda(1+z) = \psi_F(gz) \quad (z \in \mathfrak{p}_F^{\lceil m/2 \rceil}).$$

If L/F is unramified, then

$$(51) \quad (\lambda \circ N_{L/F})(1+x) = \psi_L(gx) \quad (x \in \mathfrak{p}_L^{\lceil m/2 \rceil}).$$

If L/F is totally ramified, then the same identity holds for

$$(52) \quad x \in \mathfrak{p}_L^s, \quad s = \lceil (1 + \ell(m-1))/2 \rceil.$$

Proof. In the totally ramified case, let $x \in \mathfrak{p}_L^s$. By the definition of E_j preceding (34), each summand of $E_j(x)$ is a product of j conjugates of x , so

$$v_L(E_j(x)) \geq js.$$

Since $E_j(x) \in F$ and $e(L/F) = \ell$,

$$v_F(E_j(x)) \geq \lceil js/\ell \rceil.$$

Moreover $2s \geq 1 + \ell(m-1)$, hence

$$\lceil js/\ell \rceil \geq m \quad (2 \leq j \leq \ell).$$

Equation (31) gives

$$v_F(\mathrm{Tr}_{L/F} x) \geq \lceil s/\ell \rceil \geq \lceil m/2 \rceil.$$

Thus (34) gives

$$N_{L/F}(1+x) \equiv 1 + \mathrm{Tr}_{L/F} x \pmod{\mathfrak{p}_F^m},$$

and (52) follows from (50). In the unramified case put $r = \lceil m/2 \rceil$. Then

$$v_F(E_j(x)) \geq jr \geq m \quad (j \geq 2), \quad \mathrm{Tr}_{L/F} x \in \mathfrak{p}_F^r,$$

by (31), and (34) gives (51) in the same way. $\square$

This norm-pullback calculation is [4, Lemma 5.2]. It is formalized by `Tame.covector`.

Proposition 6.6 (Stable comparison in the tame case). *If $\ell \neq p$ and $m_U(\theta_U) > 1$, then*

$$\mathcal{A}_U(\theta_U) = \mathcal{A}_E(\theta_E).$$

Proof. Use Lemma 6.4 and put $m = m_F(\lambda)$. Choose $\Gamma \in F^\times$ with

$$v_F(\Gamma) = m + n_F(\psi_F),$$

a stationary representative b for λ , and put

$$g = b/\Gamma.$$

Since E/F is tamely totally ramified, (33) gives

$$m_E(\lambda \circ N_{E/F}) = 1 + \ell(m-1) =: M,$$

and (31) gives $n_E(\psi_E) = \ell n_F(\psi_F) + \ell - 1$. Hence $v_E(\Gamma) = M + n_E(\psi_E)$, so the same Γ is admissible over E . Thus (51)–(52), with $s = \lceil M/2 \rceil$ in the ramified case, show that $g = b/\Gamma$ is a stationary coefficient for both norm pullbacks. The stable-twist formula (30) gives, for $L = U, E$,

$$(53) \quad \Delta_L(\theta_L, \psi_L) = \chi_L(g^{-1}) \Delta_L(\lambda \circ N_{L/F}, \psi_L).$$

Applying (13) and then (30) over F gives

$$(54) \quad \Delta_L(\lambda \circ N_{L/F}, \psi_L) \Lambda_L = \prod_{\nu \in S(L/F)} \Delta_F(\nu\lambda, \psi_F) = \epsilon_L(g^{-1}) \Delta_F(\lambda, \psi_F)^\ell.$$

Therefore

$$(55) \quad \mathcal{A}_L(\theta_L) = \chi_L(g^{-1}) \epsilon_L(g^{-1}) \Delta_F(\lambda, \psi_F)^\ell.$$

Lemma 3.6, applied to (χ_U, χ_E) , makes the right side of (55) independent of L . $\square$

Equations (53) and (54) are equations (28)–(29) of [4]. Proposition 6.6 is formalized by `Tame.comparison_of_conductor_gt_one`.

By Lemma 3.3, primitivity gives $\theta_U^\sigma/\theta_U \neq 1$, hence $m_U(\theta_U) \geq 1$. Thus Propositions 6.3 and 6.6 prove (42), hence Theorem 1.2 for $\ell \neq p$. This is [4, Theorem 5.3].

7. THE WILD ODD-PRIME CASES

Assume $\ell = p > 2$. Let (θ_L) be a primitive compatible family as in Definition 1.1. Thus $\theta_L : L^\times \rightarrow \mathbf{C}^\times$ and $\theta_L \circ N_{K/L} = \Theta$ for every degree- p intermediate field L . Write I for the inertia subgroup of $\text{Gal}(K/F)$.

The power relation (19) gives a p th root of unity. We will compare the local-constant formulas to obtain a fourth root of unity as well. Since p is odd, the two conditions force equality. For $p = 2$ they still allow -1 ; Sections 8 and 9 determine this sign by comparing critical functions.

Lemma 7.1. *For every degree- p intermediate field L , one has*

$$m_L(\theta_L) > 1, \quad \mathcal{A}_L(\theta_L) = \Delta_L(\theta_L, \psi_L).$$

Proof. Let σ generate $\text{Gal}(L/F)$. By Lemma 3.3, θ_L^σ/θ_L is a nontrivial character in $S(K/L)$. If L/F is ramified, it is totally wildly ramified. Consequently σ acts trivially on k_L and $\sigma(\pi_L)/\pi_L \in U_L^1$ for a uniformizer π_L . A character trivial on U_L^1 would therefore be σ -invariant, a contradiction. If L/F is unramified, Lemma 6.1 shows that K/L is wildly ramified. Proposition 5.2 then gives

$$1 < m_L(\theta_L^\sigma/\theta_L) \leq m_L(\theta_L).$$

Finally (12) gives $\prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F) = 1$; substitution in (4) proves the second assertion. $\square$

The conductor argument also holds for $p = 2$, but $\mathcal{A}_L$ then includes the additional factor $\Delta_F(\omega_L, \psi_F)$ in (4). The formal conjugate-twist theorem is `Characters.conjugacy`.

7.1. An unramified intermediate field. Suppose $|I| = p$, and put $U = K^I$. Then U/F is unramified of degree p . Let $E \neq U$ be a degree- p intermediate field. Then E/F is totally ramified and K/E is unramified. Moreover $K = EU$. The extensions E/F and K/U have the same lower break $t \geq 1$. Put

$$T = t + 1, \quad q_0 = \lceil T/p \rceil.$$

Lemma 7.2 (Logarithmic coordinates on unit quotients). *Let L be a local field of residue characteristic $p > 2$, and let P be the polynomial in (36). For integers $1 \leq r \leq M$ with $pr \geq M$, the map*

$$U_L^r/U_L^M \longrightarrow \mathfrak{p}_L^r/\mathfrak{p}_L^M, \quad 1 - z \longmapsto P(z),$$

is an isomorphism of abelian groups. The map $P : \mathfrak{p}_L^r \rightarrow \mathfrak{p}_L^r$ is bijective and preserves valuations.

Proof. Lemma 5.5 proves additivity on these quotients. Since $P(z) = z + z^2 R(z)$ with R integral and $P'(z)$ a unit on $\mathfrak{p}_L$, Hensel's lemma gives a unique solution of $P(z) = w$ in $\mathfrak{p}_L^r$ for every $w \in \mathfrak{p}_L^r$. The same expansion shows that $v_L(P(z)) = v_L(z)$ and that this bijection induces the stated isomorphism. This is [4, Lemma 6.2]. $\square$

The quotient isomorphism is `Odd.truncatedLogQuotientMulEquiv`. Bijectivity on the whole ideal and preservation of valuation are `Odd.truncatedLogOnLattice_bijective` and `Odd.ord_truncatedLog`, respectively.

Let ϕ be arithmetic Frobenius of U/F . By (14) and (15), norm pullback restricts to an isomorphism $S(E/F) \rightarrow S(K/U)$. Thus Lemma 3.3 gives $\tau_F \in S(E/F)$ such that

$$\tau_U := \tau_F \circ N_{U/F} = \theta_U^\phi/\theta_U.$$

Then $\tau_F \neq 1$ and $m_F(\tau_F) = T$ by Proposition 5.2.

Lemma 7.3 (Trace–norm identity for additive characters). *There is $\alpha \in F^\times$ such that*

$$\tau_F(1 - z) = \psi_F(\alpha P(z)) \quad (z \in \mathfrak{p}_F^{q_0}), \quad v_F(\alpha) = -n_F(\psi_F) - T.$$

Put $\Psi_L(x) = \psi_L(\alpha x)$ for $L = F, U, E, K$. Its largest trivial ideal is $\mathfrak{p}_L^T$, and

$$\tau_U(1 - z) = \Psi_U(P(z)) \quad (z \in \mathfrak{p}_U^{q_0}).$$

For $L/M = E/F$ or K/U and $w \in \mathfrak{p}_L^{q_0}$, one has

$$(56) \quad \Psi_M(\text{Tr}_{L/M} w + N_{L/M} w) = 1, \quad \Psi_L(w) = \Psi_M(-N_{L/M} w).$$

Proof. Choose α by Lemma 7.2 and additive duality; the exact conductor T fixes its valuation. Formula (31) gives the stated additive conductors, and (39) gives the formula for τ_U . For $L/M = E/F$ or K/U , write $w = P(z)$ using Lemma 7.2. The norm character is trivial on $N_{L/M}(1 - z)$, and (38) gives

$$1 = \Psi_M(\text{Tr}_{L/M} P(z) + N_{L/M}(P(z))).$$

This proves (56); the norm-filtration and precision checks are [4, Corollary 6.4]. $\square$

The normalized character data are constructed in `Odd.UR.oddURSetup_exists`. Given that normalization, `Odd.normPhase` proves both identities in (56). Fix α and Ψ_L as in Lemma 7.3.

The norms of a common element need not be the coefficients first chosen for the two characters.

Lemma 7.4 (Change of stationary coefficient). *Let θ have conductor $m > 1$ over a local field L of odd residue characteristic, and suppose that $\Psi_L(x) = \psi_L(\alpha x)$ has largest trivial ideal $\mathfrak{p}_L^J$. Put $d = \lfloor m/2 \rfloor$ and $s = \lceil m/2 \rceil$. There is $B \in L^\times$, of valuation $J - m$ and determined modulo $\mathfrak{p}_L^{J-d}$, such that*

$$\theta(1 - z) = \Psi_L(BP(z)) \quad (z \in \mathfrak{p}_L^d).$$

If $C = B + \eta$ and $v_L(\eta) \geq J - s$, then

$$\theta(1 - z) = \Psi_L(Cz) \quad (z \in \mathfrak{p}_L^s)$$

and

$$(57) \quad \Delta_L(\theta, \psi_L) = \theta((- \alpha C)^{-1}) \Psi_L(-C) \Psi_L\left(\frac{\eta^2}{2B}\right) g_L, \quad g_L \in \mu_4.$$

The correction factor $\Psi_L(\eta^2/(2B))$ is 1 if m is even or if $v_L(\eta) \geq J - d$; for even m one also has $g_L = 1$.

Proof. Since $pd \geq m$, Lemma 7.2 and the additive pairing give B with the stated ambiguity and valuation. On $\mathfrak{p}_L^d$ the terms of $P(z)$ of degree at least three do not contribute, because $3d \geq m$. Thus

$$\theta(1 + z) = \Psi_L(-Bz + Bz^2/2).$$

Apply (24) with coefficient $-\alpha C$. For odd $m = 2d + 1$, choose $\delta \in L$ with $v_L(\delta) = d$. The critical function on k_L is

$$\bar{z} \mapsto \Psi_L(-\eta\delta z - B\delta^2 z^2/2).$$

Here $\eta/(B\delta)$ is integral. Completing the square gives $\Psi_L(\eta^2/(2B))$ times the normalized pure quadratic sum, which belongs to μ_4 by (29). For even conductor the critical quotient is trivial. Finally $v_L(\eta^2/B) \geq J$ in the two stated cases, so $\Psi_L(\eta^2/(2B)) = 1$. This calculation is [4, Lemma 6.5]. $\square$

The coefficient, its ambiguity, and the full formula (57), including the assertions in even conductor, are proved by `Odd.enhancedStationary`.

For $p = 2$ and odd $m = 2d + 1$, the first step fails because $2d < m$: the unit linearization does not reach the critical ideal $\mathfrak{p}_L^d$. The completion of the square here therefore cannot replace the full critical-function comparison required in Section 9.

Proposition 7.5 (Compatible characters of conductor T). *There are $c \in \mathcal{O}_F^\times$, $d \in \mathcal{O}_U^\times$ with $d^p - d = c$, and characters χ_U^0, χ_E^0 of conductor T , forming a primitive compatible pair, such that*

$$(\chi_U^0)^\phi / \chi_U^0 = \tau_U,$$

$$\chi_U^0(1 - z) = \Psi_U(d^p P(z)) \quad (z \in \mathfrak{p}_U^{q_0}), \quad \chi_E^0(1 - z) = \Psi_E(cP(z)) \quad (z \in \mathfrak{p}_E^{q_0}).$$

One may choose $\bar{d}^{|k_F|} - \bar{d} = 1$; then $\bar{c}$ has absolute trace 1.

Proof. This is the construction of [4, Proposition 6.6] for τ_F chosen above. It first extends the displayed character on U^{q_0} to $U^\times$ with the prescribed conjugate quotient. Its pullback by $N_{K/U}$ is invariant under $\text{Gal}(K/F)$, and the descent of Lemma 3.2 gives χ_E^0 . The congruences (38)–(39) and (56) give its displayed formula. The unit coefficients d, c and the conductor of Ψ_L give exact conductor T on both fields. If the common pullback were $\xi \circ N_{K/F}$, then $\chi_U^0 / (\xi \circ N_{U/F}) \in S(K/U)$ would be ϕ -invariant by Lemma 3.1. This would make χ_U^0 invariant, contrary to its conjugate quotient $\tau_U \neq 1$. $\square$

The compatible characters, their exact conductors, and their principal-unit formulas are constructed by [Odd.UR.models](#).

The next decomposition is the wild analogue of (48): the fixed conductor- T pair (χ_U^0, χ_E^0) replaces the conductor-one tame pair.

Proposition 7.6 (A common base-field twist). *After changing χ_E^0 by an unramified character in $S(K/E)$ if necessary, there is a quasi-character λ of $F^\times$ such that*

$$(58) \quad \theta_U = \chi_U^0(\lambda \circ N_{U/F}), \quad \theta_E = \chi_E^0(\lambda \circ N_{E/F}).$$

For some integer $h \geq 0$, their conductors are

$$m_U(\theta_U) = T + h, \quad m_E(\theta_E) = T + ph.$$

If $h > 0$, then $m_F(\lambda) = T + h$; if $h = 0$, then $m_F(\lambda) \leq T$.

Proof. The matching conjugate quotients make θ_U / χ_U^0 invariant. The descent argument of Lemma 3.2 gives $\theta_U / \chi_U^0 = \lambda \circ N_{U/F}$. Compatibility puts $\theta_E / (\chi_E^0(\lambda \circ N_{E/F}))$ in $S(K/E)$, whose characters are unramified, so it is absorbed by the stated adjustment.

The conductor of θ_U is at least the conductor T of its conjugate quotient. Write it as $T + h$. Equation (16) makes every twist by $S(K/U)$ a conjugate, so no conductor drop occurs at T . Proposition 5.2 gives conductor $T + ph$ for the common pullback to K . Since K/E is unramified, this equals $m_E(\theta_E)$. The conductor of χ_U^0 is T and unramified norm pullback preserves conductors; comparison in (58) gives the assertions about λ . These are [4, Proposition 6.7]. $\square$

Once the models are constructed, this descent and conductor argument does not use $p > 2$; its quadratic form is Lemma 9.3. The common twist is formalized by [Odd.UR.twist](#).

For larger h , both pullbacks of the twisting character have half-conductor at least T . The conductor- T characters can therefore be removed by the stable-twist formula.

Proposition 7.7 (Comparison in the stable range). *If $h \geq T$, then*

$$\Delta_U(\theta_U, \psi_U) = \Delta_E(\theta_E, \psi_E).$$

Proof. Put $m = m_F(\lambda) = T + h \geq 2T$. Choose an admissible $\Gamma \in F^\times$ and a stationary representative $\beta \in \mathcal{O}_F^\times$ for λ . Since $\lfloor m/2 \rfloor \geq T$, the same Γ, β apply to both norm pullbacks by [2, Proposition 6.8(a)–(b)]. Equations (33) and (31) give

$$m_E(\lambda \circ N_{E/F}) + n_E(\psi_E) = T + p(m - T) + pn_F(\psi_F) + (p - 1)T = p(m + n_F(\psi_F)) = v_E(\Gamma).$$

The pullback conductors are m over U and $T + p(m - T) \geq m$ over E . Both are at least $2T$, so (30) gives

$$(59) \quad \Delta_L(\theta_L, \psi_L) = \chi_L^0(\Gamma/\beta) \Delta_L(\lambda \circ N_{L/F}, \psi_L), \quad L = U, E.$$

The multipliers agree by Lemma 3.6, since $\epsilon_U = \epsilon_E = 1$ by (20). Each $\nu \in S(L/F)$ has conductor at most $T \leq \lfloor m/2 \rfloor$. Equations (13) and (30) over F give

$$\Delta_L(\lambda \circ N_{L/F}, \psi_L) = \prod_{\nu \in S(L/F)} \Delta_F(\nu\lambda, \psi_F) = \Delta_F(\lambda, \psi_F)^p.$$

Here the product of the ν is trivial by (20), and their local-constant product is 1 by (12). Substitution in (59) proves the assertion. This is the stable range in [4, Section 6]. $\square$

The common stationary coefficient is constructed in `Odd.UR.stable_common_covector`; the complete stable-range comparison is `Odd.UR.oddUR_stable_comparison`.

For $p = 2$, the base-field coefficient need not be stationary over E . Proposition 9.7 instead uses a sufficiently close coefficient and retains the quadratic norm-character factors.

For $0 \leq h \leq t$, put

$$d_U = \lfloor (T + h)/2 \rfloor, \quad d_E = \lfloor (T + ph)/2 \rfloor.$$

Lemma 7.8 (Norm representatives for the twist coefficient). *Suppose $0 \leq h \leq t$. There are $A_* \in F$ and $x \in E$ such that*

$$\lambda(1 - z) = \Psi_F(A_* P(z)) \quad (z \in \mathfrak{p}_F^{d_U}).$$

Put $A = N_{E/F}(x)$. These choices can be made so that

$$(60) \quad A_* = A \quad (h < t), \quad A_* = A + \varepsilon, \quad \varepsilon \in \mathcal{O}_F \quad (h = t).$$

With $\varepsilon = 0$ when $h < t$, put

$$B_U = A + d^p + \varepsilon, \quad B_E = A - x + c.$$

Then

$$(61) \quad \theta_U(1 - z) = \Psi_U(B_U P(z)) \quad (z \in \mathfrak{p}_U^{d_U}), \quad \theta_E(1 - z) = \Psi_E(B_E P(z)) \quad (z \in \mathfrak{p}_E^{d_E}),$$

and $v_U(B_U) = -h$, $v_E(B_E) = -ph$. If $h > 0$, then $v_E(x) = v_F(A) = -h$. If $h = 0$, then x is integral; the zero coefficient class may be represented by $A = x = 0$.

Proof. Lemma 7.2 and additive duality give A_* modulo $\mathfrak{p}_F^{T-d_U}$. For $0 < h < t$, the required relative norm precision is $T + h - d_U \leq t$, so Lemma 5.3 supplies a norm in this class. The integral classes at $h = 0$ are treated by the same lemma, with $x = 0$ for the zero class. At $h = t$, precision t leaves an integral error, which is the ε in (60). These choices are [4, Lemma 6.8].

The unramified norm-logarithm formula and the formula for χ_U^0 give B_U . For the ramified pullback, the essential conversion is

$$\Psi_F(A N_{E/F}(P(z))) = \Psi_E(-x P(z)),$$

by (56), since $A = N_{E/F}(x)$ and $x P(z) \in \mathfrak{p}_E^{q_0}$. Together with (38), this gives coefficient $A - x$ for the twist and hence $B_E = A - x + c$ for θ_E . At $h = t$, the integral error contributes trivially on the required ideal. The precise depth checks and the coefficient valuations are [4, Proposition 6.9]; in particular they include this endpoint. $\square$

The norm choice is `Odd.UR.ActualTwist.normChoice`; the formulas and valuations for the prescribed characters are `Odd.UR.coefficients`.

Put

$$C = x + d, \quad Z_U = N_{K/U}(C), \quad Z_E = N_{K/E}(C), \quad \eta_U = Z_U - B_U, \quad \eta_E = Z_E - B_E,$$

and $S = \text{Tr}_{U/F} Z_U - \text{Tr}_{E/F} Z_E$. Here $C \neq 0$, since $x \in E$, $d \in U \setminus F$, and $E \cap U = F$.

Lemma 7.9 (Local constants through common norms). *For $0 \leq h \leq t$, one has*

$$(62) \quad \frac{\Delta_U(\theta_U, \psi_U)}{\Delta_E(\theta_E, \psi_E)} \in \mu_4 \Psi_F(-S) \frac{\Psi_U(\eta_U^2/(2B_U))}{\Psi_E(\eta_E^2/(2B_E))},$$

and

$$S = pA - \text{Tr}_{E/F}(x^p) + p \text{Tr}_{E/F}(x).$$

Proof. The estimates of [4, Lemma 6.10] give

$$v_U(\eta_U) \geq T - \lceil (T + h)/2 \rceil, \quad v_E(\eta_E) \geq T - \lceil (T + ph)/2 \rceil.$$

Thus (57) applies to (61), with $C = Z_U, Z_E$. The factors at $-\alpha \in F^\times$ cancel by Lemma 3.6 and (20), while compatibility gives

$$\theta_U(Z_U) = \Theta(C) = \theta_E(Z_E).$$

Dividing (57) for the two fields gives (62).

If $a_j = E_j^{E/F}(x)$, the exact norm identities are

$$Z_U = d^p + \sum_{j=1}^{p-1} a_j d^{p-j} + A, \quad Z_E = x^p - x + c.$$

Newton's identities for $X^p - X - c$ give $\text{Tr}_{U/F}(d^j) = 0$ for $1 \leq j < p-1$, $\text{Tr}_{U/F}(d^{p-1}) = p-1$, and $\text{Tr}_{U/F}(d^p) = pc$. Taking traces in the two norm identities proves the formula for S . These are [4, Lemma 6.11]. $\square$

The displacement bounds used in this construction are proved in [Odd.UR.normDepths](#). The formal theorem [Odd.UR.scalarPhase](#) constructs the representatives and packages the ratio and trace identity as `ScalarPhaseAt`; it also establishes the bounds permitting the change of coefficient.

Lemma 7.10 (Cancellation of additive factors). *Suppose $0 \leq h \leq t$. If $h < t$, or if $h = t$ and $p > 3$, then*

$$S \in \mathfrak{p}_F^T, \quad \Psi_U(\eta_U^2/(2B_U)) = \Psi_E(\eta_E^2/(2B_E)) = 1.$$

If $p = 3$ and $h = t$, put

$$a = \text{Tr}_{E/F} x, \quad b = E_2^{E/F}(x), \quad \eta_E = x^3 - A = ax^2 - bx,$$

$$\Lambda = 3a - a^3 + 3ab - \frac{b^2}{A} + \frac{\text{Tr}_{E/F}(\eta_E^2)}{2A}.$$

Then the factor multiplying μ_4 in (62) is $\Psi_F(-\Lambda) = 1$.

Proof. For $h < t$, [4, Lemma 6.10] strengthens the error bounds to $v_U(\eta_U) \geq T - d_U$ and $v_E(\eta_E) \geq T - d_E$. Lemma 7.4 gives $\Psi_U(\eta_U^2/(2B_U)) = \Psi_E(\eta_E^2/(2B_E)) = 1$. The bound on S is [4, Lemma 6.12]. At $h = t$ and $p > 3$, the same source gives the bound on S , and the stronger error bound over E still holds. Write $b = E_{p-1}^{E/F}(x) \in \mathcal{O}_F$. The boundary calculation gives

$$\Psi_U(\eta_U^2/(2B_U)) = \Psi_U((bd - \varepsilon)^2/(2A)) = \Psi_F(p\varepsilon^2/(2A)) = 1,$$

since $\text{Tr}_{U/F} d = \text{Tr}_{U/F}(d^2) = 0$ and the last argument lies in $\mathfrak{p}_F^T$. This is the boundary calculation following [4, Lemma 6.12].

For $p = 3$, the same calculation and the norm identity for x give

$$S = 3a - a^3 + 3ab, \quad \Psi_U(\eta_U^2/(2B_U)) = \Psi_F(b^2/A),$$

$$\Psi_E(\eta_E^2/(2B_E)) = \Psi_F(\text{Tr}_{E/F}(\eta_E^2)/(2A)).$$

Their substitution in (62) gives $\Psi_F(-\Lambda)$. Newton's identities give

$$\Lambda = 3a + a^3 - \frac{b^2 + b^3}{A} + \frac{a^2(a^2 - 3b)^2}{2A}.$$

The symmetric estimate (35) gives $v_F(a) \geq \lceil t/3 \rceil \geq 1$ and $v_F(b) \geq 0$. Thus the last term lies in $\mathfrak{p}_F^T$; also $v_E(a) \geq 3 \lceil t/3 \rceil \geq t \geq q_0$ and $v_E(b/x) \geq t \geq q_0$. Apply (56) to these two elements to obtain

$$\Psi_F(3a + a^3) = 1, \quad \Psi_F((b^2 + b^3)/A) = 1.$$

Thus $\Psi_F(\Lambda) = 1$. The valuation bounds and the cubic identity are [4, Lemma 6.13]. $\square$

The ordinary trace bound is `Odd.UR.traceDepth`. At $h = t$, the unramified factor for $p > 3$ is evaluated by `Odd.UR.unramifiedBoundaryCorrection`. The cubic identity in Lemma 7.10 is formalized by `Odd.UR.cubicIdentity`.

This vanishing is not a quadratic identity: at $h = 0$ and even T , the trace difference in Proposition 9.5 is $S \equiv 1 \pmod{\mathfrak{p}_F^T}$, not 0.

Substitution of Lemma 7.10 into (62) now gives

$$(63) \quad \frac{\Delta_U(\theta_U, \psi_U)}{\Delta_E(\theta_E, \psi_E)} \in \mu_4 \quad (0 \leq h \leq t).$$

Proposition 7.11. *If $|I| = p$, then $\mathcal{A}_U(\theta_U) = \mathcal{A}_E(\theta_E)$ for every degree- p intermediate field $E \neq U$.*

Proof. Proposition 7.6 gives $h \geq 0$. Since $T = t + 1$, the ranges of (63) and Proposition 7.7 exhaust these integers. In the first range the quotient belongs to μ_4 ; its p th power is 1 by (19) and Lemma 7.1. Since $(p, 4) = 1$, it is 1. In the second range equality is already proved. Use Lemma 7.1 to replace the local constants by $\mathcal{A}_U, \mathcal{A}_E$. $\square$

This is [4, Theorem 6.1]. It is formalized by `Odd.UR.comparison`.

7.2. Totally ramified extensions. Suppose $|I| = p^2$, so K/F is totally ramified. Choose distinct degree- p intermediate fields L_1, L_2 . For $i = 1, 2$, let t_i be the lower ramification break of L_i/F , choosing L_1 so that t_1 is minimal among the lower breaks of all degree- p intermediate extensions of F in K . Put

$$t = t_1 \leq t_2, \quad \delta = t_2 - t, \quad t' = t + p\delta.$$

The lower breaks of K/L_2 and K/L_1 are t and t' , respectively.

In mixed characteristic, (31) applied to $\text{Tr}_{L_2/F}(1) = p$ gives $v_F(p) \geq \lceil (p-1)t_2/p \rceil$, hence

$$(64) \quad v_K(p) \geq p(p-1)t_2.$$

This bound is `Odd.Total.oddTotal_primeValuationBound`. The best-approximation argument in [4, Proposition 7.4] also gives $p \nmid t$; in Lean, this is part of `Odd.Total.oddAS_approximate_generator`.

Lemma 7.12 (Artin–Schreier coordinate and norm estimates). *There is $\Delta \in K$ such that*

$$\Delta^p - \Delta = a \in L_2, \quad v_K(\Delta) = -t, \quad v_{L_2}(a) = -t, \quad K = L_2(\Delta).$$

So $N_{K/L_2}(\Delta) = a$, since p is odd. Put

$$b = N_{K/L_1}(\Delta), \quad s = -E_{p-1}^{K/L_1}(\Delta).$$

Then

$$(65) \quad \text{Tr}_{L_1/F}(b) - \text{Tr}_{L_2/F}(a) \in \mathfrak{p}_F^{t_2+1}, \quad v_K(b-a) \geq -t, \quad v_{L_1}(s) \geq (p-1)\delta.$$

Proof. The coordinate is constructed in [4, Proposition 7.4], using Lemma 5.4. The trace–norm congruence is [4, Theorem 7.7]; the estimate for s is [4, Lemma 7.6]. For the middle estimate, the characteristic-polynomial identity and (35) give

$$v_K(\Delta^p - N_{K/L_1}(\Delta)) \geq -pt + (p-1)t' \geq -t.$$

Since $a = \Delta^p - \Delta$, it follows that $v_K(b-a) \geq -t$. $\square$

The coordinate is constructed by `Odd.Total.aSCoordinate`. The trace–norm congruence is `Odd.Total.traceNorm`; the estimate for s follows from `Odd.Total.twistedEstimates`. The power–norm estimate used for $b-a$ is `Odd.Total.powerNorm`. For $p = 2$, Lemmas 9.13, 9.22, and 9.31 also use norms of traces to supply the norm-character coefficients.

Put

$$m_1 = 1 + t + t', \quad m_2 = 1 + t + t_2, \quad q_i = \lceil m_i/p \rceil.$$

Choose $1 \neq \tau \in S(L_2/F)$ and $\alpha \in F^\times$ such that

$$\tau(1-z) = \psi_F(\alpha P(z)) \quad (z \in \mathfrak{p}_F^{\lceil (t_2+1)/p \rceil}), \quad v_F(\alpha) = -n_F(\psi_F) - t_2 - 1.$$

The choice follows from Lemma 7.2, Proposition 5.2, and additive duality. Write

$$\Psi_L(x) = \psi_L(\alpha x).$$

Equation (31) gives largest trivial ideals $\mathfrak{p}_F^{t_2+1}$, $\mathfrak{p}_{L_1}^{t'+1}$, and $\mathfrak{p}_{L_2}^{t_2+1}$.

Lemma 7.13 (Compatibility of the unit models). *The formulas*

$$R_1(1-z) = \Psi_{L_1}(bP(z)) \quad (z \in \mathfrak{p}_{L_1}^{q_1}), \quad R_2(1-z) = \Psi_{L_2}(aP(z)) \quad (z \in \mathfrak{p}_{L_2}^{q_2})$$

define characters on $U_{L_1}^{q_1}$ and $U_{L_2}^{q_2}$. For every $u \in U_K^{q_1}$ their norm pullbacks agree:

$$R_1(N_{K/L_1}(u)) = R_2(N_{K/L_2}(u)).$$

Proof. This is [4, Theorem 8.5]. The proof descends through the unit filtration from a depth where both norm pullbacks are trivial. If their quotient is trivial on U_K^{r+1} , with $r \geq q_1$, its value at $1-z$ is additive in $z \in \mathfrak{p}_K^r$. After a suitable unramified base extension, the polynomial rigidity lemma [4, Lemma 8.3] reduces this value to the degree-one and degree- p contributions of the norm-logarithm polynomial in a scalar parameter. These are the trace and norm contributions identified in [4, Lemma 8.4]. For a monomial $x\Delta^j$, with $x \in L_2$, $0 \leq j < p$, and $v_K(x\Delta^j) \geq r$, both contributions lie in the trivial ideal when $j \neq p-2$; for $j = p-2$ they cancel under the additive character [4, Proposition 7.12 and Lemma 8.2]. Since $p \nmid t$, the nonzero terms in the expansion $z = \sum_{j=0}^{p-1} x_j \Delta^j$ have distinct valuations modulo p , so each term belongs to $\mathfrak{p}_K^r$. Additivity therefore gives triviality on U_K^r , completing the induction. $\square$

The polynomial-rigidity step is `Finite.polynomialRigidity`. The nonexceptional trace estimate and exceptional trace formula are `Odd.Total.secondDefect_nonexceptional_pairing` and `Odd.Total.secondDefect_exceptional_pairing`, respectively. The complete compatibility statement is `Odd.Models.compatibility`.

For the nonexceptional trace estimate, the original proof split $j = 0$, $1 \leq j \leq p-3$, and $j = p-1$, assuming $v_K(x\Delta^j) \geq 1 + \delta$. The trace calculation used here treats all these exponents by one expansion requiring only $x \in \mathfrak{p}_{L_2}$; $j = p-2$ still needs its separate calculation [4, Remark 7.13]. For $p = 2$, the unit formulas use $P(z) = z$ at stationary depth; compatibility instead uses Lemma 8.2, as in Propositions 9.15, 9.24, and 9.33.

Lemma 7.14 (Conductor lower bounds). *For every primitive compatible pair (θ_1, θ_2) over L_1, L_2 , one has $m_{L_i}(\theta_i) \geq m_i$.*

Proof. Let σ_i generate $\text{Gal}(L_i/F)$ and put $u_1 = t'$, $u_2 = t$. Lemma 3.3 and Proposition 5.2 give a nontrivial conjugate quotient $\mu_i = \theta_i^{\sigma_i}/\theta_i$ of conductor $u_i + 1$. Choose $y = 1 + z \in U_{L_i}^{u_i}$ with $\mu_i(y) \neq 1$. The ramification bound gives

$$\sigma_i^{-1}(y)/y \in U_{L_i}^{t_i+u_i}, \quad \theta_i(\sigma_i^{-1}(y)/y) = \mu_i(y) \neq 1.$$

Hence $m_{L_i}(\theta_i) > t_i + u_i = m_i - 1$. The ramification bound is [4, Lemma C.3], and this deduction is [4, Lemma 8.7]. $\square$

The conductor bound for a prescribed compatible pair is `Odd.Conductors.lower`. This ramification argument does not require odd p . In the quadratic cases the numerical bounds are obtained from their own tower breaks.

Proposition 7.15 (Extension and realization of the unit models). *The characters R_1, R_2 of Lemma 7.13 extend to primitive compatible characters of $L_1^\times, L_2^\times$, still denoted R_1, R_2 , of conductors m_1, m_2 satisfying*

$$R_1(1-z) = \Psi_{L_1}(bP(z)) \quad (z \in \mathfrak{p}_{L_1}^{q_1}), \quad R_2(1-z) = \Psi_{L_2}(aP(z)) \quad (z \in \mathfrak{p}_{L_2}^{q_2}).$$

More generally, suppose (θ_1, θ_2) is primitive and compatible and $\theta_2|_{U_{L_2}^{m_2}} = 1$. Then $m_{L_i}(\theta_i) = m_i$, and there are $\alpha' = j\alpha$, $1 \leq j < p$, and $\Delta' \in K$ such that

$$(\Delta')^p - \Delta' = \alpha' \in L_2, \quad v_K(\Delta') = -t.$$

For some $\rho \in \text{Gal}(L_1/F)$, one has

$$\begin{aligned}\theta_2(1-z) &= \psi_{L_2}(\alpha' a' P(z)) & (z \in \mathfrak{p}_{L_2}^{q_2}), \\ \theta_1^\rho(1-z) &= \psi_{L_1}(\alpha' N_{K/L_1}(\Delta') P(z)) & (z \in \mathfrak{p}_{L_1}^{q_1}).\end{aligned}$$

Proof. Lemma 7.13 gives the compatible principal-unit characters. They extend to the first primitive compatible pair by [4, Theorem 8.6]; the prescribed-pair assertion is [4, Theorem 8.9]. The conductor equalities need only the stated upper bound: Lemma 7.14 gives $m_{L_2}(\theta_2) = m_2$, and (33) applied to the common pullback gives

$$p m_{L_1}(\theta_1) - (p-1)(t'+1) = p m_2 - (p-1)(t+1).$$

Substitution of $t' = t + p\delta$ and $t_2 = t + \delta$ gives $m_{L_1}(\theta_1) = m_1$. The construction supplies a first character with the prescribed formula and the same pullback as θ_1 . Their quotient is in $S(K/L_1)$, so (16) identifies it with a conjugate θ_1^ρ . $\square$

The extension of the principal-unit prescriptions to the first compatible pair is `Odd.Models.globalExtension`. The original prescribed-pair statement assumed both equalities $m_{L_i}(\theta_i) = m_i$. Proposition 7.15 assumes only $\theta_2|_{U_{L_2}^{m_2}} = 1$: the lower bound of Lemma 7.14 forces the second equality, and (33) for the common pullback forces the first. Thus both exact conductors are now conclusions, rather than hypotheses [4, Remark 8.10]. The prescribed-pair result is formalized by `Odd.Models.realization`, and compatibility by `Odd.Models.compatibility`.

Proposition 7.16 (Comparison at minimal conductors). *If a primitive compatible pair (θ_1, θ_2) has $m_{L_i}(\theta_i) = m_i$, then $\mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_2}(\theta_2)$.*

Proof. Write $\Theta = \theta_1 \circ N_{K/L_1} = \theta_2 \circ N_{K/L_2}$. Use Proposition 7.15 and put $b' = N_{K/L_1}(\Delta')$. Since $p \nmid t$ and $v_K(\Delta') = -t$, one has $\Delta' \notin L_2$. Its minimal polynomial is therefore $X^p - X - a'$, giving $N_{K/L_2}(\Delta') = a'$. Since $q_i \leq \lfloor m_i/2 \rfloor$ and $3 \lfloor m_i/2 \rfloor \geq m_i$, only $z + z^2/2$ in $P(z)$ contributes on the critical ideal. Equation (24), with coefficient $-\alpha'b'$ or $-\alpha'a'$, together with (23) and (29), gives

$$(66) \quad \frac{\Delta_{L_2}(\theta_2)}{\Delta_{L_1}(\theta_1^\rho)} = \frac{g_2 \theta_1^\rho(\alpha'b')}{g_1 \theta_2(\alpha'a')} \psi_F\left(\alpha' [\text{Tr}_{L_1/F}(b') - \text{Tr}_{L_2/F}(a')]\right), \quad g_i \in \mu_4.$$

Here the additive character in Δ_{L_i} is ψ_{L_i} . Lemma 3.6 and (20) give $\theta_1^\rho(\alpha') = \theta_2(\alpha')$; their values at -1 agree as well. Compatibility, preserved by conjugation, gives

$$\theta_1^\rho(b') = \Theta(\Delta') = \theta_2(a').$$

The estimate (65) applies to Δ' , and $\psi_F(\alpha' \cdot)$ is trivial on $\mathfrak{p}_F^{t_2+1}$. Thus the quotient belongs to μ_4 . Lemma 2.2 removes the conjugation by ρ ; Lemma 7.1 identifies each local constant with $\mathcal{A}_{L_i}$. Equation (19) gives p th power 1. Since $(p, 4) = 1$, the quotient is 1. This is [4, Theorem 9.34]. $\square$

The minimal-conductor comparison, including realization of the prescribed pair, is `Odd.Conductors.minimal`. For $p = 2$ and odd conductor, the unit-model depth $\lceil m_i/2 \rceil$ exceeds the critical depth $\lfloor m_i/2 \rfloor$; moreover, $\mu_2 \cap \mu_4 = \mu_2$. Thus this final argument cannot determine the sign. The corresponding critical-function comparisons are listed in Section 9.2.

The lower bounds just proved ensure that the common-twist reduction ends at m_1, m_2 , rather than at a smaller pair. Its quadratic counterparts are Lemmas 9.16, 9.25, and 9.34; their conductor alternatives must be recomputed.

Lemma 7.17 (Reduction to minimal conductors). *Every primitive compatible pair (θ_1, θ_2) can be written*

$$\theta_1 = \chi_1(\lambda \circ N_{L_1/F}), \quad \theta_2 = \chi_2(\lambda \circ N_{L_2/F}),$$

where (χ_1, χ_2) is primitive and compatible, $m_{L_1}(\chi_1) = m_1$, and $m_{L_2}(\chi_2) = m_2$. If $m_{L_2}(\theta_2) > m_2$, then $m_F(\lambda) = 1 + r$ with $r > t_2$ and $p(r - t_2) > t$.

Proof. The norm-kernel calculation of [4, Lemma 9.32], using Lemma 5.4, shows that θ_1 on $U_{L_1}^{m_1}$ factors through its norm image. The resulting character is continuous, since this norm map is a quotient of a compact group onto its image. Extend it to λ on $F^\times$ by the character-extension argument in Lemma 3.2. Removing $\lambda \circ N_{L_i/F}$ preserves compatibility and primitivity and makes $m_{L_1}(\chi_1) \leq m_1$. Lemma 7.14 gives equality. Equation (33) then gives

$$p m_{L_1}(\chi_1) - (p-1)(t'+1) = p m_{L_2}(\chi_2) - (p-1)(t+1),$$

so $m_{L_2}(\chi_2) = m_2$. If $m_{L_2}(\theta_2) > m_2$, then $m_{L_2}(\lambda \circ N_{L_2/F}) = m_{L_2}(\theta_2)$. The same conductor formula forces $m_F(\lambda) = 1+r > 1+t_2$ and

$$m_{L_2}(\theta_2) = 1+t_2+p(r-t_2) > 1+t+t_2.$$

The last inequality is $p(r-t_2) > t$. This is [4, Proposition 9.33]. $\square$

Triviality on the deep norm kernel is [Odd.Conductors.descent](#); the resulting common twist and its conductor conditions are [Odd.Conductors.reduction](#).

Proposition 7.18 (Comparison after a higher-conductor twist). *Let (χ_1, χ_2) be a primitive compatible pair of conductors m_1, m_2 , and let λ be a character of $F^\times$ with $m_F(\lambda) = 1+r$, $r > t_2$, and $p(r-t_2) > t$. Then*

$$\Delta_{L_1}(\chi_1(\lambda \circ N_{L_1/F}), \psi_{L_1}) = \Delta_{L_2}(\chi_2(\lambda \circ N_{L_2/F}), \psi_{L_2}).$$

Proof. Put $\theta_1 = \chi_1(\lambda \circ N_{L_1/F})$ and $\theta_2 = \chi_2(\lambda \circ N_{L_2/F})$. Apply Proposition 7.15 to the minimal pair, conjugating the first character if necessary; this does not change its local constant. For this proof, rename the resulting α' and Δ' as α and Δ , and set $b = N_{K/L_1}(\Delta)$ and $a = \Delta^p - \Delta$. As in Proposition 7.16, $N_{K/L_2}(\Delta) = a$. The character coefficients are now $\alpha b, \alpha a$. Choose a nontrivial $\nu \in S(L_1/F)$ and coefficients $\beta, \gamma \in F^\times$ such that

$$\begin{aligned} \nu(1-z) &= \psi_F(\beta P(z)) \quad (z \in \mathfrak{p}_F^{[(t+1)/p]}), \\ \lambda(1-z) &= \psi_F(\gamma P(z)) \quad (z \in \mathfrak{p}_F^{[(r+1)/p]}). \end{aligned}$$

The analogous formula for a nontrivial character of $S(L_2/F)$ has coefficient α and depth $[(t_2+1)/p]$.

The norm choices in [4, Lemmas 8.17–8.18] allow these coefficients to be adjusted without changing the character formulas so that

$$N_{L_1/F}(w_1) = \gamma/\beta, \quad N_{L_2/F}(w_2) = \gamma/\alpha.$$

The analogous normalization changes in the quadratic proof are Lemmas 9.26 and 9.35: there one rescales α to make the twisting coefficient a norm from the third quadratic field. Put

$$C_1 = \gamma - \beta w_1 + \alpha b, \quad C_2 = \gamma - \alpha w_2 + \alpha a, \quad A_1 = \gamma + \alpha b.$$

The norm-pullback calculation [4, Proposition 8.20] gives

$$m_{L_1}(\theta_1) = 1+t'+p(r-t_2), \quad m_{L_2}(\theta_2) = 1+t_2+p(r-t_2),$$

$$\theta_i(1-z) = \psi_{L_i}(C_i P(z)) \quad (z \in \mathfrak{p}_{L_i}^{[m_{L_i}(\theta_i)/p]}).$$

The inequality $p(r-t_2) > t$ makes γ the term of strictly smallest valuation in A_1, C_1, C_2 ; in particular these elements are nonzero.

To separate the multiplicative and additive contributions, define

$$\begin{aligned} Q_1 &= \frac{\chi_1(A_1)}{\chi_2(C_2)}, & Q_2 &= \lambda\left(\frac{N_{L_1/F}(A_1)}{N_{L_2/F}(C_2)}\right), \\ Q_3 &= \psi_{L_1}(-\beta w_1)\theta_1(C_1/A_1), & Q_4 &= \psi_{L_2}(\alpha w_2). \end{aligned}$$

Apply (57) with zero displacement to these truncated-logarithm coefficients. The values at -1 agree by Lemma 3.6. Separating the remaining factors gives

$$(67) \quad \frac{\Delta_{L_2}(\theta_2, \psi_{L_2})}{\Delta_{L_1}(\theta_1, \psi_{L_1})} = \frac{g_2}{g_1} Q_1 Q_2 Q_3 Q_4 \psi_F\left(\alpha(\mathrm{Tr}_{L_1/F}(b) - \mathrm{Tr}_{L_2/F}(a))\right), \quad g_i^4 = 1.$$

Indeed $Q_1Q_2 = \theta_1(A_1)/\theta_2(C_2)$, and multiplication by $\theta_1(C_1/A_1)$ gives the character quotient from Lamprecht's formula. The traces of γ cancel, leaving the additive factors shown above. This is [4, Proposition 8.22].

The substantial calculation concerns the product, not its individual factors. The linear and quadratic contributions are evaluated over F ; [4, Theorem 9.29 and Corollary 9.30] give

$$(Q_1Q_2Q_3Q_4)^{-1} = \psi_F(\alpha\phi), \quad \phi \in \mathfrak{p}_F^{t_2+1}.$$

Thus their combined phase is trivial. The estimates use the trace and symmetric-function bounds of Section 5, the interpolation identities (40)–(41), and the cancellation between the linear and quadratic terms. The detailed expansions are in [4, Section 9]. Also (65) makes the final factor in (67) equal to 1. The quotient therefore lies in μ_4 . Twisting preserves compatibility and primitivity, so (19) and Lemma 7.1 give its p th power equal to 1. Since $(p, 4) = 1$, it is 1. This is [4, Corollary 9.31]. $\square$

The coefficient choices and their preservation are `Odd.Parameters.normChoice` and `Odd.Parameters.preservation`; the pullback formula is `Odd.Parameters.pullback`. The four-factor identity is `Odd.Parameters.fourFactors`. The combined phase is evaluated by `Odd.Transition.phase` and shown to be trivial by `Odd.Transition.cancel`. For the constructed coefficients, `Odd.Transition.actual_local_factors` then proves equality of the actual local constants.

Proposition 7.19. *If K/F is totally ramified and $\ell = p > 2$, then $\mathcal{A}_L(\theta_L) = \mathcal{A}_{L'}(\theta_{L'})$ for every pair L, L' of degree- p intermediate fields.*

Proof. Retain L_1 of minimal lower break and let $L_2 \neq L_1$ vary. Lemma 7.14 gives $m_{L_2}(\theta_{L_2}) \geq m_2$. If equality holds, the common-pullback conductor formula gives $m_{L_1}(\theta_{L_1}) = m_1$, and Proposition 7.16 applies. Otherwise Lemma 7.17 gives the hypotheses of Proposition 7.18; Lemma 7.1 then gives equality of $\mathcal{A}$. For a pair L, L' not containing L_1 , Lemma 3.2 supplies a character of $L_1^\times$ with the same pullback Θ . The two comparisons with that character give the asserted equality. $\square$

This is [4, Theorem 9.35], formalized by `Odd.Total.comparison`.

7.3. Conclusion for odd residue characteristic. The preceding calculations apply in both characteristics. In equal characteristic,

$$\text{char } F = p \implies p = 0 \text{ in } F.$$

In mixed characteristic the terms divisible by p in (62) and (66) are retained until their valuations are estimated, using (31) and (64), respectively.

Corollary 7.20. *The Second Main Lemma holds when $\ell = p > 2$.*

8. BIQUADRATIC EXTENSIONS AND THE SIGN

Assume $\ell = p = 2$. Let (θ_L) be a primitive compatible family as in Definition 1.1. Write L_1, L_2, L_3 for the three quadratic intermediate fields and $\theta_i = \theta_{L_i}$, so $\theta_i \circ N_{K/L_i} = \Theta$. Let ω_i be the nontrivial character in $S(L_i/F)$.

Put

$$\mathcal{A}_i := \mathcal{A}_{L_i}(\theta_i) = \Delta_{L_i}(\theta_i, \psi_{L_i}) \Delta_F(\omega_i, \psi_F).$$

Lemma 3.1 gives

$$S(K/F) = \{1, \omega_1, \omega_2, \omega_3\}, \quad \omega_1\omega_2 = \omega_3.$$

Equation (18) gives

$$\mathcal{A}_i^2 = \Delta_K(\Theta, \psi_K) \prod_{j=1}^3 \Delta_F(\omega_j, \psi_F),$$

so

$$\mathcal{A}_i/\mathcal{A}_j \in \{1, -1\}.$$

This is [4, Proposition 4.4], formalized by `Characters.quadraticSquare`.

8.1. Conductors and local-constant factors.

Lemma 8.1. *For every primitive compatible biquadratic family,*

$$m_{L_i}(\theta_i) > 1 \quad (i = 1, 2, 3).$$

Proof. Let σ_i be the nontrivial element of $\text{Gal}(L_i/F)$. By Lemma 3.3, $\mu_i = \theta_i^{\sigma_i}/\theta_i$ is the nontrivial character of $S(K/L_i)$. If L_i/F is ramified, σ_i acts trivially on $L_i^\times/U_{L_i}^1$, so a character of conductor at most one would be invariant. If L_i/F is unramified, K/L_i is wildly ramified and Proposition 5.2 gives $1 < m_{L_i}(\mu_i) \leq m_{L_i}(\theta_i)$. This is the conductor argument of Lemma 7.1; it does not use oddness of p . Its formal ingredients are linked there. $\square$

By Lemma 8.1, $m_{L_i}(\theta_i) > 1$, so Lamprecht's formula applies to every θ_i . Write

$$m_i = m_{L_i}(\theta_i) = 2d_i + \varepsilon_i, \quad \varepsilon_i \in \{0, 1\}.$$

Choose an admissible element Γ_i and a stationary representative b_i for (θ_i, ψ_{L_i}) , and let H_i, g_i be the critical function and normalized sum in (22)–(23). Substituting (24) gives

$$(68) \quad \mathcal{A}_i = \Delta_F(\omega_i, \psi_F)\theta_i(\Gamma_i/b_i)\psi_{L_i}(b_i/\Gamma_i)g_i.$$

Thus (5) in the biquadratic case is equivalent to

$$(69) \quad \Delta_F(\omega_i, \psi_F)\theta_i(\Gamma_i/b_i)\psi_{L_i}(b_i/\Gamma_i)g_i = \Delta_F(\omega_j, \psi_F)\theta_j(\Gamma_j/b_j)\psi_{L_j}(b_j/\Gamma_j)g_j.$$

8.2. Norms of one element of K .

Lemma 8.2 (Biquadratic trace–norm identity). *For $C \in K$ let $E_2(C)$ be the second elementary symmetric function of its four conjugates. Then*

$$E_2(C) = \text{Tr}_{L_i/F}(\text{N}_{K/L_i}(C)) + \text{N}_{L_i/F}(\text{Tr}_{K/L_i}(C)) \quad (i = 1, 2, 3).$$

Proof. If the involution fixing L_i pairs the four conjugates as c_1, c_2 and c_3, c_4 , the two terms on the right are

$$c_1c_2 + c_3c_4, \quad (c_1 + c_2)(c_3 + c_4),$$

whose sum is $E_2(C)$. $\square$

This is [4, Lemma 4.5]. It is formalized by [Algebra.biquadraticE2](#).

Lemma 8.3 (Norms as stationary coefficients). *Fix $\alpha \in F^\times$ and, for $L = F, L_1, L_2, L_3$, put*

$$\Psi_L(x) = \psi_L(\alpha x).$$

Let

$$D = \theta_i|_{F^\times}\omega_i,$$

which is independent of i by Lemma 3.6. Put

$$T_i = m_F(\omega_i),$$

and assume $T_i > 1$. Let $C \in K^\times$ satisfy $\text{Tr}_{K/L_i}(C) \neq 0$ and set

$$Z_i = \text{N}_{K/L_i}(C), \quad W_i = \text{N}_{L_i/F}(\text{Tr}_{K/L_i}(C)).$$

Assume

$$\begin{aligned} \theta_i(1 - z) &= \Psi_{L_i}(Z_i z) & (z \in \mathfrak{p}_{L_i}^{[m_i/2]}), \\ \omega_i(1 - z) &= \Psi_F(W_i z) & (z \in \mathfrak{p}_F^{[T_i/2]}). \end{aligned}$$

Let g_i, h_i be the corresponding normalized critical sums, with value 1 in even conductor. Then

$$(70) \quad \mathcal{A}_i = D(-\alpha^{-1})\Theta(C)^{-1}\Psi_F(-E_2(C))g_ih_i.$$

Proof. From $\theta_i(1 - z) = \Psi_{L_i}(Z_i z)$, replacing z by $-x$ gives $\theta_i(1 + x) = \psi_{L_i}(-\alpha Z_i x)$; thus in (21) one may take $b/\Gamma = -\alpha Z_i$. Similarly, for ω_i one may take $b/\Gamma = -\alpha W_i$. Hence (24) gives

$$\begin{aligned}\Delta_{L_i}(\theta_i) &= \theta_i(-\alpha^{-1})\theta_i(Z_i)^{-1}\Psi_{L_i}(-Z_i)g_i, \\ \Delta_F(\omega_i) &= \omega_i(-\alpha^{-1})\omega_i(W_i)^{-1}\Psi_F(-W_i)h_i.\end{aligned}$$

Compatibility gives

$$\theta_i(Z_i) = \Theta(C),$$

and $W_i = N_{L_i/F}(\text{Tr}_{K/L_i}(C))$ gives

$$\omega_i(W_i) = 1.$$

Multiplying the two formulas, using $D = \theta_i|_F \omega_i$, and applying Lemma 8.2 gives (70). $\square$

This is [4, Lemma 4.6]. The formal counterpart is [Stationary.commonOrigin](#).

Lemma 8.4 (Products of critical functions). *Let $C, C' \in K^\times$ satisfy $\text{Tr}_{K/L_i}(C), \text{Tr}_{K/L_i}(C') \neq 0$, and put*

$$\begin{aligned}Z_i &= N_{K/L_i}(C), & W_i &= N_{L_i/F}(\text{Tr}_{K/L_i}(C)), \\ 1 + z_i &= \frac{N_{K/L_i}(C')}{N_{K/L_i}(C)}, & 1 + w_i &= \frac{N_{L_i/F}(\text{Tr}_{K/L_i}(C'))}{N_{L_i/F}(\text{Tr}_{K/L_i}(C))}.\end{aligned}$$

Then

$$\Psi_{L_i}(-Z_i z_i)\theta_i(1 + z_i)^{-1}\Psi_F(-W_i w_i)\omega_i(1 + w_i)^{-1} = \Theta(C'/C)^{-1}\Psi_F(-E_2(C') + E_2(C)),$$

which is independent of i .

If, in addition, $m_i = m_{L_i}(\theta_i) > 1$, $T_i = m_F(\omega_i) > 1$, Z_i, W_i are stationary coefficients for θ_i, ω_i , respectively, and

$$z_i \in \mathfrak{p}_{L_i}^{\lfloor m_i/2 \rfloor}, \quad w_i \in \mathfrak{p}_F^{\lfloor T_i/2 \rfloor},$$

then the left side is the product of the corresponding critical functions in (22).

Proof. Compatibility and norm triviality give

$$\theta_i(1 + z_i) = \Theta(C'/C), \quad \omega_i(1 + w_i) = 1.$$

The additive exponent is

$$-\text{Tr}_{L_i/F}(N_{K/L_i}(C') - N_{K/L_i}(C)) - N_{L_i/F}(\text{Tr}_{K/L_i}(C')) + N_{L_i/F}(\text{Tr}_{K/L_i}(C)) = -E_2(C') + E_2(C)$$

by Lemma 8.2. $\square$

This is [4, Lemma 4.7]. The pointwise identity is formalized by [Stationary.commonFunctionProduct_eq](#).

8.3. Equality of the finite sums. A pointwise identity gives equality of sums by reindexing when the induced norm maps are bijective. At equal breaks their images can have index two. The next lemma shows why the unrepresented coset contributes zero when the kernels are distinct.

Lemma 8.5 (Cancellation from a common pullback). *Let I be an index set with at least two elements. For $i \in I$, let V_i be finite abelian groups of the same order and let $H_i : V_i \rightarrow \mathbf{C}^\times$ satisfy $H_i(0) = 1$. Suppose*

$$B_i(x, y) = \frac{H_i(x + y)}{H_i(x)H_i(y)}$$

is a nondegenerate bicharacter. Let Z be a finite abelian group and let $p_i : Z \rightarrow V_i$ be group homomorphisms with kernels of order two, images of index two, and pairwise distinct kernels. If

$$H_i \circ p_i = Q$$

for one function $Q : Z \rightarrow \mathbf{C}^\times$, then

$$\sum_{x \in V_i} H_i(x) = \frac{1}{2} \sum_{z \in Z} Q(z) \quad (i \in I).$$

In particular, the normalized sums $|V_i|^{-1/2} \sum_{x \in V_i} H_i(x)$ are independent of i .

Proof. Fix i and choose $j \neq i$. Let a be the nonzero element of $\ker p_j$. Since the kernels are distinct,

$$w_i := p_i(a) \neq 0, \quad H_i(w_i) = Q(a) = H_j(0) = 1.$$

For $x = p_i(z) \in \operatorname{im} p_i$,

$$H_i(w_i + x) = Q(a + z) = H_j(p_j(a + z)) = H_j(p_j z) = H_i(x),$$

so

$$B_i(w_i, x) = 1 \quad (x \in \operatorname{im} p_i).$$

Nondegeneracy makes $B_i(w_i, \cdot)$ the nontrivial character of $V_i / \operatorname{im} p_i$. Since $w_i \in \operatorname{im} p_i$, translation by w_i preserves its complement and multiplies each summand there by $H_i(w_i)B_i(w_i, x) = -1$. Therefore

$$\sum_{x \notin \operatorname{im} p_i} H_i(x) = 0.$$

Every element of $\operatorname{im} p_i$ has two preimages under p_i , hence

$$\sum_{z \in Z} Q(z) = 2 \sum_{x \in \operatorname{im} p_i} H_i(x) = 2 \sum_{x \in V_i} H_i(x).$$

□

This is [4, Lemma 4.8].

The original equal-break proofs identified the character on the unrepresented coset by explicit coefficient calculations. Here the common pullback, distinct kernels, and nondegeneracy give a translation multiplying every summand there by -1 , proving cancellation without those calculations; see [4, Remark 4.9].

The translation and fiber-counting step is `Finite.missingCoset`. It takes the element w_i and its two polar-pairing values as inputs. The common pullback and distinct kernels supply these inputs in `Dyadic.Equal.OneBreakResidue.hasse_sum_eq_half_common` and `Dyadic.Nonmaximal.equalBreaks.hasseCancellation`. Thus these declarations together implement the argument above.

8.4. Changing the common element. The following identities separate the common algebra from the case-specific valuation estimates for $C = Y - X$ in Section 9.

Lemma 8.6 (Norms after a quadratic shift). *Let $Y \in K$, $X \in L_3$, and let σ_i be the nontrivial automorphism of K/L_i . For $i = 1, 2, 3$, put*

$$a_i = N_{K/L_i}(Y), \quad h_i = \operatorname{Tr}_{K/L_i}(Y), \quad \beta_i = N_{L_i/F}(h_i).$$

Set $A = N_{L_3/F}(X)$, $C = Y - X$, and $Q_i = \operatorname{Tr}_{K/L_i}(Y\sigma_i X)$ for $i = 1, 2$. Primes denote the nontrivial conjugation of L_3/F . Then, for $i = 1, 2$,

$$N_{K/L_i}(C) = a_i + A - Q_i, \quad \operatorname{Tr}_{K/L_i}(C) = h_i - \operatorname{Tr}_{L_3/F}(X),$$

$$N_{L_i/F}(Q_i) - \beta_i A = (X' - X)(a_3 X' - a'_3 X),$$

$$N_{L_i/F}(\operatorname{Tr}_{K/L_i}(C)) - \beta_i = (\operatorname{Tr}_{L_3/F}(X))^2 - \operatorname{Tr}_{L_3/F}(X) \operatorname{Tr}_{K/F}(Y).$$

In particular, the last two differences are independent of $i = 1, 2$.

Proof. Expand $(Y - X)(\sigma_i Y - X')$ to obtain the first norm identity; trace linearity gives the other. Expanding the two conjugates of $Q_i = YX' + (\sigma_i Y)X$ and subtracting $\beta_i A$ gives $a_3(X'^2 - XX') + a'_3(X^2 - XX')$, proving the next identity. Finally, expand the norm of $h_i - \operatorname{Tr}_{L_3/F}(X)$ and use $\operatorname{Tr}_{L_i/F}(h_i) = \operatorname{Tr}_{K/F}(Y)$. □

These are the common algebraic identities specialized in [4, equations (218)–(219) and Lemmas 12.9, 13.9]. Their case-specific Lean proofs are linked with (84) and Lemmas 9.27 and 9.36 below.

9. THE QUADRATIC RAMIFICATION CASES

Continue to assume $\ell = p = 2$. We construct compatible characters, express the prescribed family as a common twist, and compare the resulting stationary formulas. The conductor ranges determine which critical sums remain; the pointwise identities of Section 8 determine their sign. For each coefficient specified below, H and g denote the critical function and normalized sum of (22)–(23). Subscripts i , U , and E refer to θ_i , θ_U , and θ_E ; a subscript ω_i refers to the norm character.

9.1. An unramified quadratic intermediate field. Suppose $|I| = 2$. Let U/F be the unramified quadratic intermediate field and let E/F be ramified quadratic of lower break $t \geq 1$. Put $T = t + 1$. This is the field configuration of Section 7.1 with $p = 2$. Let σ generate $\text{Gal}(U/F)$. Write $k = k_F = k_E$ and $\kappa = k_U = k_K$. For the ramified quadratic extension E/F one has

$$n_E(\psi_E) = 2n_F(\psi_F) + T, \quad \text{Tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}_F^{\lfloor (j+T)/2 \rfloor}, \quad v_F(N_{E/F}(x)) = v_E(x).$$

The first two formulas are (31), with $D_{E/F} = T$ by (32); the last is (6). These are the formulas used in [4, Section 10].

Put $q = \lceil T/2 \rceil$. By Proposition 5.2, $m_F(\omega_E) = T > 1$. Theorem 4.2 therefore supplies $\alpha \in F^\times$. Define $\Psi_L(z) = \psi_L(\alpha z)$ so that

$$(71) \quad \omega_E(1 - z) = \Psi_F(z) \quad (z \in \mathfrak{p}_F^q), \quad v_F(\alpha) = -n_F(\psi_F) - T.$$

Lemma 9.1 (Quadratic trace–norm identity). *For every $z \in \mathfrak{p}_E^q$, one has*

$$(72) \quad \Psi_E(z) = \Psi_F(N_{E/F}(z)).$$

Proof. As in Lemma 7.3, use norm triviality, now with the exact quadratic identity

$$N_{E/F}(1 - z) = 1 - \text{Tr}_{E/F}(z) + N_{E/F}(z),$$

and (31) gives $v_F(\text{Tr}_{E/F}(z)) \geq \lfloor (q + T)/2 \rfloor \geq q$, while $v_F(N_{E/F}(z)) = v_E(z) \geq q$. Thus (71) applies to $\text{Tr}_{E/F}(z) - N_{E/F}(z)$. Since $\omega_E(N_{E/F}(1 - z)) = 1$,

$$1 = \Psi_F(\text{Tr}_{E/F}(z) - N_{E/F}(z)),$$

which is (72). The identity (72) is [4, Lemma 10.1]. $\square$

The identity on the whole stated ideal is `Dyadic.UR.normPhase`.

Choose $c \in \mathcal{O}_F$ with residue of absolute trace one and $d \in \mathcal{O}_U$ such that

$$d^2 - d = c, \quad \text{Tr}_{U/F}(d) = 1, \quad N_{U/F}(d) = -c.$$

Proposition 9.2 (Compatible characters of conductor T). *There are characters χ_U^0, χ_E^0 of conductor T forming a primitive compatible pair and satisfying*

$$\begin{aligned} (\chi_U^0)^\sigma / \chi_U^0 &= \omega_E \circ N_{U/F}, \\ \chi_U^0(1 - z) &= \Psi_U(d^2 z) & (z \in \mathfrak{p}_U^q), \\ \chi_E^0(1 - z) &= \Psi_E(cz) & (z \in \mathfrak{p}_E^q). \end{aligned}$$

Proof. Use the extension, descent, and primitivity argument of Proposition 7.5, now with the linear prescription. Here $2q \geq T$ and $\sigma(d^2) - d^2 = 1 - 2d$, together with (31), verify the unit prescription and its conjugate quotient. The exact quadratic norm and (72) replace the norm–logarithm congruence. The unit coefficients give conductor T ; see [4, Theorem 10.4]. $\square$

The compatible characters and both conductor- T formulas are constructed by `Dyadic.UR.models`. Fix characters as in Proposition 9.2.

Lemma 9.3 (A common base-field twist). *For every primitive compatible pair there is a character λ of $F^\times$ such that, after an unramified adjustment of χ_E^0 ,*

$$\theta_U = \chi_U^0(\lambda \circ N_{U/F}), \quad \theta_E = \chi_E^0(\lambda \circ N_{E/F}),$$

and

$$m_U(\theta_U) = T + h, \quad m_E(\theta_E) = T + 2h, \quad h \geq 0.$$

For $h > 0$, $m_F(\lambda) = T + h$; for $h = 0$, $m_F(\lambda) \leq T$.

Proof. The descent and conductor argument of Proposition 7.6 applies with $p = 2$ to the models of Proposition 9.2. Their conjugate quotients already match those of the prescribed pair, since $S(K/U)$ has a unique nontrivial character. See [4, Proposition 10.5]. $\square$

The prescribed pair, common twist, and exact conductor alternatives are treated by `Dyadic.U.R.twist`.

For $0 \leq h < T$, put

$$s_U = \lceil (T + h)/2 \rceil, \quad s_E = \lceil T/2 \rceil + h.$$

As in Lemma 7.8, choose the twist coefficient as a norm. Here the formulas are asserted at the stationary depths s_U, s_E , which need not equal the critical depths.

Lemma 9.4 (Stationary coefficients from a norm). *Suppose $0 \leq h < T$. There is $x \in E$, with $A = N_{E/F}(x)$, such that*

$$\lambda(1 - z) = \Psi_F(Az) \quad (z \in \mathfrak{p}_F^{s_U}).$$

If $h > 0$, then $v_E(x) = v_F(A) = -h$; if $h = 0$, then x is integral. Put

$$\begin{aligned} B_U &= A + d^2, & B_E &= A - x + c, & C &= x + d, \\ Z_U &= N_{K/U}(C), & Z_E &= N_{K/E}(C), & b &= \text{Tr}_{E/F}(x). \end{aligned}$$

Then

$$\theta_U(1 - z) = \Psi_U(B_U z) \quad (z \in \mathfrak{p}_F^{s_U}), \quad \theta_E(1 - z) = \Psi_E(B_E z) \quad (z \in \mathfrak{p}_E^{s_E}),$$

with coefficient valuations $-h$ and $-2h$, respectively. Moreover

$$\begin{aligned} Z_U &= A + bd + d^2, & Z_E &= x^2 + x - c, \\ S &:= \text{Tr}_{U/F}(Z_U) - \text{Tr}_{E/F}(Z_E) = 1 + 4c - \mathcal{D}_x, & \mathcal{D}_x &= b^2 - 4A = (2x - b)^2. \end{aligned}$$

Proof. The coefficient of λ on $\mathfrak{p}_F^{s_U}$ is determined modulo $\mathfrak{p}_F^{T-s_U}$. The required relative precision is at most $\lfloor (T + h)/2 \rfloor \leq t$, so Lemma 5.3 gives a norm representative $A = N_{E/F}(x)$, with $x = 0$ for the zero class. The same norm-to-trace argument as in Lemma 7.8 now gives

$$\lambda(N_{E/F}(1 - z)) = \Psi_F(A \text{Tr}_{E/F}(z) - A N_{E/F}(z)) = \Psi_E((A - x)z),$$

using (72) on xz . Multiplying by the formulas of Proposition 9.2 gives B_U, B_E . The precision checks and nonvanishing of their leading coefficients are [4, Lemmas 10.6–10.7]. The identities for Z_U, Z_E, S follow by expanding the quadratic norms. $\square$

The representative is constructed by `Dyadic.UR.normChoice`; `Dyadic.UR.coefficients_exists` provides the resulting coefficients for the prescribed pair.

The next two comparisons replace Lemmas 7.9 and 7.10: the norm-character factors must be retained, and the remaining critical sums compared exactly.

Proposition 9.5 (Minimal-conductor comparison). *If $h = 0$, then*

$$(73) \quad \Delta_U(\theta_U, \psi_U) \Delta_F(\omega_U, \psi_F) = \Delta_E(\theta_E, \psi_E) \Delta_F(\omega_E, \psi_F).$$

Proof. Use the elements of Lemma 9.4. When T is even, (24), compatibility, and Lemma 3.6 express the quotient as $(-1)^T \Psi_F(1 - S)$. Here $1 - S = \mathcal{D}_x - 4c$. The trace-zero element $2x - b$ has valuation at least T , so (72) gives $\Psi_F(\mathcal{D}_x) = 1$; also $4c \in \mathfrak{p}_F^T$. Thus the quotient is 1. When T is odd, put $W_U = N_{U/F}(\text{Tr}_{K/U}(C)) = N_{U/F}(b + 2d)$. Applying (24) and Lemma 8.2, and writing g_U, g_E, g_{ω_E} for the normalized critical sums, gives

$$\frac{\Delta_U(\theta_U) \Delta_F(\omega_U)}{\Delta_E(\theta_E) \Delta_F(\omega_E)} = (-1)^T \Psi_F(W_U) \frac{g_U}{g_E g_{\omega_E}},$$

Here $T = 2e + 1$, where $e = v_F(2)$, and $b/2 \in \mathcal{O}_F$. Since $W_U = 4N_{U/F}(d + b/2)$ and the residue of $N_{U/F}(d + b/2)$ has absolute trace one, [4, Lemma 10.3] gives $\Psi_F(W_U) = -1 = (-1)^T$. For the critical sums, identify the critical quotients with $\kappa = k_U$ and $k = k_F = k_E$ using the

critical elements in [4, Theorem 10.8]. The common-function calculation of Lemma 8.4, with the trace-norm term over U constant modulo the trivial ideal, gives

$$H_U(z^2) = H_E(\mathrm{Tr}_{\kappa/k} z) H_{\omega_E}((\mathrm{Tr}_{\kappa/k}(\bar{C}z))^2).$$

Writing $z = u + v\bar{d}$ and $\bar{x} = \xi \in k$, the two arguments on the right are v and $(u + (\xi + 1)v)^2$. Thus they give a bijection $\kappa \rightarrow k \times k$, and squaring is a bijection of κ . Summing and using $|\kappa| = |k|^2$ gives $g_U = g_E g_{\omega_E}$. Hence (73) holds. This is [4, Theorem 10.8]. $\square$

The odd-conductor shape and the residue character used for the sign are `Dyadic.UR.oddConductor`. The complete minimal-conductor equality is `Dyadic.UR.minimal`.

The conjugated norm contributes a sign, which must be retained along with the critical sums.

Proposition 9.6 (Conjugated stationary norms). *If $1 \leq h < T$, then $\mathcal{A}_U(\theta_U) = \mathcal{A}_E(\theta_E)$.*

Proof. Use the elements of Lemma 9.4. Let x' be the nontrivial conjugate of x over F and put

$$\hat{Z}_U = A + bd + d^2, \quad \hat{Z}_E = \frac{x'}{x}(x^2 + x - c), \quad \mathfrak{u} = b^2/A.$$

The corrected coefficients satisfy

$$\begin{aligned} \hat{Z}_U - B_U &= bd, \\ \hat{Z}_E - B_E &= b(1 - c/x), \\ \mathrm{Tr}_{U/F} \hat{Z}_U - \mathrm{Tr}_{E/F} \hat{Z}_E &= 1 + cu, \\ \frac{\theta_E(\hat{Z}_E)}{\theta_U(\hat{Z}_U)} &= (-1)^h. \end{aligned}$$

By [4, Lemma 10.10], these corrected coefficients satisfy the stationary formulas on $\mathfrak{p}_U^{s_U}$ and $\mathfrak{p}_E^{s_E}$. Write g_U, g_E, g_{ω_E} for the normalized critical sums for $\hat{Z}_U, \hat{Z}_E, 1$, respectively. Equation (24) and Lemma 3.6 then give

$$(74) \quad \frac{\Delta_U(\theta_U) \Delta_F(\omega_U)}{\Delta_E(\theta_E) \Delta_F(\omega_E)} = (-1)^{T+h} \Psi_F(-cu) \frac{g_U}{g_E g_{\omega_E}} = \frac{g_U}{g_E g_{\omega_E}}.$$

The last equality is the parity calculation in [4, Proposition 10.12]. The unramified sum satisfies $g_U = 1$ by [4, Lemma 10.13]. If T is even, both g_E and g_{ω_E} are 1 by (23).

Suppose T is odd. Then $T = 2e + 1$, where $e = v_F(2)$, and the critical ideals for θ_E and ω_E are $\mathfrak{p}_E^{e+h}$ and $\mathfrak{p}_F^e$. On these ideals write their critical functions as

$$H_E(v) = \Psi_E(-\hat{Z}_E v) \theta_E(1 + v)^{-1}, \quad H_{\omega_E}(u) = \Psi_F(-u) \omega_E(1 + u)^{-1}.$$

At depth e , one cannot apply (72), whose domain begins at $e + 1$. Instead the exact norm factorization in [4, Lemma 10.14] gives

$$\omega_E(1 + N_{E/F}(w)) = \Psi_E(w) \quad (w \in \mathfrak{p}_E^e).$$

Using $w = xv$, the coefficient formulas then give

$$H_E(v) = \Psi_E(xv) \Psi_F(A N_{E/F}(v)) = H_{\omega_E}(N_{E/F}(xv))^{-1} \quad (v \in \mathfrak{p}_E^{e+h}).$$

The estimates justifying this identity are [4, Lemma 10.15]. Multiplication by x and the norm at depth $e < 2e = t$ induce bijections on the respective critical quotients. Reindexing therefore gives $g_E = \overline{g_{\omega_E}}$, and (24) gives $g_E g_{\omega_E} = 1$. Thus (74) is 1 in both parities. Substitution in (4) proves the assertion, which is [4, Theorem 10.16]. $\square$

The corrected coefficients and scalar quotient are `Dyadic.UR.conjugatedNorm` and `Dyadic.UR.criticalRatio`. The unramified sum is `Dyadic.UR.unramifiedSum`. The critical-depth norm identity is `Dyadic.UR.criticalReciprocity`; `Dyadic.UR.ramifiedSum` uses it to compare the two remaining sums. The full result is `Dyadic.UR.criticalComparison`.

Proposition 9.7 (Comparison by stable twisting). *If $h \geq T$, then $\mathcal{A}_U(\theta_U) = \mathcal{A}_E(\theta_E)$.*

Proof. This follows the stable-twist argument of Proposition 7.7, but allows the stationary coefficient over E to change. Put $n = m_F(\lambda) = T + h$ and choose $b_F \in F^\times$ satisfying

$$\lambda(1 + z) = \psi_F(b_F z) \quad (z \in \mathfrak{p}_F^{\lceil n/2 \rceil}).$$

Set

$$c_F = b_F^{-1}, \quad D = \chi_U^0|_{F^\times \omega_U} = \chi_E^0|_{F^\times \omega_E}.$$

The equality defining D follows from Lemma 3.6. The pullback conductors are n over U by [2, Lemma 3.11] and $2n - T$ over E by (33); both are at least $2T$. Choose a stationary coefficient b_E for $\lambda \circ N_{E/F}$. The coefficient comparison in [4, Lemma 10.17] gives $b_E/b_F \in U_E^{n-T}$. Since $n - T \geq T = m_E(\chi_E^0)$, $\chi_E^0(b_E^{-1}) = \chi_E^0(c_F)$. Over U , one may use b_F itself by (51). Thus (30) gives, for $L = U, E$,

$$\Delta_L(\theta_L) = \chi_L^0(c_F) \Delta_L(\lambda \circ N_{L/F}).$$

Equations (13) and (30) over F , where $m_F(\omega_L) \leq T \leq \lfloor n/2 \rfloor$, give

$$\Delta_L(\lambda \circ N_{L/F}) \Delta_F(\omega_L) = \Delta_F(\lambda) \Delta_F(\lambda \omega_L) = \omega_L(c_F) \Delta_F(\lambda)^2.$$

Multiplication yields

$$\mathcal{A}_L(\theta_L) = D(c_F) \Delta_F(\lambda, \psi_F)^2.$$

This is [4, Theorem 10.18]. □

The ramified coefficient comparison is `Dyadic.highCovectors`. The two stable-twist calculations are combined in `Dyadic.UR.stable`.

Proposition 9.8. *If $\ell = p = 2$ and $|I| = 2$, then (69) holds for every primitive compatible pair, in mixed and equal characteristic.*

Proof. Lemma 9.3 gives $h \geq 0$. Propositions 9.5, 9.6, and 9.7 cover $h = 0$, $1 \leq h < T$, and $h \geq T$, respectively. Thus

$$(75) \quad \mathcal{A}_U(\theta_U) = \mathcal{A}_E(\theta_E) \quad (h \geq 0).$$

For two ramified fields, extend their common pullback to U by Lemma 3.2 and use (75) twice. Lemma 3.3 makes the result independent of the choice over U . Equation (68) then gives (69). □

Proposition 9.8 is [4, Theorem 10.20]. The theorem `Dyadic.UR.comparison` proves it.

9.2. Breaks in a totally ramified biquadratic extension. Suppose $|I| = 4$. Let t_i be the lower break of L_i/F and label the fields so that $t_1 \leq t_2 \leq t_3$.

Lemma 9.9 (Biquadratic break patterns). *One has $t_2 = t_3$. In characteristic two, all three breaks are odd. In mixed characteristic put $e = v_F(2)$. The possibilities are*

$$(76) \quad (t_1, t_2, t_3) = (2a - 1, 2e, 2e), \quad 1 \leq a \leq e,$$

$$(77) \quad (t_1, t_2, t_3) = (2a - 1, 2r - 1, 2r - 1), \quad 1 \leq a \leq r \leq e.$$

Proof. The norm characters satisfy $\omega_1 \omega_2 = \omega_3$ by Lemma 3.1. The product of two characters of unequal conductors has the larger conductor, so the maximum of the three conductors occurs at least twice. In characteristic two, reduced Artin–Schreier representatives give odd positive breaks. In mixed characteristic, [4, Lemma 14.2] gives the possible quadratic conductors $2a$ and $2e + 1$; the product of two characters of conductor $2e + 1$ has smaller conductor. Subtracting one from the conductors, by Proposition 5.2, gives exactly (76) and (77). The full classification is [4, Lemma 14.3]. □

The relation between the breaks in the field tower is `Ramification.diamondBreaks`; the mixed-characteristic conductor alternatives are `Classification.quadraticConductors`. The resulting exhaustive case selection is part of `Classification.exhaustion`.

The tower relation of Section 7.2 gives lower breaks $2t_2 - t_1, t_1, t_1$ for $K/L_1, K/L_2, K/L_3$, respectively. As there, the model characters χ_i yield the prescribed characters by a common base-field twist. The corresponding constructions and minimal-conductor comparisons are:

Step	Equal characteristic 2	Mixed characteristic
Common element Y	Lemma 9.13	Lemmas 9.22, 9.31
Model characters χ_i	Proposition 9.15	Propositions 9.24, 9.33
Common base-field twist	Lemma 9.16	Lemmas 9.25, 9.34
Adjustment from Y to C	Lemma 9.17	Lemmas 9.27, 9.36–9.37
Critical functions	Proposition 9.18	Proposition 9.28, Lemma 9.38

The coordinates and conductor ranges differ; Lemma 8.6 gives the common algebra of the adjustment. The comparisons retain the quadratic norm-character factors and use Section 8 to determine the sign.

9.3. Equal characteristic two. Assume $\text{char } F = 2$ and K/F is totally ramified with $\text{Gal}(K/F) \simeq C_2^2$. Let k be their common residue field. Choose a uniformizer π of K and put $\Pi_i = N_{K/L_i}(\pi)$ and $\varpi = N_{K/F}\pi$. Identify $F = k((\varpi))$ and use these compatible uniformizers for the critical quotients. Such a presentation is constructed by `Residues.existsEqualCharacteristicPresentation`. By Lemma 3.6, it suffices to use

$$\psi_F(x) = (-1)^{\text{Tr}_{k/\mathbb{F}_2} \text{Res}_F(x d\varpi)},$$

whose largest trivial ideal is $\mathcal{O}_F$; the passage to arbitrary additive characters is also recorded in [4, Theorem 11.14]. Let σ_i be the nontrivial element of $\text{Gal}(K/L_i)$. Let ω_i be the nontrivial character of $F^\times / N_{L_i/F}(L_i^\times)$. For a ramified quadratic extension E/D of break t ,

$$\text{Tr}_{E/D}(\mathfrak{p}_E^a) = \mathfrak{p}_D^{\lfloor (a+t+1)/2 \rfloor}, \quad v_D(N_{E/D} x) = v_E(x), \quad m_D(\omega_{E/D}) = t + 1.$$

The trace formula is (31); the conductor formula is the first assertion of Proposition 5.2; the valuation formula is (6).

The residue formula identifies the norm characters compatibly with trace and norm.

Lemma 9.10 (Artin–Schreier norm characters). *Let f represent a nonzero class in $F/\{a^2 + a : a \in F\}$. The norm character of $F(x)/F$, $x^2 + x = f$, is*

$$(78) \quad \omega_f(u) = (-1)^{\text{Tr}_{k/\mathbb{F}_2} \text{Res}_F(f du/u)} \quad (u \in F^\times).$$

For a finite separable extension E/F , a differential η over E , and $u \in E^\times$, one has

$$(79) \quad \text{Tr}_{k_E/k} \text{Res}_E(\eta) = \text{Res}_F(\text{Tr}_{E/F} \eta), \quad \frac{dN_{E/F}(u)}{N_{E/F}(u)} = \text{Tr}_{E/F} \left(\frac{du}{u} \right).$$

Proof. The norm-character formula is [4, Lemma 11.2]. The trace–residue identities are proved in [4, Appendix B]. $\square$

The identification with the actual norm character is `Characters.artinSchreierNormCharacter_eq_normCharacter`. The two identities in (79) are `Residues.traceCompatibility` and `Residues.logNorm`, respectively.

Lemma 9.11 (Artin–Schreier invariance). *For $a \in F$ and $u \in F^\times$,*

$$(80) \quad \text{Tr}_{k/\mathbb{F}_2} \text{Res}_F \left((a^2 + a) \frac{du}{u} \right) = 0.$$

Proof. The Artin–Schreier residue pairing vanishes on the zero class represented by $a^2 + a$. This is equation (198) of [4], proved with the norm-character formula in [4, Lemma 11.2]. $\square$

Artin–Schreier invariance of the residue exponent is `EqualChar.residueExponent_artinSchreier`.

Corollary 9.12 (Rational residue identity). *For $f, w \in F$ and $H = 1 + \varpi w^2$, one has*

$$(81) \quad \text{Tr}_{k/\mathbb{F}_2} \text{Res}_F \left(\frac{\varpi f^2}{H} d\varpi \right) = \text{Tr}_{k/\mathbb{F}_2} \text{Res}_F \left(\frac{f}{H} d\varpi \right).$$

Proof. The valuation of ϖw^2 is odd when $w \neq 0$, so $H \neq 0$. Apply (80) with $a = \varpi f$ and $u = \varpi H$. Since $du = d\varpi$, the two terms in (80) are exactly the two residues in (81). $\square$

The original proof split the rational expression into two square terms and applied the Cartier operator. Here the same identity follows from Artin–Schreier invariance (80), with $a = \varpi f$ and $u = \varpi H$, so no separate Cartier calculation is needed [4, Remark 11.3]. In the formal character construction, this substitution in the invariance theorem of Lemma 9.11 is used in `Dyadic.Equal.commutator`. The separate proof of the original rational identity remains `Dyadic.Equal.cartierRational`.

Choose reduced Artin–Schreier generators $L_i = F(z_i)$ with

$$z_i^2 + z_i = f_i \quad (i = 1, 2), \quad z_3 = z_1 + z_2, \quad f_3 = f_1 + f_2.$$

Here each f_i is the sum of a constant in k and a finite sum of negative odd powers of ϖ . Retain $t_1 \leq t_2 = t_3$. For the model characters to be constructed, set the conductor and depth parameters as follows:

i	$T_i = m_F(\omega_i)$	m_i	$s_i = \lceil m_i/2 \rceil$
1	$t_1 + 1 = 2r_1$	$2T_2 - 1$	T_2
2, 3	$t_2 + 1 = 2r_2 = 2r_3$	$T_1 + T_2 - 1$	$r_1 + r_2$

Put $\delta = r_2 - r_1$. Let f_i^- be the negative-power part of the reduced Laurent series f_i . Write

$$\beta_i = \varpi^{t_2} f_i^- = d_i^2, \quad d_3 = d_1 + d_2.$$

These square roots belong to F : the negative exponents of the reduced f_i are odd, t_2 is odd, and k is perfect. Put

$$Y = d_1 z_2 + d_2 z_1, \quad B = \beta_1 f_2 + \beta_2 f_1, \quad \kappa_i = d_j d_k \quad (\{i, j, k\} = \{1, 2, 3\}).$$

As in Lemma 7.12, the norms of one element supply the model coefficients. Here Y also supplies coefficients for the quadratic norm characters, through the norms of its traces.

Lemma 9.13 (Simultaneous norm and trace formulas). *For the elements just defined,*

$$\sigma_i Y - Y = d_i, \quad \text{Tr}_{K/L_i}(Y) = d_i, \quad a_i := N_{K/L_i}(Y) = \kappa_i z_i + B,$$

and

$$v_K(Y) = -t_1, \quad v_{L_i}(a_i) = -t_1, \quad N_{K/F}(Y) = N_{L_i/F}(a_i), \quad v_F(N_{K/F}(Y)) = -t_1.$$

Proof. Expanding Y^2 gives $Y^2 = \beta_1 z_2 + \beta_2 z_1 + B$; the two involutions adding 1 to z_1 or z_2 give the three trace and norm formulas. Since the negative parts in B cancel, $B \in \mathcal{O}_F$. For $i = 2, 3$, $v_{L_i}(\kappa_i) = 2\delta = t_2 - t_1$; for $i = 1$, this valuation is zero. Hence $v_{L_i}(a_i) = -t_1$. The remaining valuations follow from (6). These are equations (201)–(202) of [4]. $\square$

The common-element identities and their valuations are `Dyadic.Equal.SimultaneousASGenerators.exact_origin` and `Dyadic.Equal.SimultaneousASGenerators.origin_orders`, respectively.

Put

$$\alpha = \varpi^{-T_2}, \quad \Psi_F(x) = \psi_F(\alpha x), \quad \Psi_{L_i}(x) = \Psi_F(\text{Tr}_{L_i/F}(x)).$$

Lemma 9.14 (Simultaneous norm-character coefficients). *For $i = 1, 2, 3$, one has*

$$\omega_i(1 - z) = \Psi_F(\beta_i z) \quad (z \in \mathfrak{p}_F^{r_i}),$$

and

$$\Psi_{L_i}(\beta_i z) = \Psi_F(\beta_i N_{L_i/F}(z)) \quad (z \in \mathfrak{p}_{L_i}^{r_i}).$$

Proof. Apply (78) and integration by parts, using $df_i = \alpha \beta_i d\varpi$. On $\mathfrak{p}_F^{r_i}$ the error in replacing $(1 - z)^{-1}$ by 1 in $f_i dz/(1 - z)$ has valuation at least $-t_i + r_i + (r_i - 1) = 0$, so it has zero residue. The second formula follows from the first by the norm-triviality argument proving (72), applied with $\Psi_F(\beta_i \cdot)$. The depth checks are the norm-character calculation in [4, Section 11.3]. $\square$

The lower-character formulas are `Dyadic.Equal.dyadicEqual_lowerCharts`; the norm-phase identity is `Dyadic.Equal.originAddChar_normPhase`.

Proposition 9.15 (Model characters at minimal conductors). *There are primitive compatible characters χ_i of exact conductors m_i satisfying*

$$\chi_i(1 - z) = \Psi_{L_i}(a_i z) \quad (z \in \mathfrak{p}_{L_i}^{s_i}).$$

Proof. The extension-and-descent step is as in Proposition 7.15. Here principal-unit compatibility uses Lemmas 8.2 and 9.14, and the conjugate-quotient calculation uses (81); the checks and extension are [4, Lemmas 11.4–11.5 and Theorem 11.6]. By (31), the largest trivial ideals of Ψ_{L_i} have exponents $2T_2 - T_1, T_2, T_2$. Subtracting $v_{L_i}(a_i) = -t_1$ gives the exact conductors m_i . $\square$

The extension to actual compatible characters is [Dyadic.Equal.models](#).

Lemma 9.16 (Common twist and conductor alternatives). *For every primitive compatible family (θ_i) , there are primitive compatible characters (χ_i) satisfying Proposition 9.15 and a character λ of $F^\times$ such that*

$$\theta_i = \chi_i(\lambda \circ N_{L_i/F}).$$

Put $n = m_F(\lambda)$. Then

$$(82) \quad \begin{cases} m_{L_i}(\theta_i) = m_i, & n \leq T_2 + r_1 - 1, \\ m_{L_i}(\theta_i) = 2n - T_i, & n \geq T_2 + r_1. \end{cases}$$

Proof. Unlike the deep norm-kernel calculation of Lemma 7.17, the constructed models already make θ_1/χ_1 invariant by Lemma 3.3. Descend this quotient by Lemma 3.2. Absorb the remaining norm-character twists into χ_2, χ_3 ; their conductors are at most $T_1 \leq s_i$ for $i = 2, 3$, so the model formulas and common pullback are unchanged. The lower-bound argument of Lemma 7.14 and (33) give the alternatives: the model conductors m_i are odd and the larger pullback conductors $2n - T_i$ are even, so no equal-conductor cancellation occurs. See [4, Proposition 11.7]. $\square$

The simultaneous choice of the model family and the common twist, with these conductor alternatives, is [Dyadic.Equal.twist](#).

To incorporate the twist, use Lemma 8.6 with Y from Lemma 9.13 and $X \in L_3$. Write $a = \text{Tr}_{L_3/F}(X)$; in characteristic two $C = Y + X$. The formulas specialize to

$$(83) \quad Q_i = aY + d_i X, \quad \text{Tr}_{K/L_i}(C) = d_i + a \quad (i = 1, 2),$$

and

$$(84) \quad N_{L_i/F}(Q_i) - \beta_i A = E, \quad E = a(aa_3 + \kappa_3 X) \in F.$$

Indeed $X' + X = a$ and $a_3 + a'_3 = \kappa_3$. These are equations (218)–(219) of [4], formalized by [Dyadic.Equal.dyadicEqual_thirdNorm_identities](#).

Fix the characters and twist of Lemma 9.16. By Lemma 3.6, the characters

$$D_\theta = \theta_i|_{F^\times} \omega_i, \quad D_\chi = \chi_i|_{F^\times} \omega_i$$

are independent of i .

Lemma 9.17 (Simultaneous stationary norms). *Suppose $n \leq T_2 + r_1 - 1$. There is $X \in L_3$ such that $C = Y + X$ has $v_K(C) = -t_1$, $\text{Tr}_{K/L_i}(C) \neq 0$, and*

$$\theta_i(1 - z) = \Psi_{L_i}(N_{K/L_i}(C) z) \quad (z \in \mathfrak{p}_{L_i}^{s_i}), \quad \omega_i(1 - z) = \Psi_F(N_{L_i/F}(\text{Tr}_{K/L_i}(C)) z) \quad (z \in \mathfrak{p}_F^{r_i})$$

for $i = 1, 2$. If $t_1 = t_2$, the same C satisfies these formulas for all three indices.

Proof. If $n \leq r_1 + r_2$, take $X = 0$. Otherwise the additive coefficient A_* of λ on $\mathfrak{p}_F^{r_1+r_2}$ is determined modulo $\mathfrak{p}_F^\delta$. With $h = n - T_2$, the needed relative precision is $\delta + h \leq r_2 - 1 \leq t_2$; Lemma 5.3 therefore gives $A = N_{L_3/F}(X) \equiv A_* \pmod{\mathfrak{p}_F^\delta}$. Lemma 8.6, (84), and the trace–norm identity of Lemma 9.14 give coefficient $A + Q_i$ for the twist. Multiplying by the model formula gives $a_i + A + Q_i = N_{K/L_i}(C)$ for θ_i . For ω_i , the change is $N_{L_i/F}(\text{Tr}_{K/L_i}(C)) - \beta_i = a^2$; the trace bound puts this change in the annihilator of $\mathfrak{p}_F^{r_i}$. The exact valuation checks, nonvanishing of $\text{Tr}_{K/L_i}(C)$, and the third index when the breaks agree are [4, Lemma 11.8]. $\square$

The simultaneous stationary coefficients, including all three indices at equal breaks, are constructed by [Dyadic.Equal.minimalOrigin](#).

Proposition 9.18 (Critical-sum comparison at minimal conductors). *If $n \leq T_2 + r_1 - 1$, then $\mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_2}(\theta_2)$.*

Proof. Take C from Lemma 9.17, and use $N_{K/L_i}(C)$ for H_i, g_i . Since each T_i is even, its critical sum is 1 by (22)–(23). Equation (70) therefore reduces the assertion to $g_1 = g_2$, and at equal breaks to equality of all three sums. Identify the critical quotient for θ_i with k by $\Pi_i^{\lfloor m_i/2 \rfloor}$, where $\lfloor m_1/2 \rfloor = t_2$ and $\lfloor m_2/2 \rfloor = \lfloor m_3/2 \rfloor = (t_1 + t_2)/2$. For $z \in \mathcal{O}_K$, put $C_z = C(1 + \pi^{t_2}z)$ and define

$$p_i(\bar{z}) = \frac{N_{K/L_i}(C_z)/N_{K/L_i}(C) - 1}{\Pi_i^{\lfloor m_i/2 \rfloor}}.$$

The trace bounds in [4, Propositions 11.9–11.10] put the relative changes of $\text{Tr}_{K/L_i}(C)$ in the stationary ideals for ω_i . Lemma 8.4 therefore gives

$$H_i(p_i(\bar{z})) = Q(\bar{z})$$

for one function $Q : k \rightarrow \mathbf{C}^\times$ independent of i .

If $t_1 < t_2$, the norm expansion gives $p_1(z) = z^2$. For K/L_2 , (31) gives consecutive trace ideals of depths $(t_1 + t_2)/2$ and $(t_1 + t_2)/2 + 1$ at source depths $t_2, t_2 + 1$; the norm term is deeper. Thus p_2 is a nonzero k -linear map. Both maps are bijections, so summing gives $g_1 = g_2$.

If $t_1 = t_2 = t$, put

$$u = \overline{C\pi^t}, \quad c_i = \overline{\sigma_i(C) - C},$$

where σ_i is the automorphism fixed above. By Lemma 9.13 and the ramification bound for the adjustment X , one has $c_i = \bar{d}_i$. All three d_i are units and $d_3 = d_1 + d_2$, so the c_i are nonzero and pairwise distinct. The norm expansions give $p_i(z) = z^2 + (c_i/u)z$; their kernels $\{0, c_i/u\}$ have order two and are distinct. Lemma 4.3 gives nondegeneracy, and Lemma 8.5 gives equality of the three normalized sums. Equation (70) proves the assertion. These two cases are [4, Propositions 11.9–11.10]. $\square$

The common critical functions are `Dyadic.Equal.dyadicEqual_minimal_commonFunctions`. The two residue-map arguments are `Dyadic.Equal.unequalBreaks` and `Dyadic.Equal.equalBreaks`.

Proposition 9.19 (Comparison with even conductors). *If*

$$T_2 + r_1 \leq n \leq 2T_2 + r_1 - 2,$$

then $\mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_2}(\theta_2)$.

Proof. For this analogue of Proposition 7.18, [4, Theorem 11.11] constructs $X \in L_3$ and $C = Y + X$ with $N_{K/L_i}(C)$ stationary for θ_i . The conductors $2n - T_i$ and T_i are even, so Equation (24), with β_i for ω_i , gives

$$\mathcal{A}_{L_i}(\theta_i) = D_\theta(\alpha^{-1})\Theta(C)^{-1}\Psi_F(\text{Tr}_{L_i/F}(N_{K/L_i}(C)) + \beta_i).$$

Put $a = \text{Tr}_{L_3/F}(X)$. By Lemma 8.2 and (83),

$$\text{Tr}_{L_i/F}(N_{K/L_i}(C)) + \beta_i = E_2(C) + (d_i + a)^2 + d_i^2 = E_2(C) + a^2,$$

which is independent of i . This proves the assertion. $\square$

The coefficient construction and the common-phase equality in this range are combined in `Dyadic.Equal.intermediate`.

In the high range, stable twisting requires the values of χ_i at the changed stationary coefficients.

Proposition 9.20 (Stable twisting with a common factor). *If $n \geq 2T_2 + r_1 - 1$, then $\mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_2}(\theta_2)$.*

Proof. Put $h = n - T_2$ and choose A with $\lambda(1 - z) = \Psi_F(Az)$ on $\mathfrak{p}_F^{\lfloor n/2 \rfloor}$. Lemma 5.3, at precision t_i , gives

$$N_{L_i/F}(x_i) = (A/\beta_i)\varepsilon_i, \quad \varepsilon_i \in U_F^{t_i}.$$

Put $c_F = (\alpha A)^{-1}$ and $c_i = (\alpha(A + \beta_i x_i))^{-1}$. By [4, Lemma 11.12], c_i^{-1} is a stationary coefficient for $\lambda \circ N_{L_i/F}$, and

$$\chi_i(c_i) = \chi_i(c_F) \Psi_F(N_{K/F}(Y)/A).$$

The half-conductors of the pullbacks satisfy

$$n - r_1 \geq 2T_2 - 1 = m_1, \quad n - r_2 \geq T_1 + T_2 - 1 = m_2,$$

and $n \geq 2T_2 \geq 2T_i$. Apply (30) over L_i , then (13) and (30) over F . The result is

$$\mathcal{A}_{L_i}(\theta_i) = D_\chi(c_F) \Psi_F(N_{K/F}(Y)/A) \Delta_F(\lambda, \psi_F)^2.$$

The character D_χ is independent of i by Lemma 3.6. This is [4, Theorem 11.13]. $\square$

The coefficients and their character values are `Dyadic.Equal.highCoefficient`; the common value of the local-constant products is `Dyadic.Equal.highComparison`.

Proposition 9.21. *For a totally ramified biquadratic extension in characteristic two, (69) holds for every primitive compatible pair.*

Proof. The conductor alternatives in (82) are exhausted by Propositions 9.18, 9.19, and 9.20. They give

$$\mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_2}(\theta_2).$$

Interchanging L_2, L_3 gives the other comparison. Equation (68) then gives (69) for the chosen additive character. Every nontrivial additive character of F is its scalar multiple; Lemma 3.6 gives the assertion for all of them. This is [4, Theorem 11.14]. $\square$

The final comparison is `Dyadic.Equal.comparison`.

9.4. Mixed characteristic and a break equal to $2e$. Assume K/F is totally ramified, $F/\mathbb{Q}_2$ is finite, $e = v_F(2)$, and

$$t_1 = 2a - 1, \quad t_2 = t_3 = 2e, \quad 1 \leq a \leq e.$$

Put $T = 2e + 1$. The conductor and stationary-depth parameters are

i	$T_i = m_F(\omega_i)$	$q_i = \lceil T_i/2 \rceil$	m_i	$s_i = \lceil m_i/2 \rceil$
1	$2a$	a	$4e + 1$	$2e + 1$
2, 3	T	$e + 1$	$2e + 2a$	$e + a$

Here m_i are the conductors of the model characters constructed below. Write σ_i for the nontrivial automorphism of K/L_i . The different exponents of K/L_i are

$$D_{K/L_1} = 2T - T_1 = 4e - 2a + 2, \quad D_{K/L_2} = D_{K/L_3} = 2a.$$

These are the ramification data of [4, Section 12].

Put $b = 2e - 2a + 1$. This is the maximal-break counterpart of Lemma 7.12: Kummer generators are chosen together so that one element Y supplies both kinds of coefficients with the required valuations.

Lemma 9.22 (Aligned Kummer generators). *There are Kummer generators $L_1 = F(S)$, $L_2 = F(R)$, $L_3 = F(SR)$ with*

$$v_F(S^2 - 1) = v_F(R^2) = b, \quad (S^2 - 1)/R^2 \in U_F^a.$$

Put $Y = (S + R - 1)/2$, $a_i = N_{K/L_i}(Y)$, $h_i = \text{Tr}_{K/L_i}(Y)$, $\beta_i = N_{L_i/F}(h_i)$, and

$$B_0 = (S^2 - 1 - R^2)/4.$$

Then

$$\begin{aligned} a_1 &= (1 - S)/2 + B_0, & h_1 &= S - 1, & \beta_1 &= 1 - S^2, \\ a_2 &= -R/2 - B_0, & h_2 &= R - 1, & \beta_2 &= 1 - R^2, \\ a_3 &= -SR/2 - (S^2 - 1 + R^2)/4, & h_3 &= -1, & \beta_3 &= 1. \end{aligned}$$

Moreover

$$v_K Y = -t_1, \quad v_{L_i}(a_i) = -t_1, \quad v_{L_1}(h_1) = b, \quad v_{L_2}(h_2) = v_{L_3}(h_3) = 0.$$

Proof. The simultaneous generator choice is [4, Lemma 12.3]. Applying the three sign-changing involutions to Y gives the table; the displayed valuations follow from the alignment and (6). These computations are [4, Section 12.2]. $\square$

The generators are supplied by `Dyadic.Maximal.alignment`; the norm/trace table and valuations by `Dyadic.Maximal.dyadicMaximal_origin_data`. Fix these generators and the associated elements.

Choose α so that, with $\Psi_L(x) = \psi_L(\alpha x)$,

$$\omega_3(1 - z) = \Psi_F(z) \quad (z \in \mathfrak{p}_F^{e+1}), \quad v_F(\alpha) = -n_F(\psi_F) - T.$$

Lemma 9.23 (Simultaneous norm-character coefficients). *For $i = 1, 2, 3$, the norm characters satisfy*

$$(85) \quad \omega_i(1 - z) = \Psi_F(\beta_i z) \quad (z \in \mathfrak{p}_F^{q_i}),$$

and

$$\Psi_{L_i}(\beta_i x) = \Psi_F(\beta_i N_{L_i/F}(x)) \quad (x \in \mathfrak{p}_{L_i}^{q_i}).$$

Proof. The simultaneous coefficients are [4, Lemma 12.4]. The second formula follows from the first by the norm-triviality argument proving (72); the trace and norm lie in $\mathfrak{p}_F^{q_i}$ by (31) and (6). $\square$

The common normalization and the three norm-character formulas are `Dyadic.Maximal.lowerCharacters`.

Proposition 9.24 (Model characters at minimal conductors). *There are primitive compatible characters χ_i of conductors m_i with*

$$(86) \quad \chi_i(1 - z) = \Psi_{L_i}(a_i z) \quad (z \in \mathfrak{p}_{L_i}^{s_i}).$$

Proof. As in Proposition 7.15, extend a unit prescription and descend its invariant norm pullback. Here one starts with L_2 ; the quadratic compatibility and norm-image checks are [4, Lemma 12.5 and Theorem 12.6]. Lemma 9.22 and (31) give the exact conductors m_i from the coefficient valuations. $\square$

The compatible family with the prescribed principal-unit formulas is constructed by `Dyadic.Maximal.models`.

Lemma 9.25 (Common twist up to conjugation). *For every primitive compatible family, there are characters (χ_i) as in Proposition 9.24 and a character λ of $F^\times$ such that, after conjugating θ_1 if necessary,*

$$\theta_i = \chi_i(\lambda \circ N_{L_i/F}) \quad (i = 1, 2).$$

If $n = m_F(\lambda)$, then

$$(87) \quad \begin{cases} m_{L_i}(\theta_i) = m_i, & n \leq 2e + a, \\ m_{L_1}(\theta_1) = 2n - 2a, \quad m_{L_2}(\theta_2) = 2n - T, & n \geq 2e + a + 1. \end{cases}$$

Proof. Use the descent argument of Lemma 9.16, starting with θ_2/χ_2 . The remaining twist on L_1 is removed by conjugating θ_1 , as in Proposition 7.15; this preserves the common pullback and local constant by (16) and Lemma 2.2. For each field, the model conductor and the higher norm-pullback conductor in (33) have opposite parity. The resulting alternatives are [4, Lemma 12.7]. $\square$

The model characters and twist for the prescribed family, including the possible conjugation and the conductor alternatives, are `Dyadic.Maximal.twist`. Use the characters and twist of Lemma 9.25, and put

$$D_\theta := \theta_i|_{F^\times} \omega_i,$$

which is independent of i by Lemma 3.6.

An additive coefficient need not itself be a norm from L_3 . As in the coefficient adjustments of Proposition 7.18, one changes the normalization without changing the character formulas.

Lemma 9.26 (Norm choice by rescaling). *For every $A_* \in F^\times$, there are $u \in U_F^{2e}$ and $X \in L_3^\times$ such that*

$$A := A_*/u = N_{L_3/F}(X), \quad (\alpha u)A = \alpha A_*.$$

The lower formulas (85) and (86) are unchanged by $\alpha \mapsto \alpha u$.

Proof. Since ω_3 has conductor $2e + 1$, its restriction to U_F^{2e} is nontrivial. Choose u with $\omega_3(u) = \omega_3(A_*)$. Then A_*/u is a norm from L_3 . The coefficient valuation checks that preserve (85) and (86) are [4, Lemma 12.8]. The identity $(\alpha u)(A_*/u) = \alpha A_*$ preserves the formula for the twisting character. $\square$

Preservation of the displayed formulas and the exact norm choice are [Dyadic.Maximal.norma lization](#). The group-theoretic choice of the unit is the general lemma [Stationary.normalizati onFreedom](#).

Lemma 9.27 (Valuations after a quadratic shift). *For $X \in L_3$, use A, C, Q_i of Lemma 8.6 with Y from Lemma 9.22, and put*

$$p_X = \text{Tr}_{L_3/F}(X), \quad \mathcal{E} = (X' - X)(a_3 X' - a'_3 X), \quad \Delta = p_X^2 + 2p_X.$$

The last two differences in Lemma 8.6 are $\mathcal{E}$ and Δ , respectively. If $v_{L_3}(X) = -h$, then

$$v_F p_X \geq e - \lfloor h/2 \rfloor, \quad v_F \mathcal{E} \geq T - a - h, \quad v_{L_1} Q_1 \geq b - h, \quad v_{L_2} Q_2 \geq -h.$$

Proof. Lemma 9.22 gives $\text{Tr}_{K/F}(Y) = -2$, so Lemma 8.6 gives Δ . The trace bound is (31); the other bounds are [4, Lemma 12.9]. $\square$

The specialized identities and their valuation bounds are [Dyadic.Maximal.commonError](#).

Proposition 9.28 (Matching the two critical sums). *If $n \leq 2e + a$, then*

$$(88) \quad \mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_2}(\theta_2).$$

Proof. For $n \leq e + a$, take $X = 0$ and $C = Y$. Otherwise choose A_* from $\lambda(1 - z) = \Psi_F(A_* z)$ on $\mathfrak{p}_F^{e+a}$. By Lemma 9.26, choose $A = A_*/u = N_{L_3/F}(X)$, replace α by αu , and redefine $\Psi_L = \psi_L(\alpha \cdot)$ with this adjusted α . Then $\lambda(1 - z) = \Psi_F(Az)$ on $\mathfrak{p}_F^{e+a}$, and $a - e \leq h = n - T \leq a - 1$. Use the notation of Lemma 9.27. By [4, Lemma 12.10], $N_{K/L_i}(C)$ and $N_{L_i/F}(\text{Tr}_{K/L_i}(C))$ are stationary coefficients for θ_i and ω_i , respectively, with $\text{Tr}_{K/L_i}(C) \neq 0$. Use these coefficients for the critical functions and sums of θ_i and ω_i . Only θ_1 and ω_2 have odd conductors; all other critical sums are 1 by (22)–(23). Equation (70) therefore reduces the assertion to $g_{\theta_1} = g_{\omega_2}$. Choose a uniformizer π of K , put $\Pi_1 = N_{K/L_1}(\pi)$ and $\varpi = N_{L_1/F}(\Pi_1)$, and set $C_z = C(1 + \pi^{2e} z)$ for $z \in \mathcal{O}_K$. The two critical maps on the common residue field k are

$$p(\bar{z}) = \frac{N_{K/L_1}(C_z)/N_{K/L_1}(C) - 1}{\Pi_1^{2e}}, \quad q(\bar{z}) = \frac{N_{L_2/F}(\text{Tr}_{K/L_2}(C_z))/N_{L_2/F}(\text{Tr}_{K/L_2}(C)) - 1}{\varpi^e}.$$

They are bijective by the trace and norm estimates in [4, Theorem 12.11]. With these identifications of the critical quotients, Lemma 8.4 gives

$$H_{\theta_1}(p(z)) = H_{\omega_2}(q(z)) \quad (z \in k).$$

Summing through the two bijections and dividing by $|k|^{1/2}$ gives $g_{\theta_1} = g_{\omega_2}$. Equation (70) now gives (88). This is [4, Theorem 12.11]. $\square$

The simultaneous stationary coefficients are [Dyadic.Maximal.minimalOrigin](#); the critical-sum comparison and the resulting equality are [Dyadic.Maximal.minimalComparison](#).

Proposition 9.29 (Reciprocal critical sums). *If $n \geq 2e + a + 1$, then*

$$(89) \quad \mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_2}(\theta_2).$$

Proof. Put $h = n - T \geq a$ and choose A_* from the stationary formula for λ on $\mathfrak{p}_F^{[n/2]}$. Use Lemma 9.26 to choose $A = A_*/u = N_{L_3/F}(X)$ with $v_{L_3}(X) = -h$. Replace α by αu and set $\Psi_L = \psi_L(\alpha \cdot)$ and $c_0 = -\alpha^{-1}$ using the adjusted α . Then $\lambda(1 - z) = \Psi_F(Az)$ on $\mathfrak{p}_F^{[n/2]}$. Use Lemma 9.27. The conductors in (87) have stationary depths

$$b_1 = T + h - a, \quad b_2 = e + 1 + h.$$

The same $C = Y - X$ satisfies

$$\theta_i(1 - z) = \Psi_{L_i}(N_{K/L_i}(C)z) \quad (z \in \mathfrak{p}_{L_i}^{b_i}),$$

by [4, Lemma 12.12]. Since $v_K(X) = -2h < -t_1 = v_K(Y)$, one has $v_K(C) = -2h$, so $C \neq 0$. Use $N_{K/L_i}(C)$ for $H_{\theta_i}, g_{\theta_i}$ and β_i from (85) for $H_{\omega_i}, g_{\omega_i}$. Lemma 8.2 and $N_{L_i/F}(\text{Tr}_{K/L_i}(C)) = \beta_i + \Delta$ give

$$-\text{Tr}_{L_i/F}(N_{K/L_i}(C)) - \beta_i = -E_2(C) + \Delta.$$

Equation (24) and Lemma 3.6 therefore give

$$\mathcal{A}_{L_i}(\theta_i) = D_{\theta}(c_0)\Theta(C)^{-1}\Psi_F(-E_2(C) + \Delta)g_{\theta_i}g_{\omega_i}.$$

Since the conductors of θ_1, ω_1 are even, $g_{\theta_1} = g_{\omega_1} = 1$. Instead of the fourth-power conclusion in Proposition 7.18, we must prove the exact identity $g_{\theta_2}g_{\omega_2} = 1$. Write $\text{gr}_j L = \mathfrak{p}_L^j / \mathfrak{p}_L^{j+1}$. For a norm N , $\text{gr } N$ is the map induced by $x \mapsto N(1 + x) - 1$ on the indicated quotients. Use the two residue maps

$$(90) \quad \text{gr}_b K \xrightarrow{/C} \text{gr}_{b+2h} K \xrightarrow{\text{gr } N_{K/L_2}} \text{gr}_{e+h} L_2$$

and

$$(91) \quad \text{gr}_b K \xrightarrow{\text{Tr}_{K/L_2}} \text{gr}_e L_2 \xrightarrow{/h_2} \text{gr}_e L_2 \xrightarrow{\text{gr } N_{L_2/F}} \text{gr}_e F,$$

Denote these maps by p and q . In the first, division by C is an isomorphism since $v_K(C) = -2h$; the norm map is an isomorphism since $b + 2h = 2(e + h) - (2a - 1)$ and $e + h > 2a - 1$. In the second, (31) gives trace-ideal depths $e, e + 1$ at source depths $b, b + 1$, so the induced trace is an isomorphism. The element h_2 is a unit, and the final norm map is an isomorphism at depth $e < 2e$. Both norm assertions are [2, Theorem 3.7].

For $V \in \mathfrak{p}_K^b$, put $C' = C + V$ and $Y' = Y + V$. The quadratic norm expansion and Lemma 8.2 give

$$\text{Tr}_{L_i/F}(N_{K/L_i}(C') - N_{K/L_i}(C)) + N_{L_i/F}(\text{Tr}_{K/L_i}(Y')) - \beta_i = E_2(C') - E_2(C) + p_X \text{Tr}_{K/F} V,$$

independent of i . The multiplicative values are the common $\Theta(C'/C)$ and the norm-character value 1. For θ_1 , the norm expansion and (31) put $N_{K/L_1}(C')/N_{K/L_1}(C) - 1$ in $\mathfrak{p}_{L_1}^{b_1}$: the trace and norm terms have depths at least $3e - 2a + 1 + h \geq b_1$ and $b + 2h \geq b_1$. Also $\text{Tr}_{K/L_1}(V)/h_1 \in \mathfrak{p}_{L_1}^e$, so the corresponding argument for ω_1 lies in $\mathfrak{p}_F^e$. Their critical functions are therefore 1 on these arguments, by their stationary formulas and even conductors. Consequently

$$H_{\theta_2}(p(v))H_{\omega_2}(q(v)) = 1 \quad (v \in \text{gr}_b K).$$

The critical functions have values of modulus one. Reindexing by the two bijections therefore gives $g_{\theta_2} = \overline{g_{\omega_2}}$; (24) gives $|g_{\omega_2}| = 1$, hence $g_{\theta_2}g_{\omega_2} = 1$. Substitution gives (89). This is [4, Theorem 12.13]. $\square$

The higher-conductor coefficient formulas are `Dyadic.Maximal.higherCoefficients`. The reciprocal product of critical sums and the full comparison are combined in `Dyadic.Maximal.higherComparison`. The original proof expanded both residue maps and checked directly that their leading terms were nonzero. In (90)–(91), the same maps are compositions of graded trace and norm isomorphisms and multiplication by nonzero elements, so bijectivity follows without those expansions [4, Remark 12.14]. This factorization is formalized by `Dyadic.Maximal.higherResidueBijections`.

Proposition 9.30. *For the break pattern (76), (69) holds for every primitive compatible family.*

Proof. Equation (87) gives the two consecutive conductor ranges. Propositions 9.28 and 9.29 prove the comparison in those ranges. Repeating the same construction with L_3 in place of L_2 gives

$$\mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_3}(\theta_3),$$

so

$$\mathcal{A}_{L_2}(\theta_2) = \mathcal{A}_{L_3}(\theta_3).$$

□

This is [4, Theorem 12.1]. The Lean theorem is `Dyadic.Maximal.comparison`.

9.5. Mixed characteristic and breaks less than $2e$. Assume K/F is totally ramified, $F/\mathbb{Q}_2$ is finite, and

$$t_1 = 2a - 1, \quad t_2 = t_3 = 2r - 1, \quad 1 \leq a \leq r \leq e = v_F(2).$$

Put $\delta = r - a$ and $T = 2r$. The conductor and stationary-depth parameters are

i	$T_i = m_F(\omega_i)$	$q_i = \lceil T_i/2 \rceil$	m_i	$s_i = \lceil m_i/2 \rceil$
1	$2a$	a	$4r - 1$	$2r$
2, 3	T	r	$2a + 2r - 1$	$a + r$

Here m_i are the conductors of the model characters constructed below. Write σ_i for the nontrivial automorphism of K/L_i . The different exponents of K/L_i are

$$D_{K/L_1} = 4r - 2a, \quad D_{K/L_2} = D_{K/L_3} = 2a.$$

These are the ramification data of [4, Section 13].

Lemma 9.31 (Aligned quadratic generators). *There are generators $L_1 = F(x)$ and $L_2 = F(y)$ with*

$$x^2 + x = f, \quad y^2 + y = g, \quad v_F(f) = 1 - 2a, \quad v_F(g) = 1 - 2r,$$

and $d \in F$ such that

$$v_F(d) = \delta, \quad E := f + d^2g \in \mathfrak{p}_F^{1-a}, \quad k_0 := 1 + d \in \mathcal{O}_F^\times.$$

Put

$$z = x + y + 2xy \in L_3, \quad Y = x + dy, \quad a_i = N_{K/L_i}(Y), \quad h_i = \text{Tr}_{K/L_i}(Y), \quad \beta_i = N_{L_i/F}(h_i).$$

Then

$$\begin{aligned} z^2 + z &= f + g + 4fg, \\ h_1 &= 2x - d, & h_2 &= 2dy - 1, & h_3 &= -k_0, \\ \beta_1 &= d^2 + 2d - 4f, & \beta_2 &= 1 + 2d - 4d^2g, & \beta_3 &= k_0^2, \\ a_1 &= 2f - E - k_0x, & a_2 &= E - 2f - dk_0y, & a_3 &= -E - dz. \end{aligned}$$

Moreover

$$v_K Y = v_{L_i}(a_i) = -t_1, \quad v_{L_1}(h_1) = 2\delta, \quad v_{L_2}(h_2) = v_{L_3}(h_3) = 0.$$

Proof. The simultaneous choice is [4, Lemma 13.2]. The involutions $x \mapsto -1 - x$ and $y \mapsto -1 - y$ give the norm and trace table by expansion. The valuations follow from the alignment and (6); see [4, Section 13.2]. □

The aligned generators are constructed by `Dyadic.Nonmaximal.alignment`; the complete table and its valuations are `Dyadic.Nonmaximal.dyadicNonmaximal_origin_data`. Fix these generators and the associated elements.

Lemma 9.32 (Simultaneous norm-character coefficients). *There is $\alpha \in F^\times$ with $v_F(\alpha) = -n_F(\psi_F) - T$ such that, writing $\Psi_L(u) = \psi_L(\alpha u)$, one has*

$$(92) \quad \omega_i(1 - u) = \Psi_F(\beta_i u) \quad (u \in \mathfrak{p}_F^{q_i}),$$

and

$$\Psi_{L_i}(\beta_i v) = \Psi_F(\beta_i N_{L_i/F}(v)) \quad (v \in \mathfrak{p}_{L_i}^{q_i}).$$

Proof. The simultaneous coefficients are [4, Lemma 13.3]. The second identity follows by the norm-triviality argument proving (72), applied to (92); the trace and norm are in $\mathfrak{p}_F^{q_i}$. $\square$

The simultaneous normalization and the norm-character formulas are `Dyadic.Nonmaximal.lowerCharacters`. Fix α as in Lemma 9.32.

Proposition 9.33 (Model characters at minimal conductors). *There are primitive compatible characters χ_i of conductors m_i with*

$$(93) \quad \chi_i(1 - v) = \Psi_{L_i}(a_i v) \quad (v \in \mathfrak{p}_{L_i}^{s_i}).$$

Proof. Repeat the extension-and-descent construction of Proposition 9.15. Its mixed-characteristic conjugate-quotient and compatibility checks are [4, Lemmas 13.4–13.5 and Theorem 13.6]. The coefficient valuations in Lemma 9.31 and (31) give the exact conductors as before. $\square$

The compatible character construction with these exact conductors is `Dyadic.Nonmaximal.models`.

Lemma 9.34 (Common twist and conductor alternatives). *For every primitive compatible family (θ_i) , there are primitive compatible characters (χ_i) as in Proposition 9.33 and a character λ of $F^\times$ such that*

$$\theta_i = \chi_i(\lambda \circ N_{L_i/F})$$

If $n = m_F(\lambda)$, then

$$(94) \quad \begin{cases} m_{L_i}(\theta_i) = m_i, & n \leq 2r + a - 1, \\ m_{L_i}(\theta_i) = 2n - T_i, & n \geq 2r + a. \end{cases}$$

Proof. Repeat the descent-and-twist argument of Lemma 9.16. The twists absorbed into χ_2, χ_3 have conductors at most $2a \leq a + r = s_i$, so their principal-unit formulas are unchanged. As there, odd model conductors and even higher pullback conductors exclude equal-conductor cancellation; (33) gives the threshold $n = 2r + a$. See [4, Lemma 13.7]. $\square$

The simultaneous model family and base twist, with these conductor alternatives, are constructed by `Dyadic.Nonmaximal.twist`. Use the twist of Lemma 9.34, and put

$$D_\theta := \theta_i|_{F^\times} \omega_i,$$

which is independent of i by Lemma 3.6.

Lemma 9.35 (Norm choice by rescaling). *For every $A_* \in F^\times$, there are $u \in U_F^{T-1}$ and $X \in L_3^\times$ such that*

$$A := A_*/u = N_{L_3/F}(X), \quad (\alpha u)A = \alpha A_*.$$

The formulas (92) and (93) are unchanged by $\alpha \mapsto \alpha u$.

Proof. As in Lemma 9.26, choose u with $\omega_3(u) = \omega_3(A_*)$, now using nontriviality on U_F^{T-1} . Then A_*/u is a norm. The valuation checks preserving both character formulas are [4, Lemma 13.8], and $(\alpha u)(A_*/u) = \alpha A_*$ preserves the twisting coefficient. $\square$

Preservation of the character formulas and the exact norm choice are `Dyadic.Nonmaximal.normalization`. It applies the general unit choice `Stationary.normalizationFreedom`.

Lemma 9.36 (Valuations after a quadratic shift). *For $X \in L_3$, use A, C, Q_i of Lemma 8.6 with Y from Lemma 9.31, and put $p_X = \text{Tr}_{L_3/F}(X)$. The last two differences in Lemma 8.6 are*

$$(95) \quad \mathcal{E} = (X' - X)(a_3 X' - a'_3 X),$$

$$(96) \quad \Delta = p_X^2 + 2k_0 p_X.$$

If $v_{L_3}(X) = -h$, then

$$v_F p_X \geq r - \lceil h/2 \rceil, \quad v_F \mathcal{E} \geq T - a - h, \quad v_{L_1} Q_1 \geq 2\delta - h, \quad v_{L_2} Q_2 \geq -h.$$

Proof. Lemma 9.31 gives $\text{Tr}_{K/F}(Y) = -2k_0$, so Lemma 8.6 gives (96). The trace bound is (31); the bounds for Q_i and $\mathcal{E}$ in (95) are [4, Lemma 13.9]. $\square$

The specialized identities and their valuation bounds are `Dyadic.Nonmaximal.commonError`.

Lemma 9.37 (Simultaneous stationary norms). *Suppose $n \leq 2r + a - 1$. After a normalization change allowed by Lemma 9.35, with α, Ψ_L denoting the adjusted data, there is $X \in L_3$ such that $C = Y - X$ has $v_K(C) = -t_1$, $\text{Tr}_{K/L_i}(C) \neq 0$, and*

$$\theta_i(1 - v) = \Psi_{L_i}(\text{N}_{K/L_i}(C)v) \quad (v \in \mathfrak{p}_{L_i}^{s_i}), \quad \omega_i(1 - u) = \Psi_F(\text{N}_{L_i/F}(\text{Tr}_{K/L_i}(C))u) \quad (u \in \mathfrak{p}_F^{q_i})$$

for $i = 1, 2$. If $a = r$, the same C works for all three indices.

Proof. If $n \leq r + a$, keep α unchanged and take $X = 0$, $C = Y$. Otherwise choose a coefficient A_* for λ on $\mathfrak{p}_F^{r+a}$. By Lemma 9.35, choose $A = A_*/u = \text{N}_{L_3/F}(X)$, replace α by αu , and redefine $\Psi_L = \psi_L(\alpha \cdot)$ with this adjusted α . Then $\lambda(1 - z) = \Psi_F(Az)$ on $\mathfrak{p}_F^{r+a}$ and $h = n - T \leq a - 1$. Lemma 8.6 and the estimates in Lemma 9.36 give the coefficients $\text{N}_{K/L_i}(C)$ and $\text{N}_{L_i/F}(\text{Tr}_{K/L_i}(C))$ on the indicated ideals. The required error bounds, nonvanishing of the traces, and the third index when $a = r$ are [4, Lemma 13.10]. $\square$

The simultaneous stationary coefficients and the third-index assertion at equal breaks are `Dyadic.Nonmaximal.minimalOrigin`.

Lemma 9.38 (Common pullback of critical functions). *Use C and the adjusted α, Ψ_L from Lemma 9.37, and put $c_0 = -\alpha^{-1}$. Let H_i, g_i be the critical function and normalized sum for θ_i with coefficient $\text{N}_{K/L_i}(C)$. Then*

$$\mathcal{A}_{L_i}(\theta_i) = D_\theta(c_0)\Theta(C)^{-1}\Psi_F(-E_2(C))g_i,$$

Choose a uniformizer π of K and put $\Pi_i = \text{N}_{K/L_i}(\pi)$. For $v \in \mathcal{O}_K$, set $C_v = C(1 + \pi^{t_2}v)$ and

$$p_i(\bar{v}) = \frac{\text{N}_{K/L_i}(C_v)/\text{N}_{K/L_i}(C) - 1}{\Pi_i^{\lfloor m_i/2 \rfloor}}.$$

After identifying each critical quotient with the common residue field k by $\Pi_i^{\lfloor m_i/2 \rfloor}$, there is one function $Q : k \rightarrow \mathbf{C}^\times$ such that

$$H_i(p_i(\bar{v})) = Q(\bar{v}).$$

These assertions hold for $i = 1, 2$, and for all three indices if $a = r$.

Proof. Repeat the common-factor and pointwise arguments of Proposition 9.18, using (70) and Lemma 8.4. The conductors T_i are even, so the norm-character critical sums are 1. Here the required stationary-depth bounds for the lower norm changes are [4, Propositions 13.11–13.12]. $\square$

The factorization uses (70), and its pointwise part uses `Stationary.commonFunctionProduct_eq`. The norm-coordinate and precision checks are supplied by `Dyadic.Nonmaximal.unequalBreaks_residualFactors` for unequal breaks and `Dyadic.Nonmaximal.equalBreaks_commonCoordinate` for equal breaks. These results supply different parts of the displayed comparison, rather than a single declaration with all its conclusions.

Proposition 9.39 (Unequal breaks: bijective critical maps). *If $n \leq 2r + a - 1$ and $a < r$, then*

$$(97) \quad \mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_2}(\theta_2).$$

Proof. Use the unequal-break argument of Proposition 9.18 with the common pullback in Lemma 9.38. Here p_1 is squaring because its trace term is deeper. For p_2 , (31) gives trace depths $a + r - 1, a + r$ at source depths $2r - 1, 2r$, and the norm term is deeper. Thus both maps are bijective and $g_1 = g_2$, proving (97). See [4, Proposition 13.11]. $\square$

The unequal-break comparison is `Dyadic.Nonmaximal.unequalBreaks`.

Proposition 9.40 (Equal breaks: distinct kernels). *If $n \leq 2r + a - 1$ and $a = r$, then*

$$(98) \quad \mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_2}(\theta_2) = \mathcal{A}_{L_3}(\theta_3).$$

Proof. Use the common element and maps of Lemmas 9.37 and 9.38. Put

$$u = \overline{C\pi^{t_1}} \in k^\times, \quad c_i = \overline{\sigma_i(C) - C} \in k^\times.$$

Here $c_1 = \bar{d}$, $c_2 = 1$, and $c_3 = 1 + \bar{d}$ are nonzero and pairwise distinct, since d and $1 + d$ are units. The norm expansion gives

$$p_i(v) = v^2 + (c_i/u)v, \quad \ker p_i = \{0, c_i/u\}.$$

Lemma 9.38 supplies the common pullback. The equal-break argument of Proposition 9.18, using Lemmas 4.3 and 8.5, therefore gives equality of all three normalized sums. Substitution in Lemma 9.38 proves (98); see [4, Proposition 13.12]. $\square$

The residual cancellation is `Dyadic.Nonmaximal.equalBreaks_hasseComparison`; the full comparison for the prescribed characters is `Dyadic.Nonmaximal.equalBreaks`.

Proposition 9.41 (Higher conductors: trivial critical sums). *If $n \geq 2r + a$, then*

$$(99) \quad \mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_2}(\theta_2).$$

Proof. Put $h = n - T \geq a$ and

$$b_1 = 2r + h - a, \quad b_2 = r + h.$$

Equation (94) gives $m_{L_i}(\theta_i) = 2n - T_i = 2b_i$ for $i = 1, 2$. Choose A_* for λ on $\mathfrak{p}_F^{[n/2]}$. By Lemma 9.35, choose $A = A_*/u = N_{L_3/F}(X)$ with $v_{L_3}(X) = -h$; replace α by αu and set $\Psi_L = \psi_L(\alpha \cdot)$ and $c_0 = -\alpha^{-1}$ using the adjusted α . Then $\lambda(1 - z) = \Psi_F(Az)$ on $\mathfrak{p}_F^{[n/2]}$. Use Lemma 9.36. The norm and trace estimates in [4, Theorem 13.13], together with (93) and Lemma 8.6, show that $C = Y - X$ satisfies

$$\theta_i(1 - v) = \Psi_{L_i}(N_{K/L_i}(C)v) \quad (v \in \mathfrak{p}_{L_i}^{b_i}).$$

Since $v_K(X) = -2h < -t_1 = v_K(Y)$, one has $v_K(C) = -2h$, so $C \neq 0$. Unlike Proposition 9.29, all four conductors are even, so every critical factor is 1 by (22)–(23). Repeat its Lamprecht factorization, with $N_{K/L_i}(C)$ for θ_i and β_i from (92) for ω_i . Lemma 8.2 and (96) give the additive factor $\Psi_F(-E_2(C) + \Delta)$, hence

$$\mathcal{A}_{L_i}(\theta_i) = D_\theta(c_0)\Theta(C)^{-1}\Psi_F(-E_2(C) + \Delta),$$

which is independent of i and proves (99). This is [4, Theorem 13.13]. $\square$

The higher-conductor stationary formulas and the common-phase comparison are proved in `Dyadic.Nonmaximal.higher`.

Proposition 9.42. *For the break pattern (77), (69) holds for every primitive compatible family.*

Proof. The first conductor range in (94) is covered by Propositions 9.39 and 9.40; Proposition 9.41 treats the second. The same construction with L_3 in place of L_2 gives $\mathcal{A}_{L_1}(\theta_1) = \mathcal{A}_{L_3}(\theta_3)$. Together with the comparison of L_1, L_2 , this gives every pair. Use (68) to obtain (69). $\square$

This is [4, Theorem 13.1]. The Lean theorem is `Dyadic.Nonmaximal.comparison`.

10. COMPLETION OF THE MATHEMATICAL PROOF

It remains to check that the ramification cases exhaust all extensions and that comparisons through a distinguished intermediate field imply the comparison for every prescribed pair. No further local-constant calculation is required.

Proof of Theorem 1.2. For $L_1 = L_2$ there is nothing to prove. Assume $L_1 \neq L_2$.

If $\ell \neq p$, Lemma 6.1 gives the unique unramified U of degree ℓ . Lemma 3.3 excludes conductor zero for θ_U , since such a character is Galois invariant. Propositions 6.3 and 6.6 give

$$\mathcal{A}_U(\theta_U) = \mathcal{A}_E(\theta_E) \quad ([E : F] = \ell, E \neq U),$$

for conductor 1 and conductor > 1 , respectively. Hence

$$\mathcal{A}_{L_1}(\theta_{L_1}) = \mathcal{A}_U(\theta_U) = \mathcal{A}_{L_2}(\theta_{L_2}).$$

If $\ell = p > 2$, Corollary 7.20 gives the required equality.

Suppose $\ell = p = 2$. Lemmas 6.1 and 9.9 give the four possibilities

$ I = 2$	Proposition 9.8
$ I = 4, \text{ char } F = 2$	Proposition 9.21
$(t_1, t_2, t_3) = (2a - 1, 2e, 2e)$	Proposition 9.30
$(t_1, t_2, t_3) = (2a - 1, 2r - 1, 2r - 1)$	Proposition 9.42.

In each case (69) and (68) give

$$\mathcal{A}_{L_i}(\theta_i) = \mathcal{A}_{L_j}(\theta_j).$$

The two mixed-characteristic break patterns are (76) and (77). Thus (5) holds in every case. $\square$

The theorem [Classification.exhaustion](#) proves that the following seven possibilities exhaust the field data. It also supplies the ramification hypotheses required by each comparison theorem. This is the formal counterpart of [4, Lemma 14.3].

Hypotheses	Lean comparison theorem
$\ell \neq p$	Tame.comparison
$\ell = p > 2, I = p$	Odd.UR.comparison
$\ell = p > 2, I = p^2$	Odd.Total.comparison
$\ell = p = 2, I = 2$	Dyadic.UR.comparison
$\ell = p = 2, I = 4, \text{ char } F = 2$	Dyadic.Equal.comparison
Mixed characteristic, (76)	Dyadic.Maximal.comparison
Mixed characteristic, (77)	Dyadic.Nonmaximal.comparison

When a comparison is first proved through a distinguished field, [Characters.distinguishedComparisons_imply_family](#) uses Lemmas 3.2 and 3.3 to extend the comparison to every prescribed $\theta_L : L^\times \rightarrow \mathbf{C}^\times$ with $\theta_L \circ N_{K/L} = \Theta$ and to every pair of intermediate fields. The theorem [secondMainIdentity_of_actualData](#) applies the seven comparison theorems. The exported [secondMainLemma](#) applies this result to the extension and family in its statement.

11. THE EXACT LEAN THEOREM

We now identify the hypotheses and conclusion of Theorem 1.2 with the actual Lean structures. This distinguishes the inputs to the exported theorem from the coordinates, model characters, and conductor ranges constructed within its proof.

11.1. The fields and the Galois group. The ambient assumptions give fields F, K , their valuation relations and topologies, and an algebra structure expressing $F \rightarrow K$. The typeclass [IsNonarchimedeanLocalField](#) expresses the local field condition, [ValuativeExtension F K](#) expresses compatibility of the valuations, and [Module.Finite F K](#) expresses finiteness of the extension.

The remaining field hypotheses are collected in [Basic.PrimeSquareExtension](#). Its definition is:

```
structure PrimeSquareExtension
  (F K : Type) [Field F] [Field K] [Algebra F K]
  [Module.Free F K] [Module.Finite F K] (ℓ : ℕ) : Prop where
  prime : ℓ.Prime
  isGalois : IsGalois F K
  galoisGroupEquiv : Nonempty
    (Gal(K/F) ≃* (Multiplicative (ZMod ℓ) ×
      Multiplicative (ZMod ℓ)))
```

Here [Multiplicative \(ZMod ℓ\)](#) is the additive cyclic group $\mathbf{Z}/\ell\mathbf{Z}$ written multiplicatively. Thus the last field of this structure is the Galois-group condition in Section 1.

An intermediate field is an element of [IntermediateField F K](#). The condition [Module.finrank F L = ℓ](#) expresses $[L : F] = \ell$. To apply the local-constant theorems to L , Lean must first supply its local-field structure and both cyclic extensions in the tower. The declarations [Basic.inte](#)

`rmmediateFieldValuativeRel` and `Basic.intermediateFieldTopology` construct the restricted valuation and its topology on L . The theorem `Basic.intermediateField_localField` proves that L is a nonarchimedean local field. The theorem `Basic.intermediateField_tower_compatible` then supplies the compatible field tower and proves that L/F and K/L are cyclic of degree ℓ .

11.2. The compatible family. The type `ContinuousQuasiChar E` is the type of continuous group homomorphisms $E^\times \rightarrow \mathbf{C}^\times$. The structure `Basic.PrimitiveCompatibleFamily` has the following fields:

Lean field	Mathematical content
<code>topCharacter</code>	A quasi-character Θ of $K^\times$.
<code>invariant</code>	Θ is invariant under $\text{Gal}(K/F)$.
<code>not_normPullback</code>	There is no quasi-character λ of $F^\times$ with $\lambda \circ N_{K/F} = \Theta$.
<code>inducingCharacter</code>	A quasi-character $\theta_L : L^\times \rightarrow \mathbf{C}^\times$ for every $F \subset L \subset K$ with $[L : F] = \ell$.
<code>compatible</code>	$\theta_L \circ N_{K/L} = \Theta$ for every such L .

The explicit field invariant is supplied by Lemma 3.2; it does not strengthen Definition 1.1. The theorem `Characters.invariantCharacter_inducing_exists` constructs such a family from an invariant character and extends a prescribed compatible pair.

11.3. The local constant and the exported statement. The type `NormCharacter F L` is identified by `Basic.normCharacter_eq_continuousHom` with $S(L/F)$ as defined in (1). The function `Basic.inductionExpression` is

$$\Delta_L(\theta_L, \psi_F \circ \text{Tr}_{L/F}) \prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F).$$

Both occurrences of Δ are the imported function `LanglandsFirstMainLemma.localConstant`. The equality with the finite-sum definition (9) is `Basic.localConstant_eq_finite`. Thus `Basic.SecondMainIdentity` asserts equality of these expressions for every pair L_1, L_2 of degree ℓ over F . In particular, the local constant in the statement is the finite-sum constant of Section 2.2, not an abstract function assumed to satisfy the desired identities.

With these definitions, the exported theorem, in the namespace `LanglandsSecondMainLemma` and with `LanglandsFirstMainLemma` open, is:

```
theorem secondMainLemma
  (F K : Type)
  [Field F] [ValuativeRel F] [TopologicalSpace F]
  [IsNonarchimedeanLocalField F]
  [Field K] [ValuativeRel K] [TopologicalSpace K]
  [IsNonarchimedeanLocalField K]
  [Algebra F K] [ValuativeExtension F K]
  [Module.Free F K] [Module.Finite F K]
  {ℓ : ℕ} (extension : Basic.PrimeSquareExtension F K ℓ)
  (family : Basic.PrimitiveCompatibleFamily F K extension)
  (ψF : ContinuousAddChar F) (hψF : ψF ≠ 1) :
  Basic.SecondMainIdentity F K extension family ψF hψF :=
  secondMainIdentity_of_actualData F K extension family ψF hψF
```

The declaration is `secondMainLemma`. The equality $\psi_F = 1$ would mean that the additive character is trivial, so $h\psi_F$ is the hypothesis imposed in Section 1. With the preceding identifications, this is Theorem 1.2. There is no assumption on the characteristic of F in this statement. The characteristic, ramification, and conductor divisions occur in its proof, as in Section 10.

11.4. Comparison with the companion proof. The mathematical references in this paper are to the arXiv version of [4]. It already incorporates the changes listed below. The comparison is with the arguments before these Lean-suggested revisions, not with the cited arXiv version.

Each row records the original step and its replacement. The Lean declarations for each change are linked at the indicated discussion; all refer to the fixed revision in Section 12.

Remark in [4]	Change	Discussion here
4.9	Explicit coefficient calculations on the unrepresented coset are replaced by translation, using the common pullback, distinct kernels, and nondegeneracy.	Lemma 8.5
7.3	Closedness of the subfield is replaced by an explicit bound on the integral approximation valuations, which guarantees a maximum.	Lemma 5.4
7.13	Three exponent cases with $v_K(x\Delta^j) \geq 1 + \delta$ are replaced by one expansion with $x \in \mathfrak{p}_{L_2}$ for $j \neq p - 2$.	Proof of Lemma 7.13
8.10	Both exact conductors were hypotheses; now only $\theta_2 _{U_{L_2}^{m_2}} = 1$ is assumed, and both equalities are derived.	Proposition 7.15
11.3	A two-term Cartier calculation is replaced by one substitution in Artin–Schreier invariance.	Equation (81)
12.14	Direct nonvanishing checks of leading terms are replaced by factoring the same maps through graded trace and norm isomorphisms.	Equations (90)–(91)

The changes concern the intermediate proofs and, in the conductor statement, a weaker sufficient hypothesis. The statement of Theorem 1.2 is unchanged. The theorem names in Section 10 give the formal comparisons for the original characters, after all auxiliary choices have been made within their proofs.

12. SOURCE AND REPRODUCIBILITY

The fixed revisions below make the declaration links reproducible and allow the exported theorem to be checked with the same dependencies. The formalization [5] used here is the repository https://github.com/fukubillueda-web/epsilon_SML at commit

c2f3764588ceac54e49d20ed6492ac1a0c11acdb.

It uses Lean 4.32.2, Mathlib revision

905b95818eb32af7874a58b427f50c1711a5e96c,

and the First Main Lemma formalization [6] at revision

b91ad93900a5e6d9207a4e6166ac94bb3bbcf3ed.

The source is distributed under the Apache License 2.0.

With Git and the indicated Lean toolchain installed, a fresh checkout can be built and checked by running

```
git clone https://github.com/fukubillueda-web/epsilon_SML.git
cd epsilon_SML
git checkout c2f3764588ceac54e49d20ed6492ac1a0c11acdb
lake exe cache get
lake build
lake env leanchecker --fresh LanglandsSecondMainLemma.Main
```

The Lean source files contain no sorry, admit, or additional axiom declarations. The transitive axiom dependencies of the exported theorem can be inspected with

```
cat > VerifySML.lean <<'EOF'
import LanglandsSecondMainLemma.Main
#print axioms LanglandsSecondMainLemma.secondMainLemma
EOF
```

lake env lean VerifySML.lean

The file `SOURCE_MAP.md` is a label-to-file index for the completed formalization and can be used to locate the Lean source corresponding to results in the mathematical paper.

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- [3] F. Ueda, *Formalization of Langlands’s First Main Lemma for local epsilon factors*, manuscript, 2026. Available in the GitHub repository https://github.com/fukubillueda-web/epsilon_FML.
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- [5] F. Ueda, *epsilon_SML: Lean 4 formalization of Langlands’s Second Main Lemma for local epsilon factors*, version 1.0.0, GitHub release at commit `c2f3764` (full hash recorded in Section 12), Apache License 2.0, 2026; repository https://github.com/fukubillueda-web/epsilon_SML.
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